An Archival ALMA Search for Unresolved Spectral-Carrier Technosignatures toward Stars within 40 pc at 114–873 GHz

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ABSTRACT

We search ALMA archives for narrowband drifting carriers toward stars within 40 pc: 405 fine-channel windows toward 61 systems, 391 resolving a drift, 47.7 GHz of union bandwidth in 26 disjoint intervals, 114.0–873.1 GHz. 1258 coarser windows constrain only a steady unresolved excess. Spectra are de-drifted to 12.0–13.3 nHz, 6 per cent of the ceiling Sheikh et al. (2019) recommend; a crossing is one channel and drift rate at 5 times the control scatter. The median system's best window triggers on 1.5×10^{15} W nine times in ten by injection, for a carrier narrower than the instrumental channel. These limits omit atmospheric decorrelation, making them underestimates, and an opposing channel-response term; which dominates varies by window. No confirmed carrier is found. Of 56 crossings, 18 are attributed to circumstellar or interstellar lines. Of the 38 unattributed – four failing a quality criterion, two outside the control sample, one atmospheric – 31 are consistent with chance against the control positions' 24.6 ($P = 0.13$), which does not exclude up to 17 further crossings; over a narrower field, 35 against 24.1 ($P = 0.027$). None of the 33 testable crossings returned to its predicted stellar-frame cell, half a channel wide; for 25 that excludes a carrier radiating through both observations – not a persistent transmitter of any kind. The other 8 are untested. Coverage limits this search: 21 systems have no repeat at a covered frequency, where no depth could have produced a confirmation. Line emission explains the strongest quality-passing events.

Subject headings: extraterrestrial intelligence, radio lines: stars, submillimetre: stars, surveys, astrobiology

1. INTRODUCTION

The public ALMA archive holds thousands of hours of millimetre and submillimetre observations taken for other reasons, mostly studies of dust discs and planet formation. Their visibilities allow a spectrum at the position of the star at the centre of each field, at a resolution and depth set by the original observer's correlator setup. This paper searches those spectra for a narrowband transmitter at the star's own position. No such signal was found.

Searches for *technosignatures* – detectable imprints of technology, intended or not (Sheikh 2020; Vidal et al. 2026) – have examined many thousands of stars over six decades, yet the last published accounting of that effort covers only 6×10^{-18} of the space of transmitter position, power, frequency, bandwidth, polarisation and repetition rate (Wright et al. 2018), and essentially all of it lies below ∼30 GHz. Two targeted searches have reached nearby stars above that frequency. Steffes & DeBoer (1994) observed 40 solar-type stars within 23 pc at 203 GHz, and Mauersberger et al. (1996) included τ Cet and ϵ Eri among 17 targets at the same frequency. Each covered a single line, over no more than 35 MHz at narrowband resolution, with no search over drift rate, and no epoch-to-epoch test of persistence; Mason et al. (2025) has since searched more distant stars in Band 3. This survey covers 47.7 GHz in 26 disjoint intervals, searches a grid of drift rates rather than a fixed frequency, tests every crossing against the 512 control positions of its own field, and uses repeat observations to ask whether an event returns at the same frequency in the star's frame.

The band has physical advantages and physical costs. Interstellar dispersion and scintillation are negligible at these frequencies, so a carrier arrives essentially undistorted, and a given aperture delivers far more gain per watt supplied. Against that, receiver system temperatures are higher, the atmosphere is opaque in places at both ends of the link, and the same high gain narrows the beam: the 12 m benchmark of §6.3 radiates into a beam 25 arcsec wide at half power, a solid angle of 2.1×10^{-8} sr, so the Earth must lie inside a beam that was deliberately aimed. The band's other difficulties belong to the archive and not to the sky, and §2 sets that out.

The signal sought is an *unresolved spectral carrier* – emission confined to a few channels, unresolved at the stellar position, and present at more than one epoch. Recurrence, and not brightness, separates an interesting event from a detection, and a signal of interest needs detailed follow-up (Sheikh et al. 2021), and the archive holds a second, separated epoch for some of these systems and not for others, so the extent that could be searched and the extent that could be confirmed differ. The frequency is allowed to drift, as it must for a transmitter carried on a rotating, orbiting body (§4.1). "Narrowband" describes the assumed transmitter and not the resolving power of the archived channels. The targets were chosen by other programmes for other purposes, so every statistical statement here is about an archive-selected sample (§3.1).

2. BACKGROUND

Radio searches for technosignatures began at centimetre wavelengths and have largely stayed there. The first proposals and targeted experiments (Cocconi & Morrison

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1959; Drake 1961), the programmes reviewed by Tarter (2001) and Project Phoenix (Backus & Project Phoenix Team 2002) all worked below a few gigahertz. The modern generation divides into targeted and commensal surveys at centimetre and metre wavelengths (Zhang et al. 2020; Czech et al. 2021; Li et al. 2026; Johnson et al. 2023; Choza et al. 2024), several of them riding back-ends built for other science, such as the MWA's correlator (Morrison et al. 2023), with wide-field low-frequency coverage from that array (Tremblay & Tingay 2024) and single-dish C- and K-band work at 6 and 18 GHz with the Sardinia Radio Telescope (Manunza et al. 2025). Machine-learning interference rejection is now central to that effort (Ma et al. 2023; Jacobson-Bell et al. 2025; Pardo et al. 2025). The classical threshold search used here lacks that capability and instead uses a spatial test on the star against control positions in the same field, which a single dish cannot make. A narrowband search also reaches only one class of signal: broadband planetary-scale leakage (Garrett 2026) is outside it by construction.

Above about 30 GHz the record is nearly empty, and that is what makes this frequency range new. ALMA's 10 receiver bands cover 35 to 950 GHz, and one of them, Band 3, has been searched for technosignatures. The lowest, 35–50 GHz, does hold observations of 4 stars of this sample, but all from one programme and all proprietary until 2027. They are also in a continuum mode whose 31.25 MHz channels are 2048 times coarser than the narrowest channel searched here: the coverage exists and cannot carry a carrier search. The band above it holds none at all. For continuity with the centimetre literature, the Arecibo planetary radar had an EIRP of 2.0×10^{13} W at 2.4 GHz (Sheikh et al. 2025); an aperture of that diameter is not realisable in the millimetre band, so the illustrative benchmark of §6.3 is instead a dish of 12 m. The limits of this paper are stated as EIRP and need no transmitter model.

Mason et al. (2025) is the only prior millimetre search to cover more than a single line, and the only archival one. Two Band 3 spectral windows, 117 MHz between them against the 47.7 GHz union of this survey, were searched toward 28 Gaia DR3 stars lying in the fields of 4 ALMA calibrators, observed for other science and so usable for free. Their nearest target is at 1.01 kpc, where a 5σ flux threshold corresponds to $\mathrm{EIRP}_{\min} > 6.9 \times 10^{17}$ W. That target lies between 25 and 314 times further away than the stars of the carrier search here, so at a fixed received flux an EIRP ratio between the two surveys would be 3–5 orders of magnitude of d^2 and almost nothing else. The quantity that does not depend on distance is the minimum detectable *received* flux, $F_{\min} = S_{\min}\,\delta\nu$, and it has to be compared at the channel width each experiment recorded: $S_{\min}$ falls as $\delta\nu^{-1/2}$ while $F_{\min}$ rises as $\delta\nu^{1/2}$, so putting a coarse window on a fine channel arithmetically credits it with a depth it never reached. Mason et al. reach 3.8×10^{-23}–5.8×10^{-23} W m^{-2} across their four fields, on a 30.52 kHz channel at a bare 5σ flux threshold. On that same threshold convention and on native channels throughout, the best window searched here reaches 4.8×10^{-24} W m^{-2} – a factor 10 deeper, in a channel 15.3 kHz wide. The median window reaches 3.0×10^{-22} W m^{-2}, a factor 6 shallower, in a channel 488 kHz wide. Per channel this survey is therefore the shallower of the two except in its finest windows, and its advantage lies elsewhere: 61 stellar systems within 40 pc and not distant stars in four calibrator fields, and a frequency range reaching about nine times higher. Mason et al. also identified the obstacles that shape the present method: drift rates scale with frequency, smearing erodes sensitivity unless drift is searched for, and the molecular line forest raises the confusion background. Every window reported here was re-reduced and re-searched from the raw archive with the pipeline of §4.

TABLE 1
The comparison programmes of Figure 1, each on its own detection threshold and on the common 5σ basis. $\mathrm{EIRP}_{\min}$ is the deepest limit each paper reports, and the factor is 5/(S/N), the minimum detectable flux being linear in the threshold in every one of their own expressions for it.

programme	S/N	factor	$\Delta\nu_{\mathrm{ch}}$	$\mathrm{EIRP}_{\min}$ (W)	at 5σ (W)
Enriquez + 17	25	0.20	2.8 Hz	5.2×10^{12}	1.04×10^{12}
Margot + 23	10	0.50	3.0 Hz	1.35×10^{13}	6.75×10^{12}
Price + 20	10	0.50	2.8 Hz	2.1×10^{12}	1.05×10^{12}
Mason + 25	5	1.00	30.5 kHz	6.91×10^{17}	6.91×10^{17}

Figure 1(a) places this survey among published searches, in minimum detectable EIRP against distance. For a carrier narrower than every programme's channel that limit is a total radiated power and not a spectral power density, so the axis is commensurable. The detection thresholds behind the published limits are not: of the 4 programmes drawn, 3 quote their limits at a threshold other than 5σ. Table 1 gives each programme's threshold, the factor that puts it on the common basis and the channel width it used, and the figure is drawn after that rescaling, with each published value kept beside it. That puts the thresholds on one convention and not the completeness behind them, which is measured here and not established on the same basis elsewhere, so the comparison is between nominal thresholds: the limits this survey quotes elsewhere are EIRP_{90}, the power recovered nine times in ten through the single-epoch trigger stage, while the values compared here and the filled symbols of Fig. 1(a) are bare 5σ thresholds. Any EIRP advantage toward the nearest systems is a d^2 effect. Panel (b) is the other axis, the native channel width of every window searched here beside the channel of each comparison programme. This survey is distinguished by its frequency range and the proximity of its targets, and not by its depth in any one channel.

A non-detection becomes a quantitative statement only through a stated search volume. Wright et al. (2018) formalised that volume, and Margot et al. (2023) its conversion into an upper bound on the fraction of stars hosting a transmitter; Sheikh et al. (2025) asked where Earth's own technosignatures would be detectable. Such a bound is valid only if the sample represents the population it is meant to describe. This sample was assembled by other observers for other purposes, so it does not, and §3 states what it is instead.

This search needs an unusually detailed statistical treatment, and the appendices set it out in full.

3. THE SAMPLE

This search covers 1663 spectral windows toward 90 stars in 83 systems within 40 pc. They are drawn from

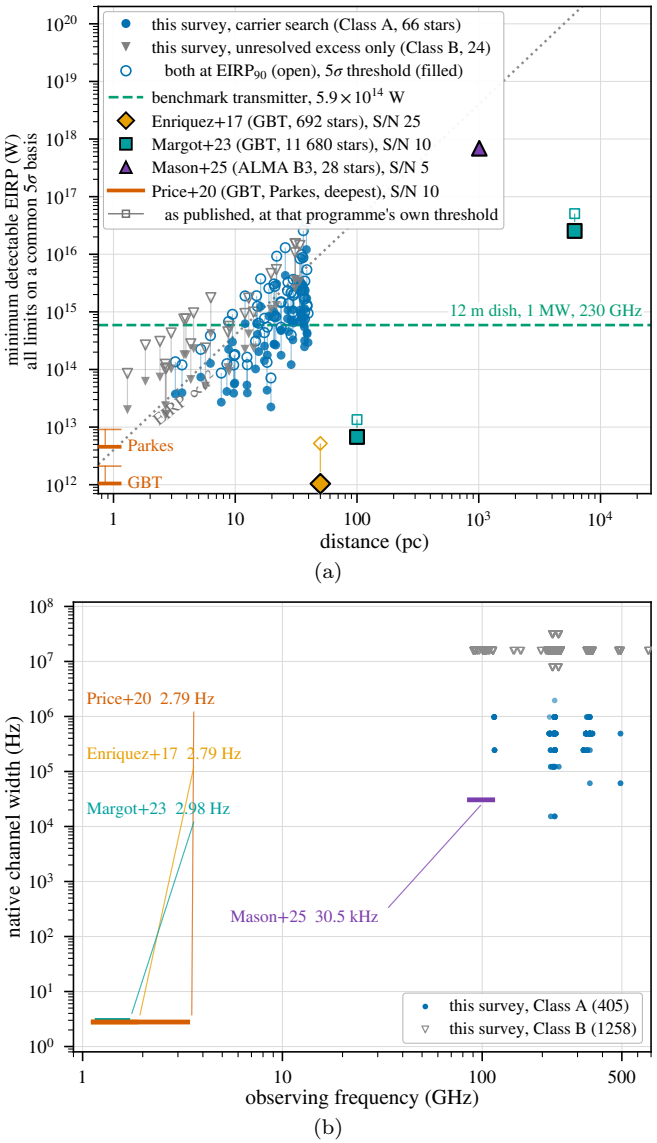


FIG. 1.— Where this survey sits among published radio technosignature searches. Every limit plotted is a total-power limit for a signal narrower than the native channel of the experiment that set it, and those channels differ by 7 orders of magnitude, from 2.79 Hz to 31.2 MHz, so a signal detectable by one programme need not be detectable by another at any power. **(a)** Minimum detectable EIRP against distance on a common 5σ basis. This survey, the deepest window per star: blue circles where that window belongs to the carrier search and grey triangles for the 24 stars whose only coverage is coarse-channel, which constrain an unresolved excess and not a drifting carrier (Table 3); *filled*, that window's bare 5σ threshold, which carries no spectral-response correction and so stands on the same footing as the others; *open*, the same window at EIRP_{90}, the single-epoch trigger completeness of §5.1, which does carry it and, like every limit here, excludes atmospheric decorrelation. Orange, Enriquez et al. (2017); aqua, Margot et al. (2023); violet, Mason et al. (2025); left-edge ticks, Price et al. (2020), each at its published limit rescaled to 5σ by the factor in Table 1, with the published value as the hollow marker above it. *Compare the hollow markers with this survey's filled ones*, each programme at its own operating threshold: the rescaling puts thresholds on one arithmetic convention and not at a matched false-alarm probability (Appendix D). Grey dotted, EIRP $\propto d^2$ at constant received flux; green dashed, the 12 m benchmark transmitter of §6.3. No symbol carries a confirmation criterion, and completeness is measured only for this survey's open symbols. **(b)** Native channel width against observing frequency for every searched window (filled, Class A; open, Class B), with the comparison programmes as horizontal segments at the widths of Table 1.

407 public ALMA execution blocks and 4.1 TB of raw data, put through one pipeline: 281 h of telescope time, and 1098 h summed over windows. That set is the *primary census*: every sensitivity, every expectation and every limit here is computed over it, and so are 50 of the 56 threshold crossings this paper reports. 4 more are coarse-channel crossings of the same census, all of them in the one execution block that fails the data-quality criterion of §4.2, and they are reported apart for that reason and not because they fall outside it. The remaining 2 lie in the *archival tail*: blocks searched after the sweep that fixed the census was closed, toward stars the census already holds, whose windows are not tabulated window by window here and enter no expectation. Three further sets were searched and enter no census statistic either – the *archival extension*, a larger harvest of blocks toward the same stars; a *hold-out*, set aside by the identifier rule of Appendix B.1 and searched as an additional sample; and a control beyond 40 pc – and each is reported with its own counts where its result is (§5.6.2). Table A1 names them in rows that sum, and Table A2 resolves every count of windows, crossings and systems into the set it is counted over.

Star means one stellar component, *system* a bound group counted once however many of its components were observed, and *target* one ALMA pointing.

The census is two experiments, divided on native channel width, which is what the search statistic and the completeness calibration both depend on. *Class A* is the 405 fine-channel windows toward 66 stars in 61 systems, and every sensitivity and exclusion quoted here refers to them unless stated otherwise; *Class B* is the remaining 1258 coarse-channel windows, which support a secondary search for an unresolved spectral excess. The division is unambiguous because the channelisation of the sample is bimodal: the widest fine channel is 1.95 MHz and the narrowest coarse one 7.81 MHz, a factor of 4 with nothing between them. Whether a window can also discriminate a drifting carrier is a separate question, and §4.1 answers it: 391 of the 405 can, 14 cannot.

Every inclusion criterion was applied by query. A star enters if its Gaia DR3 parallax is at least 25 mas, measured to better than 20 per cent, and if its proper-motion-corrected position at the observation epoch falls inside the quoted field of view of at least one public execution block. Distances are direct inversions, $d = 1/\varpi$, and the star need not sit at the pointing centre. One further rule applies to the window and not to the star: a window is kept only where the primary-beam response at the star is at least 0.22, the offset at which the Gaussian adopted here and the response of a blocked circular aperture part by 10 per cent. It removes nothing, in the census or in the archival coverage beyond it: the furthest window from a pointing centre lies at $0.82\,\lambda/D$ against a limit at $0.83\,\lambda/D$, so the limit does not stand clear of the distribution, and the windows it is nearest are the off-axis ones on ϵ Eridani, which the survey searches. The precision clause never binds: the worst fractional parallax error among the stars searched is 2.5 per cent. The query returns 168 work-list entries; the quoted field of view is optimistic, and only 115 of those stars are in fact contained by a public field, of which 90 yielded a searchable window. One target has no catalogue name – a serendipitous field star that appears

throughout as ALMA J153702653−33192492, Gaia DR3 6015967739554000768, an M5 dwarf at 35.0 pc with no measured systemic velocity.

Disc, planet and colour enter as descriptive metadata only. One exclusion, though, is a property of the catalogue and not of the sky: a Gaia DR3 parallax needs a Gaia DR3 source, and the brightest stars have none. Every one of the 22 stars in this volume brighter than second magnitude is absent from DR3 astrometry, and so is a steady 4.9 per cent of every magnitude bin below Hp 4, which saturation does not explain. Together 257 of the 4227 Hipparcos stars whose own parallaxes meet the criteria above never reached the work list: 59 of them brighter than Hp 4 and 198 fainter. Of these, seven lie inside a public field of view – Sirius A, α Centauri A, Vega, Fomalhaut, α Centauri B, β Leonis and MCC 124 – and hold 38 fine-channel and 374 coarse-channel windows. Those windows amount to 1.0 TB of raw data in 83 execution blocks, none of it searched here. Sirius shows what that costs. The three blocks that put Sirius B in this census are pointed at Sirius A, whose catalogue position lies within 0.4 arcsec of the delivered phase centre, while the companion was read 7.4–8.0 arcsec off axis in the same 12 windows. Each block's continuum map holds the primary at 27–46 times its own noise, with spectral index 2.1, and nothing at the companion. The primary is displaced 3.1–3.7 arcsec from its catalogue position by its own orbit, where the orbit of Bond et al. (2017) predicts 3.67 arcsec, in the measured direction to within the block-to-block scatter of the centroid. So the brightest star in the sky was measured three times by this survey and reported at neither position. Sirius A, α Centauri A and α Centauri B carry no fine-channel window at all, so they would add a constraint on unresolved excess rather than a carrier limit; the four that do are Vega, Fomalhaut, β Leonis and MCC 124, three of them A stars with bright debris discs. A search of this volume should admit these stars on Hipparcos astrometry where Gaia has none.

Two removals act on windows and not on stars, and account for the whole difference between the 1721 rows the extraction wrote and the 1663 retained: 45 repeated rows of one window and 13 failing a physical quality cut; the primary-beam limit above removes none. Since $\sigma\sqrt{t_{\rm on}\Delta\nu_{\rm ch}}$ is set by system temperature and collecting area, a window more than 100 times below the sample median in it is not measuring the sky; those removed lie 4.7×10^{3}–3.0×10^{4} times below and the nearest retained 24 times, so the level is not a tuned one. No integration-time cut was applied; the shortest window retained carries 21 s on source.

3.1. *The archival selection*

This sample was not designed: it is whatever previous ALMA programmes happened to observe among the stars Gaia DR3 catalogues, and 90 per cent of the stars searched were targeted under a debris-disc or planet-formation proposal. Table 2 measures the bias against a volume-limited reference census of which the searched stars are a subset, classifying both columns by one rule so that every ratio below is like against like. Both columns are Gaia-defined, so neither contains the bright stars above and the ratios are internally consistent without them.

Distance is the dominant selection effect. Stars within 5 pc are over-represented by a factor 53, and the enhancement falls monotonically with distance to a factor of two *under*-representation beyond 30 pc. Colour comes second: A stars are over-represented by about 17 and F and G stars by factors of three, because those are the stars with bright debris discs. For the A stars that factor is a floor, because the catalogue requirement is itself colour-selective: it removes 39 of the 196 stars in the volume as blue as an A star, 3.7 times its rate among all the others. M dwarfs are depleted – 41 of the 90 searched stars are M dwarfs, a factor 0.7 of their share of the reference census, and only 28 of them carry fine-channel coverage at all. Coverage is also concentrated: 285 of the 1663 windows come from 3 systems. No statement in this paper is therefore an occurrence rate for the solar neighbourhood, and the selection acts hardest on exactly the targets a designed search would pick: of the 16 sample systems within 15 pc with a confirmed planet, 6 carry no Class A coverage and so no drift-resolving limit.

3.2. *Frequency coverage and archival scope*

The Class A experiment covers $\Delta\nu_A = 47.7$ GHz of unique sky frequency, and that coverage is sparse: 26 disjoint islands between 114.0 and 873.1 GHz instead of a band, so those endpoints bound where observations exist and not what was covered. With the coarse-channel windows the archival coverage is 118.1 GHz in 46 intervals, of which 113.1 GHz survives the molecular-line mask of §4. Fig. 2 shows the distribution: a transmitter between the islands was never observable here, whatever its power.

The archive is not exhausted either. 177 further blocks lie inside the same query and were unprocessed when the census was frozen. 159 of them carry a calibration, and 144 were calibrated successfully and searched. 115 of those are reported here as repeat epochs. The remaining 29 are accounted for in Table A1 row by row: 8 archival-tail blocks, whose 2 threshold crossings enter no expectation; 8 ϵ Eri blocks, which contribute only the limits of Appendix A; and 13 from which no window reached the export. The largest single gap is the 18 blocks whose raw visibilities the archive holds but never pipeline-calibrated, 237 GB in all; no effort of ours would close that.

4. DATA AND METHODOLOGY

The search looks for one kind of signal: an unresolved carrier at the star's own position, narrow compared with a single correlator channel, and drifting in sky frequency because the transmitter accelerates along the line of sight. Over a long track its power is spread across many channels, so a search that treats each channel on its own will miss it. The analysis therefore *de-drifts* the data before it measures them: each integration's spectrum is shifted to follow a trial drift rate, and only then are they added together.

4.1. *The de-drifted search statistic*

Each window's dynamic spectrum is stacked along every trial drift rate $\dot\nu$. The search statistic is the largest stacked signal-to-noise ratio anywhere on the resulting channel×drift grid,

$$T(\mathbf{x}) = \max_{\dot\nu,\,\nu} \frac{\sum_t I(\mathbf{x}, t, \nu+\dot\nu t)\,/\sigma^2(t,\nu)}{\left[\sum_t 1/\sigma^2(t,\nu)\right]^{1/2}}, \qquad (1)$$

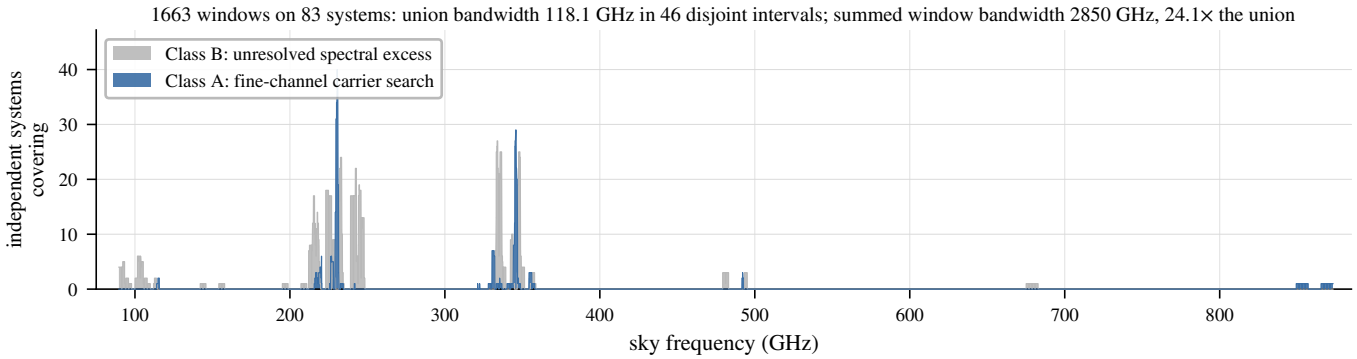


FIG. 2.— Where this survey searched: the number of independent systems covering each sky frequency, drawn separately for the two experiments – blue the fine-channel carrier search (Class A), light grey the coarse-channel spectral-excess search (Class B). Each curve steps at the searched window edges themselves and not on a grid, so an interval narrower than a plotted pixel still appears. The summed window bandwidth is 24.1× the union, so the union measures frequency coverage and the summed bandwidth search effort.

TABLE 2
SELECTION FUNCTION OF THE SEARCHED SAMPLE AGAINST A VOLUME-LIMITED REFERENCE CENSUS. BOTH COLUMNS ARE THE SAME GAIA DR3 SELECTION – $\varpi > 0$, $\sigma_\varpi/\varpi \leq 0.1$ AND $1/\varpi \leq 40$ PC, 17 566 OBJECTS – OF WHICH THE 90 SEARCHED STARS ARE ROWS. THE SURVEY'S OWN CRITERION IS THE LOOSER $\sigma_\varpi/\varpi \leq 0.2$ OF §3, AT WHICH THIS CENSUS HOLDS THE SAME 17 566 OBJECTS, SO THE TWO CUTS ARE ONE SELECTION AND NOT TWO; AND BOTH ARE CLASSIFIED BY ONE RULE, IN WHICH BP−RP COLOUR SETS THE CLASS AND THE ABSOLUTE G MAGNITUDE SEPARATES THE DEGENERATES THAT COLOUR ALONE WOULD PLACE AMONG THE A STARS. THE RULE REACHES 17 278 CENSUS OBJECTS AND 88 OF THE 90 SEARCHED STARS, THE REST CARRYING NO GAIA BP OR RP PHOTOMETRY. "RATIO" IS THE SEARCHED COLUMN FRACTION OVER THE CENSUS COLUMN FRACTION; > 1 IS OVER-REPRESENTATION. THE STRONGEST SELECTION EFFECT HERE IS DISTANCE, NOT COLOUR.

Property	Census	Searched	Ratio
Distance			
$d \leq 5$ pc	48 (0.3%)	13 (14.4%)	52.9
5–10 pc	240 (1.4%)	13 (14.4%)	10.6
10–20 pc	2069 (11.8%)	18 (20.0%)	1.7
20–30 pc	5391 (30.7%)	19 (21.1%)	0.7
30–40 pc	9818 (55.9%)	27 (30.0%)	0.5
Colour class[a]			
A	61 (0.4%)	5 (6.2%)	17.2
F	473 (2.8%)	8 (10.0%)	3.6
G	978 (5.8%)	15 (18.8%)	3.2
K	2221 (13.2%)	11 (13.8%)	1.0
M	13 082 (77.8%)	41 (51.2%)	0.7
Other axes			
Confirmed planet hosts[b]	26 (15.5%)	24 (26.7%)	1.7
White dwarfs[a]	463 (2.7%)	8 (9.1%)	3.4
Searched stars / systems	–	90 / 83	–

[a] Colour bins, not MK types. The boundaries are the colours at which the census's own temperature–colour relation crosses 7500, 6000, 5300 and 3900 K: BP−RP < 0.36 for A, 0.36–0.72 for F, 0.72–0.95 for G, 0.95–1.77 for K and > 1.77 for M. Over the 8594 census objects that also carry a Gaia temperature the two classifications agree for 98 per cent, and every disagreement is between neighbouring bins. A star is counted as degenerate when it lies more than 4 mag below the main sequence measured from the census itself: the two sequences are separated by 5 to 8 mag here, so moving the cut by a magnitude either way changes the count by under 3 per cent. The five classes are normalised over the 80 non-degenerate searched stars and the 16 815 non-degenerate census objects, and the degenerate row over the 88 classified searched stars and the 17 278 classified census objects, which is the same rule in both columns.

[b] Both columns are the same positional join, of the 168-entry ALMA-covered work list and of its searched subset, to the NASA Exoplanet Archive `pscomppars` table accessed 2026 October 4. The 24 searched hosts carry 51 confirmed planets, 22 of the 24 surviving the deuterium-burning mass limit; the Gaia census carries no planet information, so the census column in this row is the work list and not the 17 566-object Gaia census.

where $I(\mathbf{x}, t, \nu)$ is the spectrum at sky position $\mathbf{x}$ in integration t and channel ν, and $\sigma(t, \nu)$ is the noise scale there. Equation 1 is the matched filter for a constant-drift carrier: a drift-aligned, inverse-variance-weighted stack of the calibrated intensities along the track, divided by the noise of that stack, so T is an absolute signal-to-noise ratio and not a local contrast. In Equation 1 the time t runs from the first integration of the observation, so the frequency ν that labels a track is the one a carrier would occupy at the start of that block, and a repeat epoch is read at the cell the same convention fixes (§5.6). The noise scale is the median absolute deviation across 512 control positions at each integration and each channel, scaled to its Gaussian-equivalent standard deviation: local in time and frequency, global in position, and shared by the star and every control. They lie, from a fixed seed, in an annulus set by each block's own synthesised and primary beams: the field read over is that block's primary beam. The statistic is evaluated at one position at a time, identically at the star and at each control, which is what makes the two comparable. $T_\star$ denotes its value at the star.

Spectra are extracted at the star's apparent position, never from images. For each integration and each native channel the pipeline forms the weighted vector average of the phase-shifted visibilities, in total intensity: the mean $I = (XX + YY)/2$ of the two parallel-hand correlations. That average is the dirty-map value at the extraction position – an amplitude, not a deconvolved flux density, so in a field with extended emission it is not a point-source measurement. The data are used as the archive delivered them, through ALMA's own calibration and quality assurance, with their flagging carried through, every delivered channel searched, and no regridding, deconvolution or edge trimming. A per-integration median along the *frequency* axis removes the continuum; it removes a level and not a shape, and no median is taken along time, so no feature narrow against the band can be removed by it. One subtlety matters for every power quoted below: the delivered channels are smoothed, so a channel's effective noise bandwidth is not its separation and flux density times channel width is not yet a radiated power. A response correction computed from the correlator's own channel shape supplies the difference (Appendix A).

One convention holds everywhere in the paper. The search runs on the topocentric sky frequencies ALMA delivered, and every drift rate quoted is the apparent one

at the telescope, which is what the data contain and what a reader can recompute from the archived channel axes. Two steps need the star's own frame and both take it: a crossing is compared with the line catalogue, and a repeat epoch is read at the *predicted cell* – the single channel, at a single drift rate, where the first epoch's own fit says a carrier should be. Both take that frame by correcting each block's frequency axis for the Earth's motion at that epoch and then for the star's systemic velocity, which is the catalogued radial velocity from SIMBAD (Wenger et al. 2000), queried on 2026 October 4, and zero where the catalogues give none. One star departs: toward β Pic the adopted value is $20\,\mathrm{km\,s^{-1}}$, the stellar radial velocity adopted in the ALMA CO analysis of its disk (Matrà et al. 2017), which this survey's own blocks reproduce to $0.7\,\mathrm{km\,s^{-1}}$ in the barycentric frame. The catalogued value is lower by $3.16\,\mathrm{km\,s^{-1}}$, in the sense and of the size of the template-mismatch offset the *Gaia* pipeline documents for stars this hot and leaves in its archive (Blomme et al. 2023).

Two power quantities carry the main text, both defined in Table 3: the trigger level $P_{\mathrm{trig}} = 4\pi d^2 S\,\Delta\nu$ at distance d, for a carrier that makes $T_\star$ reach $T_{\mathrm{trig}} = 5$ in one native channel of width $\Delta\nu$, and the sensitivity EIRP_{90}. The second is measured, not assumed: synthetic carriers of known amplitude were injected into calibrated visibilities and recovered through the unmodified pipeline. Three campaigns did that, one for each stage of the chain the completeness depends on: the trigger completeness itself, on 49 of the 405 fine-channel windows injected directly and on 5 936 carriers of the coarse class; the cost of the spatial rank, on 8 860 carriers over 52 of 56 attempted block–window configurations; and the primary-beam correction, on a third set of 40 scored units, 10 of them also among the directly injected windows. Between them they span both ALMA arrays, four bands and the sample's full range of channel widths. Every limit quoted in this paper is the EIRP_{90} of its own window, given to two significant figures beside its native channel width, except the bare-threshold comparisons of §2 and Fig. 1(a), which are labelled as such; the measured completeness is §5.1 and its error budget is Table 4. Two comparisons bear on the amplitude scale and they are not the same kind of check. An independent reduction of the same AU Mic visibilities is reproduced to 0.994 ± 0.014. That tests the extraction and not ALMA's flux transfer, an error in which would appear in both reductions alike. The external comparison is with the published quiescent flux density of ϵ Eridani, recovered to 1.10 ± 0.31 (Burton et al. 2022) and so a 28 per cent measurement. The absolute scale is therefore carried at ALMA's own band-dependent accuracy specification. The sample as a whole carries ±11 per cent, and the 14 Class A windows in Bands 8 and 10, at 492–493 and 849–873 GHz, carry ±21.

The drift range searched is bounded by an acceleration, not by a frequency. Drift rate and line-of-sight acceleration are one quantity in two sets of units,

$$\frac{\dot\nu}{\nu} = \frac{a_{\mathrm{los}}}{c} \qquad \text{and so} \qquad a_{\mathrm{los}} = \frac{c\,\dot\nu}{\nu}, \tag{2}$$

so a ceiling expressed as $\dot\nu/\nu$ is the same physical limit in every band. The default ceiling is 12.0–$13.3\,\mathrm{Hz\,s^{-1}\,GHz^{-1}}$, the same number as the nHz the literature states drift reach in. It is the acceleration of a transmitter orbiting a solar-type star at a few hundredths of an au. It is the Mason et al. (2025) fiducial with a threefold margin, widened for a confirmed exoplanet host whenever the innermost known planet demands more; at unit margin the search would have been blind to two of the windows that produced a crossing. It is 6 per cent of the 200 nHz that Sheikh et al. (2019) recommend as a ceiling for a general search, their figure being set by the closest observed exoplanet orbit and not by an orbit of a few hundredths of an au, and the restriction here is deliberate. The grid is uniform in $\dot\nu$, and fine enough that a carrier midway between two trials loses no more than 0.26 of a native channel over the longest track; it runs from the negative ceiling to the first trial at or beyond the positive one, so the ceiling bounds the interval searched from below exactly and from above to within one step (Appendix D.8). Three limits follow: a deliberately chirped carrier is not distinguished from an accelerating platform; a transmitter need not be bound to a known planet, so drifts faster than the ceiling are unconstrained; and orbital and rotational acceleration add, so a transmitter on a rapidly rotating planet can exceed the ceiling on its spin alone – the fastest rotation measured for any planetary-mass object, that of β Pictoris b, which orbits one of the stars searched here, corresponds to 19.4 nHz (Sheikh et al. 2019). For the most demanding known case, transiting TRAPPIST-1 b, 29 per cent of the orbital phase lies above the default ceiling. Its maximum line-of-sight acceleration and the widened grid used on that star's own windows are equal to within the ±3.1 per cent that its host's mass and its orbit are known to, so the fraction of phase above that grid is anywhere between nothing and a sixth (Fig. 3).

Only part of the archive can carry a drift search at all, and the two classes reported here are that division (Table 3). Channelisation in the sample is bimodal with an empty gap between the modes, so the split is unambiguous – but it is a split on channel width, and the criterion that decides whether a window can discriminate a drifting carrier at all is whether the maximum searched drift crosses of order one channel or more over the on-source track, $\eta_{\mathrm{drift}} = |\dot\nu_{\mathrm{max}}|\tau_{\mathrm{track}}/\Delta\nu_{\mathrm{ch}} \gtrsim 1$. The two do not partition the survey identically: 391 of the 405 Class A windows satisfy it and 14 fall below – six toward ALMA J153702653−33192492, two toward HD 38858, two toward Wolf 219, one toward HR 1010, one toward PM J03433+1958, one toward WD 0407+179 and one toward ϵ Eridani – each of them a track of 60–181 s against a class median of 40 minutes, so it is the short track and not the channel that fails them. 38 of the 1258 coarse windows meet the criterion and stay in Class B, because meeting it is necessary for drift discrimination and not sufficient: their channels are 15.625 MHz, the whole searched range moves a carrier only 1.02–1.35 of one, and a channel-confined carrier there is diluted by 32 times the median fine channel, which is the Class B limit measured for them by injection. Nothing is hidden by that choice: all 6 of their systems already carry fine-channel coverage and none of the 38 holds a crossing, so no system and no crossing turns on it; they would add 8.5 GHz of frequency outside the Class A union.

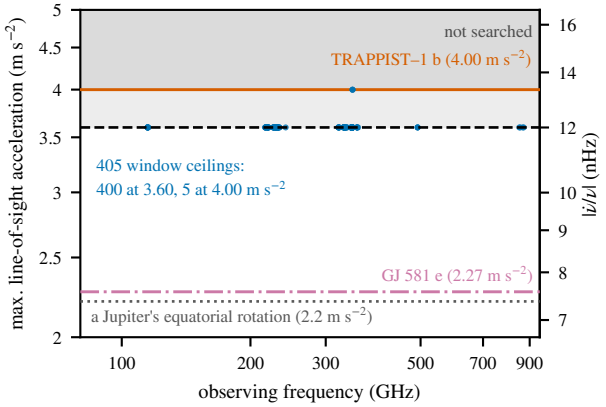


FIG. 3.— What the drift search reaches, as a physical selection function. The drift ceiling of each of the 405 Class A windows (points), converted to line-of-sight acceleration by Equation 2, against observing frequency; the shaded region above is unsearched. The right-hand axis is the same quantity as a fractional drift rate, $|\dot{\nu}/\nu| = a/c$, which is the unit the literature states drift reach in. The boundary drawn is the default ceiling, 3.60 m s^{-2} or 12.0 nHz, which is also the median; it runs to 4.00 m s^{-2} on the 5 TRAPPIST–1 windows. Horizontal lines are reference accelerations: the two most demanding known planets of any target, TRAPPIST–1 b (4.00 m s^{-2}, known to ±3.1 per cent and so equal to the widened ceiling within that) and GJ 581 e (2.27 m s^{-2}, whose host is searched in coarse-channel windows only), and the equatorial rotation of a Jupiter, 2.2 m s^{-2} or 7.4 nHz; Earth's own rotation and orbit, 0.034 and 0.0059 m s^{-2}, lie below the axis. **Transmitters whose combined orbital and rotational acceleration exceeds the ceiling are not constrained by this search, and the ceiling is a budget they share**: a transmitter on the equator of a rotating giant spends 61 per cent of it on spin alone. Nor are unbound transmitters, or any platform accelerating for its own reasons.

4.2. *From a crossing to a confirmed signal*

Every window passes through the same three tests in the same order, and three words name what comes out of them: a *crossing*, an *unattributed crossing* and, had one survived, a *confirmed signal* (Fig. 4). Each is set out below as the quantity it measures, the number that quantity is compared with, and what passing it does and does not establish. There is no fourth test, and none of the three is a significance test.

(i) *Trigger → a crossing.* The quantity is $T_\star$, the de-drifted statistic of Eq. 1 read at the stellar position; the criterion is $T_\star \geq T_{\rm trig} = 5$ in at least one channel×drift cell. $T_{\rm trig}$ is a trigger threshold and not a detection significance: T is not shown to follow any standard distribution under the null, so the level carries no tail probability, and a window holds between 112 and 6.87×10^6 of those cells, so one level buys very different protection in a coarse single-pointing window and in a finely channelised one on a long track. Every cell that reaches the level is recorded, not only the largest in each window, because a carrier is masked as effectively by a noise excursion as by a line. A crossing establishes only that it goes on to the next two tests.

(ii) *Line attribution → an unattributed crossing.* The quantity is the crossing's velocity offset from the nearest transition of a molecular-line catalogue, evaluated in the star's own rest frame; the criterion is ±50 km s^{-1}. The line catalogue is a single database query, made on 2026 October 6 and repeatable as

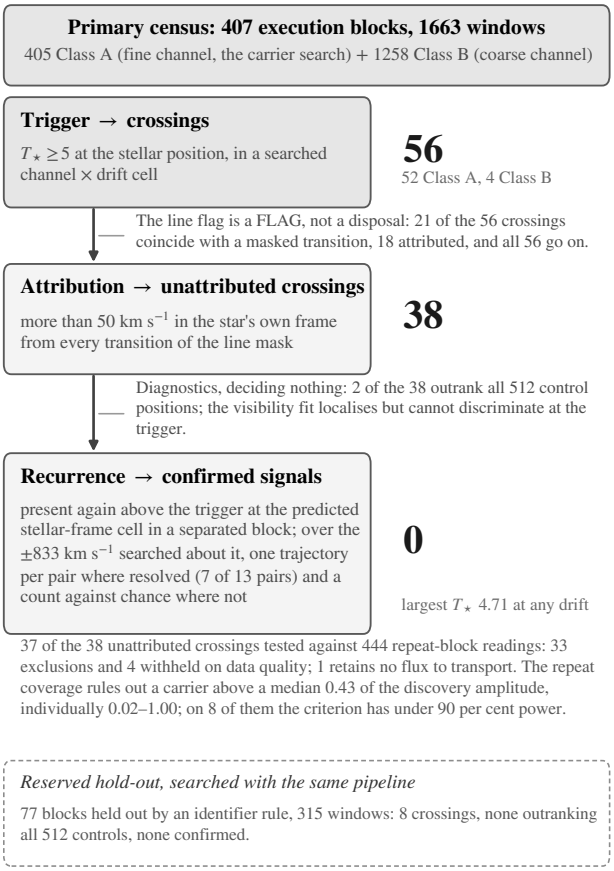


FIG. 4.— The detection logic, with the count at each level. Three levels and no more: a *crossing* is an event requiring investigation, an *unattributed crossing* is a target for follow-up, and a *confirmed signal* – recovered in a block of another observing session more than 1 day away – would be a detection. Coincidence with a catalogued transition is a flag, so all 56 crossings continue past it; the spatial rank and the visibility fit are diagnostics attached to the middle level and decide nothing. The reserved hold-out is drawn as a separate branch because a transmitter in it would matter whatever its calibration role. Every count is read from the ledger, which lists every crossing the search produced, including those in the archival-tail blocks and the quality-failing block that no expectation is computed over, so the figure, the ledger and the text cannot disagree. The counts drawn are the primary crossings, one to a window; the 21 secondary crossings of §5.3 enter the same three levels and three reach the second, none the third.

it stands. Each crossing's *disposition* – the account this paper gives of what it is – rests on the physical evidence set out beside it. The half-width follows the Keplerian line-of-sight velocities of gas surviving beyond a few au, with margin for the systemic velocity. A crossing inside that window is *attributed* where the species is one that such systems are known to carry: a coincidence with a catalogued transition makes an astrophysical origin far more likely than a transmitter, the more so where a second transition of the same species or known circumstellar emission is present. A further three crossings coincide with a species no debris disc has been shown to carry; their offsets are reported and they are counted as unattributed (Appendix C). A coincidence is in any case a statement about a catalogue and not a property of a frequency, so this is an attribution, and every crossing goes on to the recurrence test whichever side of the window it falls, wherever a later block cov-

ers the frequency. An unattributed crossing is a target for follow-up and not a detection. The window's price is that the survey constrains nothing inside it (Appendix C).

(iii) *Independent recurrence → a confirmed signal.* The quantity is $T_\star$ at the cell the first epoch predicts, read in another execution block of the same star at a single drift rate after the frame correction of §4.1; the criterion is that it reach the trigger again; and the test was specified before the repeat epochs were searched. That cell is the one a carrier would occupy at the start of that block, which is where the de-drift of Equation 1 is anchored, so the criterion is met only by a carrier whose stellar-frame frequency at the two epochs agrees to within half a native channel: it has the power to reject a carrier that has moved further and none to accept one. The recovering block must be a *separated* one: an execution of a different observing session, *and* more than 1 day from the first. One reason covers both halves. Blocks of one session share calibrators, atmosphere, correlator setup and interference environment, and the residual of §5.3.2 is what that can produce – which holds for two blocks of one session however far apart they were executed, and 31 pairs of blocks here are of one session and more than a day apart, the widest by 18.02 d. So a recovery in any block of the discovery's own session is a *within-session repeat*, whatever the interval: necessary for a confirmation and not sufficient for one. This is also the rule the confirmability census of §5.6.1 scores the archive on, so the criterion and the scoring are one rule. Two crossings of one star that no single trajectory can be shown to join are an *open* pair, which is neither a confirmation nor an exclusion and bears on the result only through the agreement of the number of such pairs with chance. It is a weak criterion, and the archive's uneven temporal sampling is why: only part of the sample has epochs separated well enough to apply it at all.

Two diagnostics hang off the second test, and the word is exact: each says which crossing deserves attention first, and neither decides anything. The first ranks $T_\star$ against the same statistic at that window's own 512 control positions (§4.1), none of which sits on the star; its lowest value, $1/(512+1)$, means the star beat every control. It is a ranking filter and not a significance test, because the stellar and control positions are not demonstrably exchangeable: on 315 reserved windows outside the census, the stellar rank has median 0.405 where one half is expected if the two were interchangeable, and 0.421 over the 28 of those blocks that share no observing session with a searched one. A parameter-free correction for the annulus geometry does not restore an exchangeable statistic (Appendix B). It also responds to compactness and not to origin, as β Pictoris shows: genuine carbon monoxide ranks below its own control ring there, because the resolved disc lifts the ring along with the star. The second diagnostic is a visibility-domain fit, which asks whether a crossing's emission is at the stellar coordinates; it cannot discriminate at the trigger either, because a noise maximum at the star has flux density at the star (§5.5). Neither gates a disposition, so neither is charged to the sensitivity, and the limits here are correspondingly deeper than a chain charging for the rank would give. Throughout, p denotes a genuine significance level, and no probability computed against the rank's own null is called one.

One execution block is not a null, and a uniform rule says so. The comparison with the control positions needs each block's ensemble to behave like noise, and a block fails when the median control statistic over its own windows, scaled by the median of that quantity over its class, exceeds twice that scale. Of the 407 primary-census blocks, 1 fails: A002_Xc7fa6f_X28a8 toward BD+05 1668, by a margin that does not depend on where the line was drawn (Appendix D.2). Its 4 windows are all coarse-channel, so the Class A results are identical with it and without it: the same 405 windows and 52 crossings either way. But it holds 4 of the 56 crossings, which is the survey's entire Class B crossing population (§5.3.2). The quality-passing data therefore hold 52 threshold crossings, all of them Class A. The criterion is applied to blocks, and 1 of those 52 lies in a window the same diagnostic flags inside a passing block (§5.3.2); it is counted here and withheld from the amplitude-dependent tests, which is stated where each is made. The 50 in the primary census are the baseline every expectation here is set against; the two in archival-tail blocks enter none of them. The 4 from the failing block are reported in full in the ledger as the record of a block whose own control ensemble does not satisfy the noise model the rest are compared with, and not as crossings of the same standing (Appendix D).

Because a fixed threshold is not a fixed false-alarm rate, the number of crossings expected by chance is computed window by window, from the exceedance rate measured at each window's own control positions; the reserved blocks then test the one assumption that makes, which is that a star behaves like a control position in its own field (§5.3.1).

Three corrections precede the statistic: the extraction field is chosen so that the star lies inside the delivered primary-beam field and not merely inside the quoted field of view; the stellar position is propagated to the observation epoch for proper motion, though not for the annual parallax or for orbital motion, whose omissions are corrected window by window in §5.1 and bounded for every component of a multiple in Appendix A.2; and on-source time is accounted per integration from the delivered exposure records and not from the scan intentions. That the chain finds what it is meant to find is shown twice over, by the injected carriers above and by β Pictoris CO – a real feature at a known stellar position, recovered and localised in two transitions and 9 execution blocks (§5.4).

5. RESULTS

The primary experiment is the Class A search for a drifting, unresolved spectral carrier: 405 fine-channel windows toward 61 systems, 391 of which are channelised finely enough on the criterion of §4.1 to tell a drifting carrier from a stationary one. The Class B search for an unresolved spectral excess is secondary and carries no drift discrimination: 1258 coarse windows. Together they are the 1663 windows of the primary census. No event satisfied the confirmation criterion set out in §4.2. Depth comes first below, then what the search found and

TABLE 3
THE QUANTITIES AND THE THREE WORDS THIS PAPER USES, DEFINED ONCE. EVERY TERM BELOW IS USED IN THIS SENSE THROUGHOUT AND IN NO OTHER. THE DISTINCTION THE FIRST THREE ROWS DRAW IS THE ONE THE LIMITS OF §5.1 TURN ON: A CROSSING IS A SINGLE-EPOCH EVENT, AND A CONFIRMED SIGNAL NEEDS A SECOND AND SEPARATED EPOCH THAT NO SENSITIVITY SUPPLIES.

Term	Definition
Crossing	A channel × drift cell at the stellar position reaching $T_\star \geq 5$ in one execution block; neighbouring channels above the trigger belong to one crossing. The largest in a window is its *primary* crossing and any other a *secondary* one, and because the largest cell is the window's own statistic a secondary crossing exists only where a primary one does. An event requiring investigation; no claimed significance.
Unattributed crossing	A crossing lying more than $50\,\mathrm{km\,s^{-1}}$ in the star's frame from every transition of the line catalogue. That is a reproducible frequency coincidence; which coincidences are astrophysically plausible is a separate judgement, stated with each one. A target for follow-up; not a detection.
Confirmed signal	An *unattributed* crossing recovered at the predicted stellar-frame frequency and drift in a *separated* block: an execution of a different observing session, more than 1 day from the discovery block. A recovery in the discovery's own session is a *within-session repeat*, which a confirmation requires and which does not establish one. The criterion is applied to every crossing, and a crossing coincident with a catalogued transition that satisfies it is a recurring molecular line: 13 of them are, which is this stage's positive control (§5.6). The reading is taken at one channel at one drift, so it holds only for a carrier whose stellar-frame frequency moved by between ±0.08 and $\pm0.63\,\mathrm{km\,s^{-1}}$ at most between the epochs compared. Over the $\pm833\,\mathrm{km\,s^{-1}}$ also searched about that cell, a pair of crossings is tested against a single bounded trajectory wherever the drift grid resolves the interval between them (7 of 13 pairs, Table 5). A pair the test cannot separate is *open*, which is neither a confirmation nor an exclusion. It can be applied only where the archive observed the same frequency again, so for the 21 Class A systems where it did not, no sensitivity makes a confirmed signal possible.
P_{trig}	The power of a carrier that makes $T_\star$ reach the trigger in one native channel. An internal pipeline level: not a sensitivity, not a significance.
EIRP_{90}	The power at which nine injected carriers in ten produce a crossing at the star: a measured single-epoch trigger completeness, not a detection probability. Measured for the 49 injected Class A windows; for the other 356 an estimate from the class-average calibration. Nominal in one respect: it excludes atmospheric decorrelation, which Table F1 gives system by system.
Class A	The primary experiment, defined by channel width: 405 fine-channel windows, median 488.3 kHz, toward 61 systems over 47.7 GHz of union bandwidth, 391 of which satisfy $\eta_{\mathrm{drift}} \gtrsim 1$ (§4.1).
Class B	The secondary experiment: 1258 coarse-channel windows, mode 15.625 MHz, constraining a persistent unresolved excess. Never a carrier search: in 1220 of them no searched drift leaves one channel, and in the other 38 the whole searched range moves a carrier 1.02–1.35 channels, which registers one but does not discriminate it (§4.1).
The sample	Neither volume-limited nor representative of the solar neighbourhood: the stars ALMA happened to observe for other reasons (§3).

how each finding was disposed of.

5.1. *Sensitivity: the power at which a carrier would have crossed the threshold*

The median system, on its best window, would have produced a threshold crossing from a carrier of 1.5×10^{15} *W.* Across the 61 systems those limits run from 7.1×10^{13} to 3.7×10^{16} W (Fig. 6a), and 26 of the systems reach 10^{15} W. The median Class A *window* reaches 2.8×10^{15} W. Over the 391 windows in which the searched drift crosses a channel (§4.1) the search covers 44.4 GHz toward 58 systems, and their median system limit is 1.8 per cent deeper than the figure above.

They are a recovery completeness of one epoch, and not a threshold. Artificial carriers were injected at a random sub-channel phase and a random drift from each window's own grid, then recovered through the unchanged pipeline; the power at which nine in ten come back is EIRP_{90}, the only sensitivity quoted here. It is measured through the single-epoch trigger and the localisation at the stellar position inside one execution block. It is therefore the power at which a carrier would have produced a crossing and not the power at which one would have been confirmed, because confirmation needs a second and separated epoch, which the archive supplies for some systems and not others (§5.6.1). The 512-position spatial rank decides nothing and is not charged against it either: charging it would cost a further factor 1.31, and in 3 bright-ring windows it could not be met at any injected power (Fig. 5).

Measured that way the completeness is one number for the class. Nine carriers in ten are recovered at $3.03\,P_{\mathrm{trig}}$, and the sampling error of that pooled value is ±2 per cent, a statement about the class average and about no individual window. It is a total power set against a nominal threshold, P_{trig} being the bare 5σ of the window itself. The loss a channel-confined carrier suffers to the correlator's spectral response is carried by the injected tones, so it sits inside the measured factor – ×2.29 of it – and is not applied to the result as well (Appendix A). Every one of the 49 directly injected Class A windows resolves a ninety per cent point of its own, and the value does not depend on the brightness of its control ring: the 5 windows whose ring carries resolved emission come back at $3.24\,P_{\mathrm{trig}}$ against 2.94 for the rest (Fig. 5). Class B carries its own factor $4.17\,P_{\mathrm{trig}}$, which sets the limits for Barnard's Star, Wolf 359 and Proxima. It is measured the same way, from 5 936 carriers injected into 35 coarse windows of 14 execution blocks.

Calibration of that factor is good to ×0.89–×1.11 (Table 4), which is an interval on the median and on the counts and not on any one window's limit. The median system's limit above carries it – 1.3×10^{15} to 1.6×10^{15} W – so the second figure of that median is set by the cali-

bration and not by the depth of any observation.

Every limit quoted here is nominal in one respect: it excludes atmospheric decorrelation. An injected carrier is added to already calibrated visibilities and so suffers none, whereas a real carrier does, and the bias runs one way only. Measured block by block from the phase rms the observatory records on each block's phase calibrator (Cortes et al. 2026, §§10.4.11, 11.3.1), through the standard phase-scatter relation (Richards et al. 2022, Eq. 3), it leaves the median limit optimistic by 1 per cent. It is least secure at the top of the frequency range: 8 per cent for the median window above 400 GHz, and 11 per cent at the ninetieth percentile of windows. Across the systems' own best windows – the limits a reader uses – the median is 3 per cent, and 13 of them carry more than a tenth. The strongest limits are among the least affected: of the ten deepest in the survey only η Crv carries more than a twentieth, at 28 per cent, and ϵ Eri carries no estimate at all, its delivery exposing neither a phase record nor a baseline to match on. Table F1 gives the figure system by system, measured from the phase calibrator of a system's own block where the delivery exposes one and imputed from the blocks matched to it in baseline length where it does not (Appendix F). *That is why this term is stated and not folded in*: the three factors that are applied are available for every window of the census, while this one is available for 279 blocks of 407 and the rest are measurably the harder ones. Folding it in where it is known would leave a set of limits that is neither corrected nor uncorrected.

Three further factors act window by window, and all three are divided out of the limits above and not quoted beside them (Appendix A). The first is the primary-beam response A at the star's own offset from the pointing centre, which is what turns a measured noise into a flux density at the star. It divides into the limit, so its cost is the factor $1/A$: nothing in the median window, where the star is all but on axis, more than $\times 2$ in 8 windows and $\times 4.3$ in the worst. It is evaluated on the full width the observatory documents, at each window's own dish diameter and centre frequency. Of the two widths available that is the one that follows a blocked circular aperture (Appendix A). Only the 12 windows furthest off axis, toward ϵ Eri and LP 736-15, would move by more than a tenth on the wider of the two, where they would be up to 23.2 per cent deeper. One of those windows is its own system's best: ϵ Eri is corrected by $\times 3.57$ at that window's response of 0.280, and that limit carries the beam-model departure as a systematic beside the calibration interval.

The second is the annual parallax, which the search's phase centre omits. It displaces the star by at most 0.67 of a synthesised beam and makes the median Class A limit shallower by 0.4 per cent, while 13 windows in 4 blocks, toward ϵ Eri and Wolf 28, have no astrometric keys and are left uncorrected. 12 of those are ϵ Eri's, where the displacement follows from the catalogued parallax instead and leaves the tabulated limit optimistic by at most 9 per cent (Appendix A.2). That correction is concentrated, which is why the median limit moves far more than the median window's own factor (Table 4): 219 of the 405 windows need none at all, while 14 lose more than a tenth and lie around the middle of the distribution of limits.

The third the injections do not reproduce at all, each carrier being laid down at one frequency per integration while a real one sweeps within it. The sweep costs the mean of the delivered channel response over it, these channels being wide enough that a sweep moves a carrier's dwell time between channels instead of destroying coherence inside one (Appendix A). It costs nothing in the median window and up to $\times 1.05$ in the 9 finest-channel ones, which are also the deepest, and it takes the best Class A window from 6.8×10^{13} to 7.1×10^{13} W.

The net bias on one window's limit is therefore of either sign. The decorrelation leaves a limit optimistic, while the injected profile falls 8 per cent short of the exact channel response and leaves it conservative by that much (Table 4). In the median window the second is the larger, so the quoted limit is conservative by 7 per cent; at the ninetieth percentile of the systems' own best windows the first is, and the quoted limit is optimistic by 21 per cent, rising to 55 per cent for the most affected system. Neither direction describes the sample, and Table F1 gives the decorrelation system by system so that a reader can apply it to the one limit in question.

Carriers were injected directly into 49 of the 405 Class A windows, 12 per cent, and the result transferred to the remaining 356. That transfer is the dominant caveat on everything above. The injected windows are the Class A windows of execution blocks drawn in a random order, balanced by band and array; they hold 7 crossings against the 6.3 the population rate predicts. *The unit is the block and not the window*, because a block's calibrated visibilities are what an injection campaign has to rebuild, so band and array were fixed by the draw and the conditions that vary within a block were compared with the population afterwards.

That comparison covers 6 properties – channel width, integration time, noise, control-ring brightness, the channels a carrier at the drift ceiling crosses in one track, and the intra-integration retention – and on every one the injected windows span the central eight-tenths of the population, five of them agreeing in the median to 6.3 per cent. The sixth, the noise level, does not: the injected windows are $\times 1.7$ noisier, which follows from drawing whole blocks, and it is also the property the measurement is built to be insensitive to, each ladder being in units of its own window's threshold: splitting the injected set at the median of each property in turn moves the ninety per cent point by at most $0.27\,P_{\rm trig}$ (Appendix F).

A window that was not injected therefore carries an estimated EIRP_{90}. One dimensionless number per class is carried across – the ninety per cent point in units of the window's own trigger power – while the channel width, the noise, the integration time, the primary-beam response, the parallax displacement and the intra-integration retention that turn it into a power are each that window's own measurement. Across the injected windows the ninety per cent point ranges from $\times 0.80$ to $\times 1.58$ of the adopted factor, with 41 of the 49 lying within a tenth of it. The upper end of that range is one window and not a typical error: the second largest is $\times 1.11$. That dispersion, and not the calibration interval above, is what one uninjected window's own limit is uncertain by. Two bands and two channelisations of the survey hold no injected window at all, 30 windows between them, and 4 systems take their best window from one: β Pic, 51 Eri, HD 172555 and HIP 10679. Their

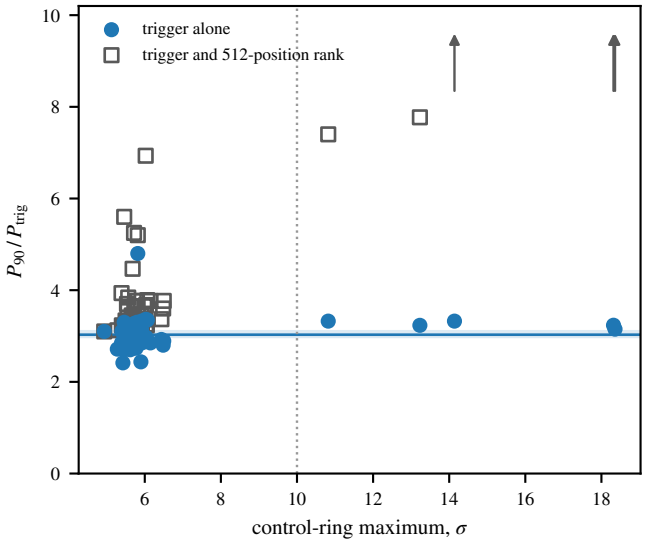


FIG. 5.— Why one completeness factor serves the whole class. Each of the 49 of the 405 Class A windows injected directly is plotted at the power that recovers nine carriers in ten, against the maximum of its own 512-position control ring. Filled circles are the criterion adopted here – the trigger and localisation at the star – with the pooled value and its uncertainty as the line and band; they show no trend with the ring. Open squares add the demand that the carrier outrank every control position, and arrows mark the windows that then reach ninety per cent at no injected power. Dotted, the 10σ ring brightness above which a ring carries resolved emission, not noise.

completeness is carried across a configuration boundary and not within one, and because no injected window lies in those configurations the range just quoted is not an empirical bound on it. The deepest limit in the survey is one of these. The best window was itself injected into for 6 of the 61 systems, marked as measured in Table F1; for the other 55 the headline limit is an estimate carrying that dispersion. Each of those six at its own measured factor instead of the class average would move by at most 10 per cent, and over the population the estimate is not biased in a direction one can name (Appendix F).

A carrier that is not always radiating is penalised in proportion. The de-drifted stack runs the length of a whole execution block, so over 3 888 trials at duty cycles down to 10 per cent the recovered amplitude followed the duty cycle to 1 per cent: a transmitter radiating a tenth of the time needs 10× the power quoted above. Those trials carry no drift, so the joint drifting and intermittent case is not measured here.

5.2. *The nearest stars: Barnard's Star and Wolf 359*

Barnard's Star is the second-nearest stellar system and Wolf 359 the third if brown dwarfs are excluded, and to our knowledge these are the first millimetre or submillimetre technosignature limits for either; radio searches of the nearest stars have worked at centimetre wavelengths (Enriquez et al. 2017; Price et al. 2020; Margot et al. 2023). Both are non-detections. *Both limits come from the secondary Class B experiment and not from the carrier search*: the archive holds only coarse Band 6 windows toward these stars, so the constraint is on an unresolved spectral excess and not on a drifting carrier. On the EIRP_{90} convention of §5.1 Barnard's Star reaches 2.7×10^{14}–4.0×10^{14} W over 12 windows. Wolf 359 reaches 3.1×10^{14}–4.5×10^{14} W over 12.

5.3. *Threshold crossings and how each was disposed of*

The result is a chain of three tests, and it is read in one direction. Of the windows the search covers – the 1663 of the primary census and the archival-tail blocks beside them – 56 hold a cell reaching $T \geq 5$ at the stellar position. 4 lie in the one execution block that fails the quality criterion of §5.3.2, which was written after the condition was seen in that block and is insensitive to where the line was drawn (Appendix D.2), so the population that characterises the search is the 52 in quality-passing data and those 4 are reported beside it. The molecular-line mask is $\pm50\,\mathrm{km\,s^{-1}}$ wide in each star's own rest frame, and it places 21 of the 52 within that distance of a catalogued transition. Of those coincidences, three are with a species no debris disc has been shown to carry, and are reported, not acted on (Appendix C). That leaves 18 crossings attributed and 34 unattributed. The attributions are kinematic and not an accident of bandwidth: in the frame the correlator delivered, the crossings with a measured peak frequency crowd onto the transitions these setups were tuned to far more tightly than a free landing on their own grids would put them (Appendix E). Of the unattributed, 2 outrank all 512 control positions of their own window, and none was found again in any other epoch covering its frequency. No crossing survives the chain, and this survey reports no detection. Figure 4 counts what survives each step and the crossing ledger lists every crossing, a sample of it in Table D2.

Anything bright in a window hides a weaker carrier behind it, so the weaker cells were sought too. The de-drifted statistic is retained in every channel, so behind a window's tabulated crossing – its largest cell – weaker above-trigger cells can be read as well, neighbouring channels counting as one. 399 of the 405 Class A windows retain such a profile and the other 6 retain none. Re-read that way they hold 49 maxima at or above the trigger – one for each of the census's 50 Class A crossings but HD 14055's, whose own window retains no spectrum – and twenty-one weaker cells besides, in 9 windows and as many as 8 in one. 18 of the weaker cells are the resolved carbon monoxide of a disc spread over channels. The three unattributed ones, toward HD 14055, 61 Vir and β Pic, each have 3–16 covering blocks, and none reaches the trigger in any of them: the deepest repeat is $T_\star = 4.4$. Applied to the window maxima the rule returns 50 of the 56 tabulated crossings to 0.002 of a channel. The 6 it misses are the 4 of the failing block, whose repaired profile was not retained, and crossings whose own window retains no usable profile. Cells are counted in the controls too, and outside the mask the star holds 1.05–1.15 times as many as a control position at every level from $T = 3$ to the trigger, with no step where a carrier search would put one; the expectation that follows is §5.3.1's.

5.3.1. *The unattributed count is what the control positions predict*

The trigger is a threshold on a statistic and not a false-alarm rate: the number of searched channel × drift cells per window spans 4.8 decades, so one level buys very unequal protection. How many crossings chance should produce is therefore measured window by window, over the

TABLE 4

What sets $EIRP_{90}$, and in which direction. The uncertainty on the calibration is an uncertainty on the class-average factor, and so on any statistic formed over many windows; its flux-scale term is band-dependent, so the submillimetre windows carry the larger combined figure. The spread between windows is an observed dispersion and is the statement about one window's own limit; the signed terms are biases, stated with their direction and not folded into an error bar, and the decorrelation term is measured in 279 of the 407 blocks and not imputed to the rest.

	value	what it does to the limit
Class-average completeness	×3.03	nine carriers in ten recovered at this multiple of a window's own trigger power. *Measured* in the 49 Class A windows injected directly and *estimated*, at this factor, in the other 356
Uncertainty on that calibration (per cent), combined in quadrature		
The factor itself	±2	sampling error of the pooled value over the 49 injected windows
Absolute flux scale	±10.2	the observatory's own band-dependent specification (ALMA Proposer's Guide (Privon et al. 2026), with the Technical Handbook's quality-assurance criterion (Cortes et al. 2026) at Band 8, where the two differ): ±5 in Bands 1–5; ±10 in Bands 6 and 7; ±20 in Bands 8–10, weighted over this sample's bands
Distance and visibility scale	±4.5	the Gaia parallaxes, which enter as d^2, and the block-to-block visibility scale
Combined	−11/ + 11	×0.89–×1.11 on the factor, and so on the median limit and on the system counts; ±21 for the 14 Class A windows in Bands 8 and 10, at 492–493 and 849–873 GHz
Spread between windows. An observed dispersion, not an error bar.		
Window to window	×0.80–×1.58	the 49 injected windows resolve 2.41–4.80 $P_{\rm trig}$ individually, 41 of them within a tenth of the pooled value and the second largest at ×1.11. This, and not the interval above, is what an estimated limit is uncertain by; it is wider, and it applies to 55 of the 61 systems
Applied. Folded into every limit quoted, and so not combined with the above.		
Intra-integration smearing	×1.00–×1.05	a carrier sweeps through part of a channel within one integration and the injected ones do not: $\eta_{\rm smear}$ at the drift that sets a ninety per cent point, from each window's own integration length where it is recorded (1054 windows) and from the longest delivered anywhere where it is not (609)
Primary beam	×1.00–×4.3	the star is not at the pointing centre, so each limit is divided by the response at its own offset, on the full width the observatory documents. Nothing in the median window; the 12 windows furthest off-axis carry the departure of that form from a blocked circular aperture, up to 6.2 per cent, as a one-sided systematic
Annual parallax	×1.00–×2.06	the phase centre omits it, so each limit is divided by the amplitude a compact source retains at its own displacement, a median 0.13 per cent
Signed terms. A bias is not an error bar and is not added to one.		
Channel response	×2.29	the share of a channel-confined carrier the peak channel keeps. Carried by the injections, so already inside the completeness above, which is this factor times a margin of ×1.325 over the nominal trigger
its residual	+8	the injected profile falls short of the exact response: *conservative*
Decorrelation	+0/ + 1	measured block by block from the phase rms the observatory records on each block's phase calibrator, which an injected carrier does not suffer and a real one does: +8 above 400 GHz, and more than +20 in 5 per cent of windows

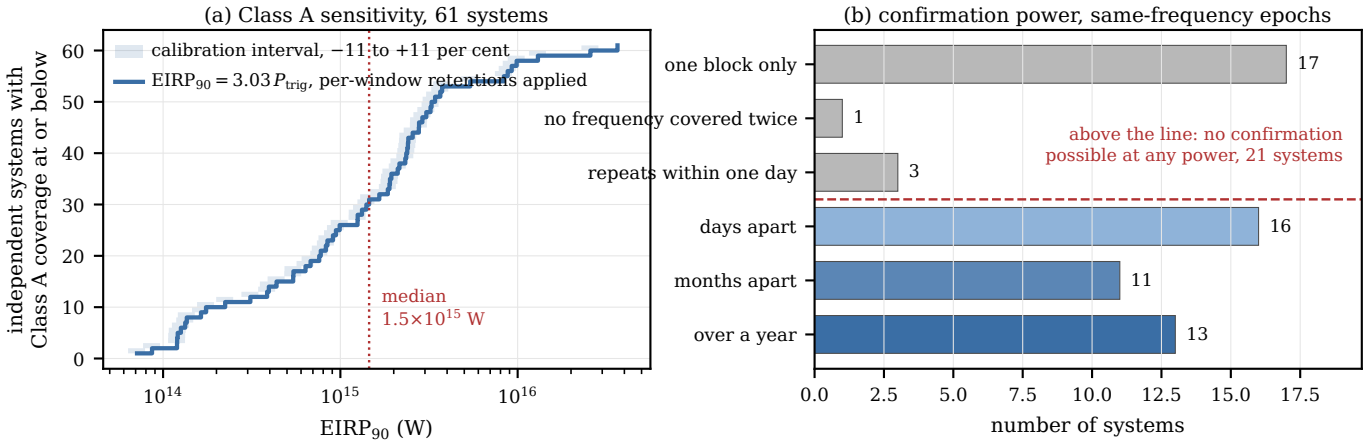


Fig. 6.— **The two completenesses of the carrier search, over the same 61 systems.** *(a)* Single-epoch trigger completeness: the cumulative number of systems whose best window reaches a given $EIRP_{90}$ (Table 3), evaluated as $3.03\,P_{\rm trig}$ with each window's own primary-beam, annual-parallax and smearing retentions applied and atmospheric decorrelation excluded, which Table F1 gives system by system; the shaded band is the calibration interval of Table 4. *(b)* Confirmation completeness, for a carrier whose stellar-frame frequency at the two epochs agrees to within half a native channel, which no amount of power buys: systems grouped by the longest separation between two execution blocks of different observing sessions covering a common frequency, which is what the criterion of §4.2 requires, over the primary census together with every repeat block the recurrence test of §5.6 reads. **Left of the dashed line, 21 systems have no such pair, and for them confirmation was not possible at any sensitivity in panel (a).** The two panels are separate statements and neither follows from the other: (a) is the power at which a carrier crosses the trigger in one block, (b) whether a crossing there could then have been tested at all.

405 Class A windows of the census and with one condition on both sides of the comparison: a cell at the stellar position that reaches the trigger and that the line mask does not attribute. A window's own 512 control positions measure how often a position in that field reaches the trigger, and the channels their exceedances land on measure how often one of them would be attributed. Both are read from the data themselves, and Appendix D gives that accounting. The control positions carry no continuum, so a calibration error varying with frequency would raise the star and not them; on the continuum these stars have, the least demanding window would need 21.4 per cent of it to reach the trigger, and the mild excess of §5.3 runs the wrong way with continuum brightness, 1.01 against 1.06 over the 350 windows with a photospheric estimate (Appendix B). These fields' own control positions predict 24.6 unattributed Class A crossings and 31 are found. A count that large or larger arises by chance with probability 0.13. Both sides are read over the whole primary beam §4.1 defines; over a narrower field, which drops the integrations whose pointing falls outside it and so is not that statistic, the same comparison gives 35 against 24.1 at $P = 0.027$ (Appendix D). The stars do carry 4.6 per cent more above-threshold cells than their own controls below the trigger, and that excess has no identified cause; the rate adopted is the control positions' own, which is the smaller of the two and the more demanding of the count under test. Carried in as a sensitivity test the excess would give 25.7 and a probability of 0.18. A further two unattributed crossings, both toward HD 14055, lie in archival-tail blocks the catalogue does not tabulate window by window, and enter neither side.

That establishes consistency with the expectation these fields' own control positions set, under the assumptions above, and not that the unattributed crossings are noise: the most this comparison could not have rejected is 17 further crossings against those same control positions, or 22 on the narrower reading, which is a bound on compatibility and not a confidence limit on a population of transmitters (Appendix D). Each crossing is tested individually by the recurrence test of §5.6.

5.3.2. *One execution block accounts for four of the crossings*

4 crossings belong to BD+05 1668 (GJ 273, Luyten's Star), all of them in one ACA execution block, `A002_Xc7fa6f_X28a8`, whose whole field is above threshold. 512 of its 512 control positions exceed $T = 5$, where the median of a window's own control ensemble is 2.74 across the survey. Its single-integration noise is as predicted, 0.1609 Jy against 0.1611 Jy, but it does not integrate down. A jackknife on the channel mean gives 0.030 Jy against 0.0048 Jy thermal, 6.2 times too large, while neighbouring channels stay uncorrelated (0.007). The two numbers describe a residual that persists from one integration to the next and differs from channel to channel. Equation 1 divides by the noise of the stacked spectrum, which falls as the root of the integration count, so a residual that does not average down accumulates along a drift track faster than the denominator grows, and it does so at every position in the field; and having no part common to the channels of one integration, it is not what the continuum median of §4.1 removes. The 24 sibling executions of the same field, array and setup reach only $T_\star = 4.26$ and produce no crossing. *The field is inflated and the star is not.* The strongest of the four reaches $T_\star = 73.4$ where its own controls peak at 74.7, and 1–421 of the 512 stand above the star. The criterion fails this block and no other, at any threshold between it and the next most extreme block (Appendix D.2), so its 4 crossings are reported apart from the 52; Appendix E carries what the block might instead be. Milder cases of the same condition are not unique to it: read window by window the same diagnostic flags 5 windows in 2 blocks, and the 1 flagged crossing outside this block – HD 48370 – is withheld for it too.

5.3.3. *Stellar flares are excluded by a broadband prediction*

Known flare stars are in the sample, but a millimetre flare raises every channel of a window alike, so one able to produce a single-channel crossing at the trigger must produce a $5\sqrt{N}\sigma$ continuum event in the same integrations. Over the 48 crossings with a recoverable continuum measurement that prediction runs 49–4 717σ. The observed continuum statistic has median -0.29, with $|Z| < 3$ in 45 of them. One block settles it: AU Mic `A002_Xa42f75_X319` holds both a crossing ($T_\star = 5.27$) and the millimetre flare of MacGregor et al. (2020), and that flare contributes 0.17σ to the single-channel statistic – raising the continuum at $Z = 10.3$ while the crossing's own de-drifted track reaches $Z_{\max} = 2.1$.

5.4. *β Pictoris: an astrophysical positive control, and the obstacle it identifies*

A search that finds nothing must show that it could have found something, and the survey's strongest features show exactly that. The pipeline recovers the β Pictoris circumstellar carbon monoxide without being told the disc is there. It recovers it in two rotational transitions and in 10 execution blocks spanning 8.4 yr. Those blocks belong to 6 observing sessions, several of them executions of one session on one tuning, and each transition is recovered in more than one session: the closest two sessions to hold the same transition are 25 d apart – `A002_X6f1341_X1484` and `A002_X7116f1_X1785` in CO(2–1) – while one session's own blocks here span 1.02 d – so a repeat between sessions is a repeat between epochs, and a repeat inside one is not a confirmation however far apart the two blocks fall. In 9 of the blocks the star also outranks every control: 5 Band 3 windows, at $T_\star$ up to 30.76, and 4 Band 6 windows at 11.68. The identification is kinematic and does not rest on those statistics. Carried through the frame chain – to the barycentre at each block's own epoch, then to the star's rest frame – all 10 crossings lie within 0.5 km s^{-1} of the stellar systemic velocity, in the two lowest CO transitions of a star with mapped circumstellar CO (Dent et al. 2014; Matrà et al. 2017). The Band 3 peaks agree across blocks to 1.00 channel once each block's own tuning is removed, so any apparent drift is far below the grid searched. The alternative reading requires a transmitter placed by coincidence at matched offsets from two CO transitions of one star.

It is also the control that shows the recurrence test of §5.6 working in the positive direction: every flagged β Pictoris carbon-monoxide feature is recovered in another execution block, and each transition in a block of another session as well, the screened Band 6 window re-

turning $T_\star = 8.95$ in the second block of its own recurrence set. A non-recurrence elsewhere means nothing unless the same machinery recovers a repeat where one exists.

One of those blocks is a counterexample, and it is the more useful half of the control. In it several of the 512 control positions exceed the stellar statistic, so the spatial rank disfavours a window holding carbon monoxide known to be real and centred on the star; the disc is resolved on the scale of the primary beam in that configuration, which is the geometry the statistic is built to reject. The rank is therefore a test of compactness and not of authenticity, and on a real extended source it errs conservatively – a stronger statement than a clean positive control, because it bounds the diagnostic's reach as well as demonstrating it (Fig. B1).

HD 48370 is the complementary case and is equally instructive. Its crossing reaches $T_\star = 26.83$, but the control ring rises with the star, to 18.36, as emission filling the primary beam does. The velocity matches the prominent CO(2→1) that Cataldi et al. (2023) report toward the source and flag as possible cloud contamination. The block's ^{13}CO(2→1) window is bright at the ring and faint at the star. The extended emission is therefore identified by the *level* of the control ring and not by the star's rank against it – the star here outranks every control – which is why the window-level flag of Appendix D.2 is defined on the ring's median.

These two results identify the dominant obstacle for millimetre technosignature searches, and it is not sensitivity. The strongest apparent signals in the quality-passing data are circumstellar and interstellar molecular emission, the 4 highest statistics in the survey belonging instead to the one block that fails the criterion of §5.3.2: 21 of all 56 crossings land on a catalogued transition and 18 of them are attributed to it. The mask that removes them covers 12 species and costs 6.4 per cent of the Class A union bandwidth, and it is nevertheless the single most productive test in the chain. The warning for future work is about line lists and not about depth, and Appendix C puts a number on it.

5.5. *The two crossings that lead their own control fields*

Frequencies in this section and the next are in the star's own rest frame, which is the frame the recurrence test registers in; the delivered ones are in Table 6 and in the ledger.

Two crossings reach a stellar statistic above every one of the 512 control positions in their own window, and neither is a detection. They are 61 Vir at 346.3232 GHz and CP−72 2713 at 344.2965 GHz, both in Band 7; both are tabulated in Table 6 and the first is drawn in Fig. 7. Their control rank is recorded because it says which crossing was examined first, and it disposes of nothing (§4). CP−72 2713's crossing lies $-12\,\mathrm{km\,s^{-1}}$ from SO(8_8–7_7), a coincidence reported as a note, not as a disposition (Appendix C). Both events go on to the recurrence test. Each was also re-fitted in the visibility domain, as 51 of the 56 crossings were, by phase rotating to the star and following the recovered drift, and 41 of those fits are consistent with a point source at the star. The fit excludes neither crossing: injected sources in the bin these two occupy return $\mathrm{Re}/\sigma = 6.58 \pm 1.97$. Its value is that a pedestal fed through a sidelobe from elsewhere in the field, which a dirty map at the star cannot tell from a source at the phase centre, is not a point source there. Of the 512 control positions the rank screen uses, 8 retain a dynamic spectrum and 12 are reachable in the visibilities.

Neither crossing is present when the same stellar-frame frequency is observed again, and Table 6 gives both readings of that silence. The nearest covering block assumes least. It lies 1.97 and 2.05 hours from each discovery, and rules out anything above 0.95 and 0.71 of the discovery amplitude. Combining every covering block instead – 3 and 5 of them – deepens that to 0.56 and 0.66. It asks the same return over baselines of 3 and 2 days, however, and not of hours. Either reading covers half a channel in the star's frame, ± 0.211 and $\pm 0.213\,\mathrm{km\,s^{-1}}$, which each carrier's own fitted drift would cross in 77 and 119 seconds: the test asks for a return to that frequency, not a stability at it (§5.6). At the faintest amplitude consistent with each crossing having been found at all, the criterion would have recovered a persisting carrier with probability 1.00 and 0.66 – and injected carriers at that amplitude are recovered 1.00 and 0.65 of the time by the same reading – so for CP−72 2713 the silence is not an exclusion and is not read as one. Appendix B follows its crossing through every step of the chain.

5.6. *Recurrence, applied to every unattributed crossing*

A transmitter on a body in orbit about its star does not arrive at the same frequency twice: its line-of-sight velocity changes between epochs, and a carrier at a fixed rest frequency changes with it. Two questions are therefore asked of every unattributed crossing that another block covers, and the broader one carries the result. It is whether anything above the trigger is present again anywhere in a covering window, searched at every drift of the grid. That window is $\pm 833\,\mathrm{km\,s^{-1}}$ wide in the median case, against the $\pm 316\,\mathrm{km\,s^{-1}}$ an orbit at the grid's own $3.6\,\mathrm{m\,s^{-2}}$ ceiling can produce about any star in this sample, whatever its eccentricity and orientation, so it spans the excursion such a carrier could have made. 29 of the 328 covering windows hold a statistic above the trigger, and that is the number chance predicts in trials of that size: 28.3. That expectation is a sum over those windows and not a control rate applied to their number: each contributes its own chance of holding one cell above the trigger, from the number of independent channel–drift cells fitted to it and the noise scale its own control positions measure (Appendix D.6). The strongest of them reaches 5.96 and the nearest sits $56\,\mathrm{km\,s^{-1}}$ from the crossing's own frequency, a median $447\,\mathrm{km\,s^{-1}}$ away. A catalogued window covers 32 of the 33 crossings that carry an exclusion and spans that excursion in 27 of them; for 5, toward β Pic, HD 110411 and HD 69830, the widest covering window is narrower than the excursion, and 1 is covered by no catalogued window at all.

The second question is far narrower, and it is the one each ledger exclusion answers: whether a carrier returned to the single cell its own discovery predicts. One trajectory model carries it – a carrier at a fixed rest frequency, whose line-of-sight velocity relative to its star is the same at the two epochs to within half a channel, and which drifts through each observation at the rate fitted there. The predicted channel is set by the Earth's motion at the repeat epoch and the star's systemic velocity and by

nothing else – that frame moves the cell by up to 116 channels, and for η Crv by −80.2 – while the fitted drift enters only as the de-drift within each observation, where half a step of the window's own grid costs 0.4 per cent of the statistic (Appendix B). The statistic is read at that one channel, maximised over nothing. The model is a strong assumption and these crossings disfavour it: every one of them carries a fitted line-of-sight acceleration of 0.5 to at least $3.6\,\mathrm{m\,s^{-2}}$, so such a carrier arrives at the same frequency twice only by coincidence of phase, which the interval the cell permits makes a fraction of order 10^{-4}–10^{-3} of the orbits that ceiling admits. A *confirmed signal* is an *unattributed* crossing that satisfies the criterion (Table 3); a line-coincident crossing that satisfies it is a recurring molecular line, and is reported as one.

None of the unattributed crossings returned to its own predicted cell, while 13 crossings attributed to circumstellar or interstellar carbon monoxide do recur, as expected. Those 13 are the positive control of this stage and they test it on both of the ways real emission arises here: 10 are the resolved circumstellar disc of β Pic and 3 are interstellar gas toward HD 285968, whose sightline carries catalogued cloud emission at the crossing's own velocity in the local standard of rest (Appendix C). No exclusion is formed for them, because what one would exclude has just been observed. Close to the trigger the control is thinner than that count suggests: every circumstellar recovery has $T_\star \geq 8.0$, and the 2 attributed crossings that are not recovered are the two faintest tested, both at $T_\star = 5.41$, so there the stage is controlled by HD 285968 alone – by interstellar gas and not by a point source. The control reaches across the baselines the exclusions are quoted over: every one of the 13 is recovered in a block more than 1 day from its own and in a different observing session, 39 to 1984 days away (Table D3). That control is a window search, so the same cell was also read for each of them in the way the exclusions are read – one channel at the frequency its own discovery predicts, through the same frame correction, start-of-block anchor and carried drift – and what it establishes divides with the width of the line it recovers: the registration of a single cell is tested by the 3 readings of HD 285968 alone, whose line is confined to one channel, while the rest test the frame correction only to the width of a resolved line (Table D3, Appendix D.5). 37 of the 38 unattributed crossings were tested, over 444 readings in 151 execution blocks. At the predicted cell the largest statistic anywhere is +2.00 for a crossing's whole repeat coverage combined at the *carried drift* – the rate fitted at the discovery epoch and carried across to the repeat. Read in one block alone it is 2.48 at that drift and 4.71 at any drift of the grid, against a trigger of 5. The exception is HD 14055 `X1001fdc_X29ae`: 30 blocks cover its frequency, but its own window retains no spectrum, so there is no discovery amplitude to transport to them and no reading is quoted for it.

Every exclusion is conditional on that model, and the condition is narrow. The reading is taken at one cell at one drift, so it constrains only a carrier whose stellar-frame frequency returned to within half that cell: half a native channel, 244 kHz in the median case, or ±0.08 to $\pm 0.63\,\mathrm{km\,s^{-1}}$ across the 33 crossings that carry an exclusion, with a median of ±0.21. It is not a requirement of frequency stability, and these crossings are not stable. Their fitted drifts are line-of-sight accelerations of 0.5 to $3.6\,\mathrm{m\,s^{-2}}$ in the observatory's frame, reaching 3.66 in the star's own, and the largest of them is a lower bound because the search found its maximum on the outermost trial of the grid; the carrier of Fig. 7 crosses 22 channels of its own window inside the discovery observation. At those rates a carrier leaves the half-channel the reading covers in 77 and 119 seconds. The single cell therefore asks only that the carrier be back where it was – stable over the gap, or periodic and sampled at the same phase – and says nothing about one its own motion has carried elsewhere. The window search above tests whether recurrences exceed chance as a population; it excludes no individual accelerating carrier, and 12 of the 13 testable pairs remain open.

Neither the registration nor the anchoring of that cell changes any of this. The reading was re-made over every cell within the measured registration residual and at every anchor along the discovery track, and no anchor recovers anything anywhere in this survey (Appendix D.5).

The amplitude each exclusion reaches is the fraction of the discovery amplitude still excluded at the trigger, $(T_{\mathrm{rep}} + 5)/T_{\mathrm{pers}}$. Of the 38 unattributed crossings, 33 carry a quoted exclusion; 4 take their discovery amplitude from the block that fails the quality criterion of §5.3.2 and 1 retains no discovery spectrum, so none can be formed for those. Over the 33 the repeat coverage rules out a carrier at a median 0.43 of the discovery amplitude, and the deepest at 0.017 of it. The individual limits run to 1.00, however, and for 3 of them – β Pic at 241.77 and 241.86 GHz and HD 110411 at 219.44 GHz – nothing below 0.84 is ruled out. Combining every covering block buys most of that depth: the nearest block alone rules out a median 0.98 of the discovery amplitude, and 15 of the 33 rule out nothing. The chance that a carrier would have satisfied the criterion is evaluated at the trigger amplitude $5S/T_\star$ and not at the discovery amplitude, which is selected as the largest of many cells at a statistic of about five and is biased high (Appendix B). It exceeds 90 per cent for 25 of the 33. For the remaining 8 it is η Crv at 357.25 GHz (0.29), β Pic at 241.86 GHz (0.35), ALMA J153702653−33192492 at 347.19 GHz (0.49), β Pic at 241.77 GHz (0.55), CP−72 2713 at 344.30 GHz (0.63), AU Mic at 230.67 GHz (0.64), CD−57 1054 at 346.53 GHz (0.74) and HD 110411 at 219.44 GHz (0.84), so those are untested and not excluded. These are conditional detection probabilities under the persistent-carrier model adopted here, not probabilities that a candidate was noise: a crossing that does not recur has not been shown to be anything else. Each was therefore measured as well as calculated, by injecting a carrier at the trigger amplitude into the retained per-integration spectra of every covering repeat block and reading it back by the same procedure (Appendix B, Table B1); the injected value is the weaker of the two readings and is the one quoted here. At η Crv it falls to 0.29 where the calculation gives one, because that repeat observation is much the longer of the two while the drift grid is sized to the discovery, so that crossing is reported as untested and not excluded and is counted so above.

Part of the power lost at a single cell is the threshold and not the archive: a bar set instead to hold the chance

of one false confirmation across every reading would be lower, and nothing would have been confirmed there either (Appendix D.5).

Each of the 29 repeat windows that hold something above the trigger holds a crossing, so a carrier present again would appear in its repeat block as one, and each pair must be asked whether one transmitter could have produced both members. Inside that interval there are 16 pairs in which at least one member is unattributed (Appendix D.6 takes the one count to the other); Table 5 asks the question of the 13 of them, across 5 stars, whose two crossings each carry an independently fitted drift and a block epoch measured from their own data. The other 3 cannot be asked: 2 involve the crossing whose discovery products do not survive and 1 a line-attributed crossing for which no drift was fitted. Both fitted drifts are carried into the star's frame first, by removing the observatory's own line-of-sight acceleration, which is diurnal here and so cannot be extrapolated across days. Two criteria then follow, and they partition the pairs differently. Constant line-of-sight acceleration means $\dot{\nu}_1 = \dot{\nu}_2$, which 13 of the 13 pairs fail by more than the grid resolves; the weakest model admitting two unequal drifts is a drift changing at a constant rate, and all 7 of the pairs whose rates the grid resolves fail that too. The claim is weaker and safer – that no trajectory inside the searched grid joins the two crossings – and it separates 1 pair, the two crossings of HD 14055 needing 25 times the grid's own ceiling between them. The remaining 12 are *open*, neither a confirmation nor an exclusion, and Table 5 marks 2 of them *edge*, where one of the two fitted rates sits at the edge of the grid and is a lower bound. The comparison has the power to reject, which it uses on all 7 resolved pairs, and none to accept a drifting carrier – and it does accept the one pair that is a single emitter, the CO(2–1) line of HD 285968, which is a line that does not drift (Appendix D.6).

The 2 attributed crossings of the 15 tested that are not recovered both remain consistent with their attribution: each is bounded only 2.1 and 1.4 standard deviations below what carbon monoxide persisting at the discovery flux density would have given, and for the serendipitous field star among them the attribution rests on a catalogued interstellar velocity and not on this test (Appendix C). Whether an interstellar line recurs at the phase centre is set by where the cloud is bright, which the control ring measures in the same map.

5.6.1. *What can be searched, and what can be confirmed*

A sensitivity limit and a confirmation limit are different statements. This experiment searched 61 systems and 405 Class A windows over 47.7 GHz of union bandwidth. For a carrier whose stellar-frame frequency at the two epochs agrees to within half a native channel it could have confirmed a signal in 40 of those systems – those observed again in another observing session more than a day later at a frequency the first block covered, counting every block the recurrence test above reads and not the census alone – and in 13 of them more than a year later (Fig. 8). The count applies the criterion of §4.2 entire, and that condition is the whole of its reach: a carrier accelerating as these crossings do is back inside the interval the reading permits for a fraction of order 10^{-4}–10^{-3} of the phases of an orbit at the searched ceiling, so under this criterion none of the 61 systems could have been confirmed except by that coincidence. Requiring the different session as well as the day moves no system, because each of the fifteen systems holding a same-session pair more than a day apart holds a pair in two sessions as well. A count taken over the census blocks alone and blind to frequency arrives at the same number over a different block set, so the cost of retuning cannot be read off the two together.

An independent execution block is also not an independent epoch. The baselines run from 0.46 hours to 1.8 yr. For 21 of the 37 crossings that carry a discovery amplitude at all, the nearest covering block falls within a day, and for 15 of those it is a block of the same observing session as well – the two events of Table 6 among them, whose nearest coverage lies 1.97 and 2.05 hours away. A recovery in a block of the discovery's own session would not have been a confirmation. Separated coverage is not merely present; it carries the depth. Read over blocks of another session more than a day away alone, 30 of the 33 keep such a block and rule out a median 0.44 of the discovery amplitude, from 0.018 to 11.79, with 23 of them still reaching 90 per cent recovery at the trigger amplitude against 26 over every covering block. Those two counts are the modelled recovery and are comparable with each other, not with the 25 the injections measure over every covering block, which is the weaker reading and the one quoted everywhere else here. The three crossings whose only coverage lies in their own session – β Pic at 241.77 GHz, β Pic at 241.86 GHz, ALMA J153702653−33192492 at 347.19 GHz – are all among the 8 reported too shallow to exclude anything. Over months and years these tests bound no duty cycle, and for the other 21 systems they bound nothing at all.

5.6.2. *The reserved hold-out and the control beyond 40 pc*

Three sets of execution blocks enter no census statistic (Table A1); each was searched with the same pipeline, each holds data on real targets and each is reported here as a search. The hold-out is 77 blocks, 315 windows, 36 stars and 214 h on source, set aside by the identifier rule of Appendix B.1 and sharing no execution block with the census. It contains 8 threshold crossings, the strongest at $T_\star = 8.64$, and none of them outranks its own control positions: the fewest controls above the star in any of them is 5 of 512. 7 were tested for recurrence on the same code path as the census, and exactly one is present again – HD 285968 at $+9\,\mathrm{km\,s^{-1}}$ from CO(2–1). Its 4 later blocks combine to $T_{\rm rep} = +6.71$ at the carried drift, the value Table 7 prints, while the strongest single block reaches 5.87 at its own best drift. It is a molecular line, so its recovery is the test working in the positive direction here as well; for every other tested crossing the repeat coverage rules out a carrier at 0.47 of the discovery amplitude or less (Table 7).

The archival extension was searched the same way. Its 463 windows in 115 blocks hold 13 threshold crossings, every one of them in a fine-channel window. 5 lie inside the mask in their own star's frame and 8 do not. Over the 67 of those windows that are fine-channel (Table A1) the control positions predict 5.6 unattributed crossings by chance (Appendix D). The chance of 8 or more is 0.20, so the extension returns what the census's own fields predict. A sample this size could have shown a shortfall and

does not, $P(0\,|\,5.6) = 0.004$. The frame these offsets are read in is approximate: no barycentric term is recorded for these blocks, and 7 of the 13 carry a measured systemic velocity. Over the whole range either term can take, the unattributed count stays at 8 or 9. 3 of the 13 outrank their own control positions, and all 3 are the β Pictoris disc's carbon monoxide.

The control beyond 40 pc is 47 blocks and 198 windows toward 22 background sources. It carries 88 h on source and holds 8 crossings, the strongest at $T_\star = 12.7$, none of which outranks its own controls. No systemic velocity is catalogued for any of them, so the set supports no limit and shows only that the pipeline behaves the same way on fields beyond the sample. The third reserved set calibrates the spatial rank and is not external at all (Appendix B).

5.7. *A stellar-frame multi-epoch stacked search*

Combining the epochs of a star in that star's own rest frame deepens the limit. 56 of the 83 surveyed systems hold more than one searched execution block, and 59 of the surveyed stars can be stacked, 5 of them from blocks beyond the census. Over 320 stacked groups, 2219 epoch-windows and 1431 h, the flux-density limit improves on each star's best single epoch by a median factor 1.52 and by 4.45 at most. 10 further stars stack too, all of them control stars beyond 40 pc, and are excluded from those figures; over all 69 stacked stars, whose stacks are mostly shallow two-epoch ones, the median gain is 1.46. Every stacked limit is measured by injection into the stack itself and never read off a stacked spectrum as a threshold.

A fixed stellar-frame frequency is not a fixed observed frequency, so the epochs are registered before they are added. The barycentric term alone swings by a median 8.2 km s^{-1} between the epochs of one star and by as much as 53.9 – many channels at the 488 kHz Class A width – so epochs added as observed would dilute. Each epoch is transformed to the stellar rest frame, divided by its own primary-beam response and combined by inverse variance over the channels of common stellar-frame coverage, keyed on the star's catalogued identity and not the directory the search ran in. Spectra with pathological archived visibility weights are excluded first (27 of 2838, which would otherwise dominate the sum). The transform is verified on a real line. Over 4 epochs spanning 44 d, HD 285968's molecular line collapses from a 19.26-channel topocentric spread to 0.798 channels.

The criterion is a stacked $Z \geq 6.0$ above every stacked control maximum, no epoch supplying more than 0.60 of the weighted sum, and at least 0.50 of the signal surviving removal of the strongest epoch. Controls are stacked identically and every clause evaluated at them, measuring the whole test's false-alarm rate at 0.21 per cent against a 1 per cent ceiling.

The gain is $\sqrt{N_{\rm eff}}$ and not $\sqrt{N}$, where $N_{\rm eff} = \sigma_{\rm min}^2 \sum_e \sigma_e^{-2}$ is the equivalent number of equal-depth epochs and the median $N_{\rm eff}/N$ is 0.65. A stack realises $\sqrt{N_{\rm eff}}$ divided by the measured noise of its own stacked spectrum, and that noise follows the inverse-variance prediction to 0.992 in the median, with a per-stack scatter of 7 per cent and nine stacks in ten between 0.81 and 1.18. Against $\sqrt{N}$ the realised gain is 0.78, the shortfall being unequal epoch depth, so an epoch count alone misleads.

TABLE 5

Can two crossings be one carrier? Every pair of crossings toward one star, in two execution blocks, lying within ± 833 km s^{-1} of each other in the star's frame – the half-width of the median spectral window a repeat could be found in – and carrying between them two independently fitted drifts and two block epochs measured from their own data: 13 pairs with an unattributed member, of the 16 inside that interval. Below the rule is the control, the 1 of 23 pairs of already attributed crossings for which a drift and an epoch were both recorded; it is a line that does not drift, so it shows the test accepting a pair at one frequency and not a drifting carrier. Stars are identified by position, so the two archival-tail crossings of HD 14055 pair with its catalogued ones. $\Delta\nu$ is the stellar-frame offset and $\dot\nu_1$, $\dot\nu_2$ the drifts fitted independently in the two blocks and carried into that frame. A carrier whose drift changes at a constant rate between the epochs would put the pair at $\Delta\nu = \frac{1}{2}(\dot\nu_1 + \dot\nu_2)\Delta t$; constant line-of-sight acceleration is the special case $\dot\nu_1 = \dot\nu_2$, which 13 of these pairs fail outright. *Miss* is how far the pair is from that prediction in units of the drift grid's resolution carried over Δt, which is also the resolution the comparison has on the mean of the two drifts; a dash means it exceeds the window searched, so the comparison has no power at all. The last column is the verdict under the weaker criterion of a trajectory whose drift stays inside the searched grid: *no* where no such trajectory joins the pair, *open* where the pair is not separable – neither a confirmation nor an exclusion – *yes* for the control row. The subscript $\star$ marks the star's frame, in which the drifts are given; the grid was stepped in the observatory's, and Table D4 gives the rates in it. 1 of the rates printed here, 4216 Hz s^{-1} toward HD 14055, is the outermost trial of its own window's grid: there the search reached the edge of the interval it covers, so the rate is a lower bound – 3.66 m s^{-2} or more – and every pair that uses it is marked *edge*, not *open*. The exception is the row the criterion separates, whose verdict uses neither rate: that pair's own offset of 825 MHz is 25 times the 33 MHz the ceiling permits over 2.23 hours, whatever the two rates are. Appendix D.8 establishes the grid's edges and lists every edge-limited crossing.

Star	blocks	Δt (d)	$\dot\nu_{1\star}$ (Hz s^{-1})	$\dot\nu_{2\star}$	$\Delta\nu$ (MHz)	miss (σ)	one carrier?
61 Vir	X822/X82f	3.00	+1092	+3221	+755.4	15	open
J1537−3319	X10359/X398	123	−2000	+2390	+223.5	3.6	open
AU Mic	X1838/X319	310	+1440	+2645	−159.0	–	open
HD 14055	X2b2ee/X2c3bb	0.09	−2390	+4216	+825.2	2447	no
	X24be0/Xd46	1.92	+1134	−3823	+63.0	41	open
	X2c3bb/X3835	2.00	+4216	+1800	−514.6	144	edge
	X2b2ee/X3835	2.09	−2390	+1800	+310.6	48	open
	X3835/Xd526	607	+1800	−2563	−555.5	–	open
	X2b2ee/Xd526	609	−2390	−2563	−244.9	–	open
	X3835/X25c5e	649	+1800	+2611	+423.3	–	open
	X2c3bb/X25c5e	651	+4216	+2611	−91.3	–	edge
	X2b2ee/X25c5e	651	−2390	+2611	+733.8	–	open
β Pic	X3a90/Xa9df	1.02	−1611	−679	+85.5	132	open
HD 285968	X108f/X5b4e	4.99	−29	−29	−0.002	0.6	yes

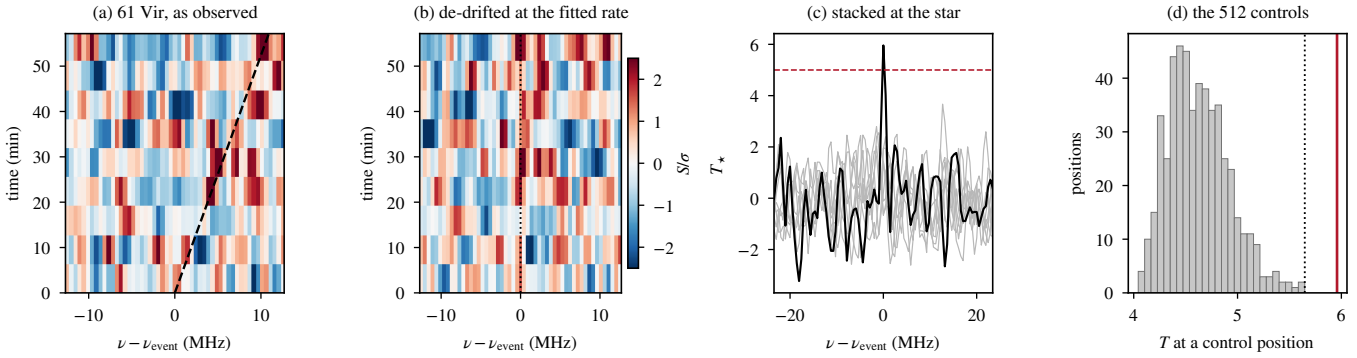


FIG. 7.— **The strongest unattributed crossing of the quality-passing data, and the control field it leads.** Frequencies are in the star's rest frame. *(a)* The dynamic spectrum at 61 Vir's position around the crossing at 346.3232 GHz, observed 2017 May 20, with the fitted drift of $+3182\,\mathrm{Hz\,s^{-1}}$ as the dashed track. The 225 integrations are binned into 9 and plotted in units of each bin's noise. At 119±20 mJy against 303 mJy per integration the carrier is 0.39 standard deviations in any one of them, and exists only in the drift-aligned stack of all 38 minutes on source. *(b)* The same rows shifted by that drift, which carries the carrier 22 channels of 488.3 kHz across the 57 minutes of elapsed time the block spans, not the 38 on source. *(c)* The de-drifted stack at the star (black) and at the 8 control positions with retained dynamic spectra (grey), in units of its own noise; dashed, the trigger. *(d)* The same statistic at each of the 512 rank-screen control positions: the star (solid) against the highest control (dotted). The recurrence test for this crossing, and for the second one – CP−72 2713 at 344.2965 GHz – is in Table 6.

HD 161868 stacks 13 epochs for a gain of only 1.15. Its $N_{\rm eff}$ is 1.7, so $\sqrt{N_{\rm eff}}$ is 1.29, and its stacked spectrum came out 1.12 times as noisy as predicted.

The deepest stacked limit is Proxima Centauri's, and it is a Class B limit. Proxima is covered by 66 blocks over 85 h spanning 6.3 yr, all of them coarse 15.625 MHz windows, so the stack constrains an unresolved spectral excess and discriminates no drift. 25 of those blocks belong to the primary census, 33 were searched afterwards from the same public archive and 8 come from the reserved hold-out, all observed over the census blocks' own years. Stacking the census blocks alone, the set every other limit here uses, gives $\mathrm{EIRP}_{90} = 2.2 \times 10^{13}$ W. Adding the rest of the archive deepens that by 1.26 to 1.73×10^{13} W. The best single epoch, on the same nominal injection-calibrated convention, gives 5.51×10^{13} W. Its $N_{\rm eff}$ is only 7.9, so $\sqrt{N_{\rm eff}} = 2.80$ of the measured 3.19 is what combining the epochs buys, the rest being this spectrum's own noise. The limit holds for an emitter steady in the stellar frame to within one 15.625 MHz channel, which is $20.9\,\mathrm{km\,s^{-1}}$ peak to peak, and Proxima Centauri b does not satisfy it. At 11.185 d about a star of $0.122\,\mathrm{M_\odot}$ – the archive's default parameter set (Akeson et al. 2013), for the planet of Anglada-Escudé et al. (2016) and a host mass agreeing with Kervella et al. (2017) – the planet moves at $47\,\mathrm{km\,s^{-1}}$, so a transmitter on it swings over $94 \sin i\,\mathrm{km\,s^{-1}}$, some 5 channels at $\sin i = 1$. The stack cancels such a carrier instead of building it unless the orbit is seen nearly face-on, $\sin i \leq 0.22$ or $i \leq 13°$, which is 2.5 per cent of randomly oriented orbits. The 66 epochs sample 98 per cent of that excursion at the orbital phase least favourable to the statement, so the shortfall is in these observations and not in an idealised orbit. Within a single block the same transmitter gains at most $1.2\,\mathrm{km\,s^{-1}}$, 6 per cent of one channel, so the single-epoch limit of 5.51×10^{13} W is the one that applies to it – 3.2 times shallower than the stack, and the deepest limit this survey places on the nearest plausible transmitter site. It also requires no inter-epoch drift above $7.8 \times 10^{-2}\,\mathrm{Hz\,s^{-1}}$, which is that same 15.625 MHz channel crossed once over the span of the epochs. At 1.30 pc that depth is mostly geometry; the gain is what stacking bought.

2 groups pass the full criterion and both are known lines. They are both β Pictoris carbon monoxide, in the star's rest frame and on the sign convention of Appendix C: CO $J = 1-0$ stacks $+0.07\,\mathrm{km\,s^{-1}}$ from the transition over 8 epochs, at $Z = 45.3$, and CO $J = 2-1$ $-0.39\,\mathrm{km\,s^{-1}}$ from it at $Z = 12.1$ – both inside the $0.5\,\mathrm{km\,s^{-1}}$ of the single-epoch crossings of the same star (§5.4). The two agree to $0.46\,\mathrm{km\,s^{-1}}$, or 0.73 of a CO(1–0) channel: the astrophysical positive control, passing in the stack as well. The survey's own $\pm 50\,\mathrm{km\,s^{-1}}$ mask leaves none of them unattributed. HD 285968 reaches $Z = 12.89$ on the line used above as the registration standard and is rejected by its own controls at 14.19: the emission is extended, as a cloud is and a transmitter is not.

Every stacked limit above holds only for a carrier that stays inside one channel. At CO(2–1) one 488 kHz channel spans $0.63\,\mathrm{km\,s^{-1}}$, the whole excursion a stacked carrier is allowed, peak to peak. Take a transmitter on a short-period giant planet, its line-of-sight velocity swinging by $\pm 30\,\mathrm{km\,s^{-1}}$. It crosses 94 of those channels peak to peak and de-registers completely, against 3.0 of a 15.625 MHz window: the coarse stacks buy reflex tolerance with bandwidth. Drift divides the same way, a stack in 488 kHz channels being confined to $1.3 \times 10^{-1}\,\mathrm{Hz\,s^{-1}}$ over the median inter-epoch span of 44 d. No crossing of §5.6 drifts that slowly. The slowest measured is $519\,\mathrm{Hz\,s^{-1}}$, which leaves one channel in 16 minutes, and in 33 of 38 cases the stack covering a crossing contains its discovery block. Toward the 20 stars carrying them the stack is a deeper limit on a non-drifting carrier, not a test of these events. Each stacked limit therefore constrains only emitters stable in the stellar frame to within its own channel; the single-epoch limits of §6.3 apply generally.

6. DISCUSSION

6.1. *What is different about a millimetre technosignature search*

This experiment's most transferable result is not the non-detection but a measurement of what limits a carrier search at these wavelengths. It is not depth: the power

TABLE 6
The two events, the crossings whose stellar statistic exceeds all 512 control positions of their own window; the first of them is Fig. 7. Repeat epochs measured are the blocks at which the predicted stellar-frame cell was read, the remainder of them coming from the reserved hold-out and the archival extension. $T_{\rm pers}$ is what a carrier persisting at the discovery amplitude S would have given there and $T_{\rm rep}$ is what is measured, combining every repeat epoch by inverse variance at the carried drift: one cell and one trial, so it may be negative, while the two indented readings beneath it are maxima and so positive on average whatever is there – the second of them the same cell at every trial of the window's own drift grid, which bounds an error in the carried drift. The nearest block alone follows, because the combination asks the model of §5.6 of every block across the whole baseline while the single reading asks it only of the two observations. The exclusion is the fraction of S still ruled out at the trigger, $(T_{\rm rep}+5)/T_{\rm pers}$; the displacement permitted is half a channel in the star's frame. The last row but one is the chance $\Phi(5T_{\rm pers}/T_\star-5)$ that a carrier at the trigger amplitude $5S/T_\star$ would have satisfied the criterion, and in brackets the same at S, which is selected as the largest of many cells and biased high.

	61 Vir	CP−72 2713
Frequency, stellar frame (GHz)	346.323236	344.296455
as delivered, topocentric	346.313811	344.269746
Drift rate ($\mathrm{Hz\,s^{-1}}$)	+3182	−2056
$T_\star$	5.958	5.809
Highest control T	5.65	5.68
Fitted amplitude (mJy)	118.8 ± 19.9	11.3 ± 2.0
Peak-channel EIRP (W)	5.1×10^{14}	8.9×10^{14}
Repeat epochs measured	3	5
of them in the primary census	3	1
census repeats with no retained spectrum	1	0
Nearest covering epoch, separation (h)	−1.97	+2.05
covering epochs more than a day away	2	4
Longest baseline (d)	3	2
T expected if persistent, $T_{\rm pers}$	9.70	6.30
T measured, matched drift, $T_{\rm rep}$	+0.39	−0.82
largest single epoch	+1.65	+0.96
maximised over every drift	2.71	3.60
Nearest block alone, $T_{\rm pers}$	5.28	6.21
$T_{\rm rep}$	−0.01	−0.57
Amplitude still excluded, $(T_{\rm rep}+5)/T_{\rm pers}$	0.56 S	0.66 S
from the nearest block alone	0.95 S	0.71 S
from separated blocks alone	0.67 S ($n=2$)	11.79 S ($n=1$)
over the matched registration tolerance	0.67 S	0.84 S
Displacement permitted ($\mathrm{km\,s^{-1}}$)	±0.211	±0.213
Recovery chance at $5S/T_\star$ (at S)	1.00 (1.00)	0.66 (0.90)
Disposition	not confirmed	not confirmed

a transmitter must radiate to cross the threshold here is a median 1.5×10^{15} W per system, and more power buys nothing at a frequency the archive never tuned to or never returned to. That inherited coverage sets the limit. Molecular-line confusion is a different matter: it costs bandwidth, and it decides how a crossing is disposed of once one is found.

Line emission, and not interference, is the astrophysical false positive here. Catalogued molecular transitions, almost all of them carbon monoxide, account for 18 of the 52 crossings in quality-passing data, and no crossing in this survey is attributed to interference (Appendix E). 16 of those attributions carry support beyond the frequency match – a second transition of the same species, catalogued gas on the sightline, or the crossing's own recurrence – while two rest on the coincidence alone, which leaves them consistent with molecular emission and not identified as it. The attributed crossings include the strongest events in the quality-passing data, but they are not the commonest: of the 34 unattributed crossings the 31 of the primary census are no more numerous than the control positions of their own fields predict (§5.3.1), so chance at a trigger of 5 is enough to account for most crossings by number while line emission accounts for the ones that stand out. 10 of the 12 windows that reached the spatial stage are disc or interstellar carbon monoxide, all but one the resolved disc of β Pictoris and the exception a cloud along the sightline to HD 48370. The remedy that works is a velocity-space mask evaluated in each star's own rest frame, and §5.4 gives what it costs in bandwidth.

Spatially extended emission is the second discriminant, and it has no analogue at centimetre wavelengths. A line filling the primary beam raises the star and the control annulus together instead of cancelling, so the spatial statistic loses power exactly where circumstellar gas is brightest. β Pictoris shows the benign face, its circumstellar carbon monoxide recovered by the same search (§5.4); HD 48370 shows the other. Its emission is an interstellar cloud and not a disc (Appendix C), it lies near a band edge, and that one window contributes 59 per cent of every near-edge threshold exceedance in the survey. Pooled, the near-edge rate is ×2.464 the interior rate; without that window it is ×1.020, and on a held-out half ×1.181 ($p=0.24$). That near-edge excess is one interstellar cloud, so no edge correction is applied to any limit.

The channelisation is inherited. In 1220 of the 1258 coarse-channel windows every drift in the grid stays inside one channel over a whole track and in 38 more it crosses barely one; none of them can distinguish a drifting carrier from a stationary one, which is why the carrier search rests on the 405 fine-channel windows. The widths on offer here run from 15.3 kHz to 31.25 MHz, about 3.3 orders of magnitude. Set beside the 2.8 Hz channels of centimetre-band carrier searches the range becomes 7 orders (Fig. 1(b)), and nothing in the search could choose them.

Repeat epochs turn a crossing into a testable claim, and an archive supplies them only incidentally. Let the *confirmation coverage* $C_{\rm conf}(\nu)$ be the number of systems toward which two execution blocks of different observing sessions more than a day apart both cover the sky frequency ν, so that a carrier found once there could have

TABLE 7

The threshold crossings of the reserved hold-out, searched with the same pipeline and disposed of by the same tests as the census. $\Delta v_\star$ is the offset from the nearest masked transition in the star's own frame and $n_{\rm rep}$ the number of covering repeat blocks; $T_{\rm pers}$ and $T_{\rm rep}$ are as in the crossing ledger. The one crossing that is present again is a carbon monoxide line. A dagger marks an offset inside the attribution interval whose species the attribution set excludes, as in the crossing ledger; such a crossing is reported and not acted on (Appendix C).

Star	block	$\nu_{\rm topo}$ (GHz)	$T_\star$	line	$\Delta v_\star$	$n_{\rm rep}$	$T_{\rm pers}$	$T_{\rm rep}$	disposition
HD 285968	Xdaef5a_Xdd9b	230.5081	8.642	CO(2–1)	+8.7	4	11.6	+6.71	attributed; present again
HD 14055	X107376c_X6126	330.7123	5.513	^{13}CO(3–2)	+112.6	30	312.0	+0.56	unattributed; does not recur
TRAPPIST−1	Xd0e450_X259b	346.2454	5.326	SO(9_8–8_7)	−306.6	5	11.1	−0.32	unattributed; does not recur
HD 172555	X107fa96_X1300f	344.6066	5.144	SO(8_8–7_7)	+251.3	–	–	–	unattributed; no repeat coverage
HD 285968	Xdb4b9a_Xa7c	229.9129	5.106	CO(2–1)	−768.0	4	11.2	+0.21	unattributed; does not recur
HR 1010	Xc6b674_X23f2	346.2301	5.044	SO(9_8–8_7)	−238.7	7	25.9	−0.80	unattributed; does not recur
HD 10647	Xff99e1_X9cdf	345.7059	5.017	SO(2_3–2_1)	+37.1†	13	142.1	−0.93	unattributed; does not recur
HD 170773	Xff294f_X1fb	345.5597	5.001	SO(2_3–2_1)	−113.8	4	59.3	+1.45	unattributed; does not recur

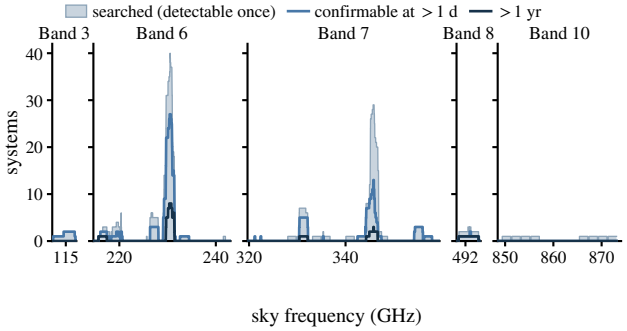


Fig. 8.— What could be searched against what could be confirmed, for a carrier whose stellar-frame frequency at the two epochs agrees to within half a native channel. For the Class A experiment, against sky frequency: the number of systems toward which a carrier could have been detected once (grey), the number for which a block of another observing session more than a day later also covers that frequency (blue), and the same beyond a year (dark). Counted over the primary census; the repeat blocks beyond it that §5.6 reads add 1.0 MHz to this support and move no curve visibly. One panel per populated ALMA band, widths in proportion to the frequency each contributes, as in Fig. 9.

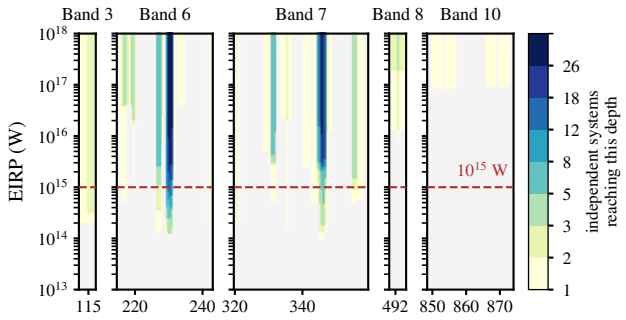


Fig. 9.— Where the experiment could have found something. Shading gives the number of independent systems in which an unresolved carrier of the plotted EIRP at the plotted sky frequency would have been recovered nine times in ten *in a single epoch*; unshaded means none. That is a recovery completeness: confirmation requires a second independent epoch covering the same cell, which Figure 8 maps and which no amount of transmitted power can supply. One panel per populated ALMA band, with panel widths in proportion to the frequency each contributes: the 47.7 GHz Class A union is 26 disjoint islands across 114.0–873.1 GHz, which a continuous abscissa would draw as hairlines. The dashed line is 1×10^{15} W.

been sought again. It must be stated against frequency and not as a fraction of the sample, because repeat visits re-tune and a second epoch is of no use at a frequency it does not cover. Its support is 24.0 GHz of union bandwidth, 50 per cent of the 47.7 GHz searched, and it falls to 8.5 GHz beyond a year. At its best, near 231 GHz, it reaches 27 systems (Fig. 8). The searched extent is about twice the extent over which a carrier returning to its predicted cell could have been confirmed, and for close to half the sample the archive constrains an intermittent emitter hardly at all, whatever its power.

6.2. *The search space this archive opens*

The searches of Steffes & DeBoer (1994) and Mauersberger et al. (1996) reached nearby stars above about 30 GHz at one frequency only, and the one other millimetre search of more than a single line, Mason et al. (2025), covered 117 MHz of Band 3 toward far more distant stars (§2); Band 3 is searched here, in 112 windows down to 90 GHz including β Pictoris' CO $J = 1-0$ control. The haystack axis of Wright et al. (2018) runs from 10 MHz to 115 GHz, and this experiment covers 19.9 GHz of it. Only 1.0 GHz of that is carrier search, because the Class A union spans 114–873 GHz, and the rest lies above the axis altogether.

Union bandwidth overstates what that buys: different frequencies were searched toward different numbers of stars and to very different limits. Figure 9 gives instead the number of independent systems that would have yielded a recovery at ninety per cent completeness, against sky frequency and transmitter power. At an EIRP of 1×10^{15} W the covered frequency falls to 19.1 GHz, which is 40 per cent of the union. It reaches 26 systems, and 18 at the best-covered frequency near 231 GHz. With no limit on power at all that frequency reaches 40 of the 61 systems, and half the sample is reached only above 3.9×10^{15} W.

On the continuous-waveform transmitter figure of merit of Enriquez et al. (2017), with the single survey-wide fractional bandwidth replaced by the star-bandwidth product an archive requires, this experiment is 3.7×10^4 on the median system and 9.5×10^5 on the sample's largest threshold power. That is a factor 2.2×10^5 above the Enriquez value once both are read at the same threshold, which splits into 2.76 decades of haystack and 2.58 decades of depth: a far smaller haystack than a wideband centimetre survey of thousands of stars, and a shallower one. Appendix G gives the terms, the published values on a common threshold, and the two conventions for the denominator.

6.3. *What this experiment excludes, and what it does not*

The signal these limits describe. Every limit here applies to one kind of signal: an unresolved carrier at the stellar position, confined to a single 15.3–1953 kHz channel, lying outside the molecular-line mask, radiating while one of the archival blocks is on sky, and changing frequency linearly at no more than its own window's ceiling, 12.0–13.3 nHz. Such a carrier would have produced a crossing nine times in ten above the nominal EIRP_{90} of §5.1, which excludes atmospheric decorrelation, and none survived toward any of the 61 systems the carrier search covers. Four conditions bound it as sharply as power does: the coverage is sparse and inherited (Fig. 9); the drift grid reaches an acceleration that contains every known planet of the sample but one, while a transmitter on a rapidly rotating planet spends most of that reach on its spin alone (Fig. 3); emission broader than one channel is retained but unmatched by the single-channel filter, which needs close to 32× the flux density for a feature as wide as the widest line these fields show, so the thresholds quoted here do not apply to it; and a carrier on for part of a block comes back at that fraction of its amplitude, while one absent throughout is not constrained at all.

Recurrence applies over a far smaller domain. Where the archive observed a frequency only once, no transmitted power makes a confirmation possible (§5.6.1). Where the test does apply it has power only against a carrier as strong as the trigger demands of the repeat, so a shallow repeat excludes nothing (§5.6).

The result is a non-detection and not an exclusion of persistent transmitters. An exclusion is read at the one cell the first epoch predicts, so the transmitter it tests is one that comes back to that cell, and four restrictions on that model are ours. A transmitter operating intermittently need not have been radiating during the second block. A second block need not cover the frequency at all, and over about half the searched bandwidth none does (§6.1). The interval that cell permits is half a channel in the star's own frame, which the drifts fitted to these crossings cross in seconds (§5.6), so a transmitter accelerating like that is in the predicted cell again only if its motion is periodic and the second epoch catches it at the matching phase. The wide search over the whole covering window at every drift bears on such a carrier only as a population, through the agreement of the number of recurrences with chance. A transmitter may also change frequency deliberately, and nothing here addresses that. Outside the adopted criterion, then, nothing is excluded: neither emission inside the $\pm 50\,\mathrm{km\,s^{-1}}$ molecular-line mask at any strength, nor any emission from the stars within 40 pc that ALMA has not observed, which is almost all of the 17 566 this volume holds.

An illustrative engineering benchmark. Take a 12 m dish radiating 1 MW at 230 GHz with an aperture efficiency of 0.7. Its directive gain is $G = \eta(\pi D\nu/c)^2 = 5.9 \times 10^8$, so its EIRP is 5.9×10^{14} W, above the single-epoch crossing limit of 17 of the 61 Class A systems. The same gain makes its beam $\Omega = 4\pi/G = 2.1 \times 10^{-8}$ sr, illuminating 1.7×10^{-9} of the sky, with a half-power width of 25 arcsec at that frequency. EIRP is not transmitter power – the same megawatt radiated isotropically would be 10^6 W – so these limits apply to beamed transmitters whose beam contains the Solar System.

Two blind spots. The search forms Stokes I as (XX + YY)/2, which retains the full intensity of a circularly polarised carrier; the loss is not sensitivity but the discrimination polarisation would offer against unpolarised astrophysical emission, because ALMA's feeds are linear and the circularity lies in the cross-hands, which this analysis does not extract. And the frequency coverage is the archive's and not a choice: the 203 GHz positronium hyperfine line that Steffes & DeBoer (1994) and Mauersberger et al. (1996) observed lies in ALMA Band 5, where this survey holds 4 coarse-channel windows, all toward α CMa B, and none of them covers it.

6.4. *What a dedicated ALMA technosignature experiment should do differently*

A purpose-built experiment on the same telescope would be far stronger, and the changes it needs follow from the limitations above. *Channelise finely everywhere:* the 1258 coarse-channel windows cannot carry a carrier search at all. *Choose the targets and the frequencies:* an archive weighted toward dusty discs is the worst case for molecular-line confusion, the M dwarfs that dominate the nearby habitable-zone planet census are the least covered, and windows placed deliberately away from the bright transitions would recover most of the 3.05 GHz of the 47.7 GHz Class A union that the mask costs – all three are decisions a proposal can simply reverse. *Repeat the epochs on purpose, at the same tuning:* two or three visits separated by days and by years turn a crossing into a testable claim, and the separation is a scheduling constraint rather than extra sensitivity; Fig. 8 is what its absence costs. Two epochs cannot test an orbit, only fit one; three separated epochs over-determine it, and 27 of the 61 systems here carry three at a common frequency. The shortfall is not coverage but a candidate seen at more than one of them. *Retain the visibilities:* rotating the phase centre to the star leaves a genuine point source entirely in the real part of the visibility. That asks whether the data contain a source where the star is, which is a stronger question than any rank against neighbouring positions in an image.

7. CONCLUSIONS

1. *What is new.* In the millimetre band the record for nearby stars is two measurements of one spectral line: Steffes & DeBoer (1994) and Mauersberger et al. (1996) each covered no more than 35 MHz at narrowband resolution around the 203 GHz positronium line, with no search over drift rate and no epoch-to-epoch test of persistence. This survey covers at least 1363 times that bandwidth, searches a grid of drift rates, and asks of every event whether it returns (§2).
2. *What was searched.* The carrier search is 405 fine-channel windows toward 61 systems within 40 pc, covering 47.7 GHz of unique frequency in 26 disjoint intervals. In 391 of those windows the track is long enough for the searched drift to cross a channel, covering 44.4 GHz toward 58 systems. A coarse-channel search for a steady unresolved excess adds 1258 windows toward 82 systems. Both come from one pass through 407 public ALMA execution blocks and together span 90–873 GHz (§3).
3. *How deep.* The median system's best window would

have produced a threshold crossing from a carrier of 1.5×10^{15} W nine times in ten, and 26 systems reach 10^{15} W on that measure. That is the completeness of a single observation and not of a confirmation (§5.6.1); for 55 of the 61 systems it is carried across from the windows that were injected into and not measured in their own; and it leaves out atmospheric decorrelation, which a real carrier suffers and an injected one does not, so the net bias on any one limit runs either way and Table F1 gives it system by system (§5.1).

4. *What was found.* The search produced 56 threshold crossings: 50 Class A in the primary census, four Class B, and two Class A in archival-tail blocks. All four Class B crossings come from the single execution block that fails the data-quality criterion of Appendix D.2, so the quality-passing data hold 52 crossings, every one of them from the carrier search. 18 of those are attributed to a catalogued molecular transition within $\pm 50\,\mathrm{km\,s^{-1}}$ of the star's velocity, circumstellar or interstellar, and 34 are not. An attribution is not an identification, and §6.1 says which of them carry evidence beyond it. Of the 34, two lie in archival-tail blocks that enter no expectation and one is atmospheric, leaving 31 to set against the 24.6 these fields' own control positions predict: a count that large or larger arises by chance with probability 0.13, or 35 against 24.1 at $P = 0.027$ over a narrower field, which drops integrations and so is not the statistic §4.1 defines. Either way this is consistency and not a demonstration that the crossings are noise, since a further 17 would have been consistent too, 22 on the narrower reading (§5.3, §5.3.1).

5. *What the repeat observations exclude.* Nothing came back. A transmitter on an orbiting body arrives at a frequency its own motion has moved, so every unattributed crossing another block covers was searched again twice over: across that whole covering window at every drift of the grid, where 29 of the 328 windows hold a statistic above the trigger against the 28.3 chance predicts in trials of that size, and at the single cell the crossing's own discovery predicts, where the largest statistic anywhere is 2.48 at the drift fitted at discovery and 4.71 at any drift of the grid, against a trigger of 5. The same reading recovers 13 of the 15 line-attributed crossings it was tested on, each in a different observing session more than a day away (Table D3), so the machinery does return a repeat where one exists. That exclusion is narrow, and the narrowness belongs to the criterion and not to the archive: a carrier radiating in both observations whose stellar-frame frequency comes back to within half a native channel. A transmitter accelerating at the 0.5 to $3.6\,\mathrm{m\,s^{-2}}$ these crossings themselves show re-enters that interval only if its motion is periodic and the second epoch catches it at the matching phase, so this is not a test of a persistent transmitter of any kind. Where two crossings of one star might have been a single carrier, 13 pairs carry enough measurement to be asked: the searched grid separates 1, and the other 12 are *open* – neither a confirmation nor an exclusion, with only the agreement of their number with chance bearing on them (§5.6).

6. *What could have been confirmed at all.* A search limit and a confirmation limit are different statements, and here the second is the binding one. Of the 61 systems the carrier search covered, 40 were observed again in another observing session more than a day later at a frequency the first block covered, and 13 more than a year later; for the remaining 21 no repeat exists, so no transmitted power would have made a confirmation possible (§5.6.1). Coverage limits this search, and no depth of observation could have removed that limit.

7. *Stacking the epochs.* Combining each star's observations in its own rest frame deepens the flux-density limit by a median factor 1.52. The deepest result is Proxima Centauri's: 2.2×10^{13} W from the census blocks and 1.73×10^{13} W once the rest of the archive is added, in coarse 15.625 MHz channels, so it bounds a steady unresolved excess and not a drifting carrier (§5.7). Stacking buys that depth with a tolerance of one channel, and Proxima Centauri b's own orbit carries a transmitter across 5 of them, so the stacked figure applies to the one body in that system only if the orbit is within 13° of face-on. The single-epoch limit, 5.51×10^{13} W, applies at any inclination.

8. *What this does not constrain, and what would change the answer.* These limits hold only inside the domain actually searched. The drift grid stops at 12.0 nHz, 6 per cent of the ceiling recommended for a general carrier search (Sheikh et al. 2019), so an unbound transmitter, one on a rapidly rotating planet, or one whose orbital and rotational accelerations together exceed that is not constrained here at all (§6.3). The obstacle transfers beyond this archive: channelisation chosen by other observers, and not collecting area, decides which search is possible – and three choices a proposal can simply make would lift it, namely fine channels everywhere, frequencies and targets chosen against the line forest and not inherited from programmes that sought it, and repeat visits at one tuning days and years apart (§6.4).

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This paper makes use of the following ALMA data: the 67 projects listed in Table A3, each cited as `ADS/JAO.ALMA#` followed by its project code. The 407 primary-census execution blocks searched span 2013 October 6 to 2025 July 11, and we thank the principal investigators and teams who proposed and obtained them. ALMA is a partnership of ESO (representing its member states), NSF (USA) and NINS (Japan), together with NRC (Canada), NSTC and ASIAA (Taiwan), and KASI (Republic of Korea), in cooperation with the Republic of Chile. The Joint ALMA Observatory is operated by ESO, AUI/NRAO and NAOJ. This work has made use of data from the European Space Agency (ESA) mission *Gaia* (Gaia Collaboration et al. 2023), processed by the *Gaia* Data Processing and Analysis Consortium (DPAC). This research has made use of the SIMBAD database (Wenger et al. 2000) (CDS, Strasbourg, France); of the NASA Exoplanet Archive (Akeson et al. 2013), which is operated by the California Institute of Technology under contract with the National Aeronautics and Space Administration under the Exoplanet Exploration Program; and of the Splatalogue spectral-line database (Remijan

et al. 2007).

Use of generative artificial intelligence. Large language models were used in this work. They wrote most of the software: the archival retrieval and calibration scripts, the search and vetting pipeline, the statistical tests, and the generators that produce every number, table and figure in this paper. They also drafted text, which the authors revised. The authors set the scientific question, chose every criterion, and are responsible for the content of this paper.

Testing and validation of that analysis. No result here is accepted on a model's assertion. No number in this paper is typed: each is written by the program that computes it from the data, so a value in the text cannot differ from the analysis it cites – which is not the same as saying that no other extraction of the same data would give a different one, and Table D3 names the one window where the archive holds two extractions that differ. A separate audit re-derives the identities that must hold among the published values – counts that must partition, quantities that must sum, numbers reached by two routes that must agree – and refuses the paper where one does not. Every assertion in the analysis was first made to fail on purpose, on deliberately wrong input, before it was relied on, since an assertion that has never fired is no evidence that anything is right.

Software. The pipeline depends on CASA 6.7.6 (CASA Team 2022), PYTHON 3 with ASTROPY (Astropy Collaboration et al. 2022), NUMPY (Harris et al. 2020) and SCIPY (Virtanen et al. 2020), and the ALMA Science Archive and SIMBAD TAP interfaces through ASTROQUERY (Ginsburg et al. 2019); all are free and open-source, at the versions cited.

Author contributions. G.J.W. conceived the survey, defined the target sample and the selection criteria, and wrote the manuscript. R.D. built and operated the archival retrieval, calibration and search pipeline, implemented the spatial control statistic and the injection–recovery campaigns, and produced the per-window catalogue and its regeneration scripts. Both authors designed the vetting procedure, chose the masked species, carried out the statistical validation reported in §4.1 and Appendix D, and agreed the dispositions of the crossing ledger.

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DATA AVAILABILITY

Public. Every observation this paper rests on is public and is identified here precisely enough to be fetched. The 407 primary-census execution blocks, and those of every other set of Table A1, are held in the ALMA Science Archive under the 67 project codes of Table A3 and are retrieved by the execution-block identifier, which the crossing ledger gives for each of the 56 threshold crossings; archival data should carry the full provenance of their origin and the credit for it (Chen et al. 2022). The astrometry is Gaia DR3, the planet counts the NASA Exoplanet Archive, and the mask one Splatalogue query of CDMS and JPL made on 2026-10-06 and repeatable as it stands.

Five plain-text files accompany the paper on the arXiv, served beside the article and not part of it, each describing its own columns in its own header: the per-window catalogue, one row for each of the 1663 searched windows, with the star, its system and the identifier it was extracted under, the execution block and ALMA band, the window's frequency limits and channel width, the on-source time, the noise and the amplitude limit it sets, the drift and acceleration ceilings searched, the adopted EIRP_{90} limit with its interval, the distance, the search class, the start epoch of its execution block, whether its completeness was measured or transferred, its control ensemble, the exceedances in it, the share of them the line mask attributes and the share the atmosphere takes, the crossing flag and the disposition of any crossing it holds; the crossing ledger, one row for each of the 56 crossings, with the star, its execution block and band, the crossing frequency in both frames, the search and control statistics, the nearest masked transition and the offset from it, the recurrence test and the disposition and sampled in Table D2; the master counts, all 21 rows of Table A2; for each crossing carrying a quoted exclusion, the probability that a carrier at the trigger amplitude would have satisfied the recurrence criterion, calculated and measured; and, for the same crossings, the exclusion depth read over separated blocks alone and over the matched registration tolerance.

What they support. With the tables here – Table A1 for the block and window partition, Table A2 for the headline counts, and Appendix F for every system of the carrier search – they rebuild the window, star and system counts; the crossing counts and their classification; the chance expectation of 24.6 crossings, as a sum over each Class A window's own control exceedance rate, masked share and atmospheric factor; the EIRP_{90} summaries, including which windows carry an injected completeness and which a transferred one; and every exclusion depth with its recovery probability. Three things they do not give on their own: which crossing the atmospheric screen of Appendix E removes, which the ledger does not mark; the sub-trigger stellar-to-control step of Appendix D, measured on above-threshold cell counts that are not released; and the interval on the chance expectation, whose resampling the per-window rates support only in part.

Not public. The calibrated visibilities, the per-window products and the code that produced them are not distributed, so this execution of the search cannot be re-run. The released tables support a narrower claim: the principal statistics can be rebuilt from them, row by row. Whether the description here suffices to implement the search afresh has not been tested.

APPENDIX

A. SPECTRAL RESPONSE, FREQUENCY CONVENTIONS AND THE PRIMARY-BEAM RESPONSE

This appendix sets out the three things every threshold quoted in this paper is checked from: the frame of each

frequency, the instrumental response, and the primary-beam factor that turns a measured noise into a flux density at the star.

A.1. *Frequency frames, linewidth and smearing*

Every frequency axis here – searched spectra, tabulated ranges and crossing frequencies alike – is *topocentric* as delivered by the ALMA correlator, and the search applies no velocity-frame shift. Velocities enter at the vetting stage, where the line mask carries every crossing frequency to the barycentre at its own block's epoch and then to the stellar frame before comparing it with a transition (Appendix C). The residual error is then the catalogue radial-velocity uncertainty alone, a few $\mathrm{km\,s^{-1}}$ within 40 pc; second-order relativistic terms at the $50\,\mathrm{km\,s^{-1}}$ scale are $\lesssim 5$ kHz at 350 GHz, below the finest native channel searched. The stacking is drift-aligned and inverse-variance weighted: coherent in frequency trajectory, not in electric-field phase.

The native channel width is also the linewidth convention for every EIRP threshold. The statistic is the single-channel flux density after drift correction, so $\mathrm{EIRP_{trig}}$ is the minimum *total* power of a channel-confined carrier, flat in the assumed linewidth below the channel and never a spectral power density. Above the channel it is not a threshold at all: spreading a fixed total power over N channels lowers the statistic as $N^{0.81}$, measured on Eq. 1 as the power recovered nine times in ten, so a feature as wide as the widest line these fields resolve (69 channels) must carry 32× the power of a carrier. A filter matched to such a feature would pay only 8.5×, going as $N^{0.50}$, the square root of its width, and this search uses no such filter. Flatness is exact only for a centred tone, and a boundary-straddling one divides its power between two channels, which is the loss the injections carry and one reason EIRP values are quoted to two significant figures. The Hz-resolution surveys' limits are total powers in the same sense, each the best case of its own channelisation, which makes the axes of Fig. 1(a) commensurable; and no reprocessing recovers resolution the correlator never recorded. Channel widths come from the measurement sets' own CHAN_WIDTH; MS v2 defines no "EFFECTIVE_BANDWIDTH" column, so the quoted channelisations are separations and are not claimed to be noise bandwidths.

ALMA applies online Hanning smoothing by default (Cortes et al. 2026), so the spectral response is ∼2 channels wide and neighbouring channels are correlated. For the 0.25/0.5/0.25 kernel the coefficients are $\rho_1 = 2/3 = 0.667$ and $\rho_2 = 0.167$, and the smoothing narrows the noise by $\sigma_{\mathrm{smoothed}}/\sigma_{\mathrm{raw}} = 0.612$. The correlation is not confined to adjacent channels: the lag-two term is a sixth of the lag-one term. The peak channel holds 0.50 of an unresolved carrier's flux at a channel centre and 0.42 on a boundary, a median 0.48 over sub-channel phase.

That loss enters the limits exactly once; the thresholds are free of it and the limits are not. A window's trigger flux is the bare 5σ of its own spectrum, so setting the *total* power of a channel-confined carrier against it would be optimistic by ×2.00–×2.36. The injected carriers are not set against it in that way: each is deposited with the correlator's own channel response, at a sub-channel phase drawn afresh, and recovered through the same trigger, which reads the peak channel. The measured completeness therefore already contains the loss, and factorises into it exactly: $3.03\,P_{\mathrm{trig}}$ is ×2.29 for the channel response and ×1.325 for the margin a peak channel needs above the nominal trigger for nine carriers in ten to return. So P_{trig}, the sensitivity S_{min} below and the filled symbols of Fig. 1(a) are nominal throughout, which is what makes them commensurable with other programmes' detection thresholds, while $\mathrm{EIRP_{90}}$ carries the correction and is not corrected again. The deposition falls 8 per cent short of the ideal response averaged over phase, so the completeness is conservative by that much; and where in a channel a carrier was placed makes no difference to what is recovered, because at a typical searched drift rate it crosses about 11 channels during one observation and samples the whole response.

Intra-integration smearing, which the injections do not model. Each injected tone is deposited at one frequency per integration, so the campaign carries the channel response and the drift between integrations but not the sweep within one. A carrier moving Δ channels during an integration spends its dwell time spread across the delivered channel response, so the amplitude the correlator accumulates is the mean of that response over the sweep: η_{smear} is that mean, and every limit in this paper is divided by it once. The response is the one that sets ×2.29 above – Hanning-smoothed, and channel-averaged where the window is one of the 115 Class A windows that are. *The sweep cannot act inside the transform*: these channels are 9×10^4 times wider than the reciprocal of an integration, so a sweep does not destroy coherence but redistributes dwell time. The transform-limited amplitude $\mathrm{sinc}(\pi\Delta/2)$, which is what a sweep leaves when the spectral resolution is set by the integration itself, is the more severe of the two wherever they differ: in the worst case, a carrier crossing 0.97 of a channel, it asks for ×1.53 where the delivered response gives ×1.05 – a correction larger by 46 per cent. The two forms part by more than one per cent in 13 windows, and the correction itself exceeds one per cent in 9. The Δ that sets a ninety per cent point is the one at the ninetieth percentile of each window's own drift draw, 89 per cent of the searched ceiling over the campaign's 10 332 tones. The integration length is recorded for 1054 of the 1663 windows, 455 of them in the delivered metadata and 599 more in the exposure records the search kept. For the other 609 the factor is *bounded* at 11.6 s, the longest integration delivered anywhere, where none falls below 1.000. Nothing moves in the median window, and 9 windows on 4 stars rise by more than one per cent, up to ×1.05: HIP 10679, HD 172555, β Pic and η Crv. Re-scoring the 49 injected windows' own ladders with every tone de-rated by its own η_{smear} gives a penalty never larger than the one applied and smaller by at most 1 per cent, so the correction errs the safe way.

A.2. *The primary-beam response*

The sensitivity quoted for a window is $S_{\mathrm{min}} = 5\sigma/A$, where A is the primary-beam response at the star's offset from the phase centre: the noise σ is measured in apparent flux density, the limit is quoted in true flux density, and the two differ by A. That offset is not a nuisance. Archival fields are pointed at a catalogue position, often decades old, while these are high-proper-motion stars, so the star can sit well off axis. Over the 1663 windows

the offset has a median of 0.089″, a 99th percentile of 7.98″ and a maximum of 17.99″. The response A runs from 0.999982 in the median down to 0.2308. It falls below 0.99 on 99 windows and below 0.9 on 46. Every one of those is a physically explicable J2000 pointing on a fast-moving star rather than an error.

The response applied is a Gaussian of the full width the observatory documents, $1.13\,\lambda/D$ (Cortes et al. 2026), evaluated at each window's *own* dish diameter. The dish matters: the 1058 rows taken on the 1.7× smaller compact-array antennas have a correspondingly wider beam, and a Gaussian forced to $D = 12$ m reproduces the per-window geometry only to 7.7×10^{-2} against 8.7×10^{-5} for the per-dish form. That width is a physical choice rather than a convention. The blocked-aperture response, which carries the shadow of the secondary and is what CASA applies (CASA Team 2022), has a half-power width of $1.15\,\lambda/D$ on a 12 m antenna and $1.14\,\lambda/D$ on a 7 m one. The Gaussian used here is that beam to 2 per cent and departs from it by at most 6.2 per cent anywhere in the sample. The other width available, a Gaussian of the Airy first-null width, is wider and departs from that beam by up to 30.8 per cent, which is why the documented width is the one adopted; what it would do to the limits of the windows furthest off axis is in §5.1.

A window is used where the two forms agree. At low response a small error in the beam model becomes a large error in inferred flux density, and the two forms part company as the offset grows, so the limit on retention is the offset at which they differ by 10 per cent: $0.83\,\lambda/D$ from the pointing centre, where the response is 0.22. The furthest window in the primary census lies at $0.82\,\lambda/D$, where the two agree to 6.2 per cent, so the criterion excludes nothing from it; it is stated because it is what admits the one star this survey covers far off axis.

That star is ϵ Eridani, at 3.22 pc the nearest in the sample with Band 6 coverage. 20 of its 44 windows, in 11 execution blocks, place it at the pointing centre and need no correction at all. The other 24 lie 0.75–$0.82\,\lambda/D$ off it, where the response is 0.23–0.29. Those offsets are inside the main lobe – the blocked-aperture response does not reach its first null until $1.36\,\lambda/D$ – and there the two beam forms agree to 7 per cent, which is the extra uncertainty those limits carry. No window reaches the trigger, the largest statistic being 4.7. Over 6.6 window-hours the deepest of the 11 fine windows reaches 1.2×10^{14} W at 488–977 kHz, in `A002_Xe59f51_X1822`; the deepest that lies inside the primary census is in `A002_Xe539c7_X12189` at 1.4×10^{14} W, and that is the limit Table F1 gives for this star. The 33 coarse ones reach 4.7×10^{14}–4.7×10^{15} W. 12 of these windows fall inside the primary census and carry its statistics – 8 of them off axis and 4 at the pointing centre – counted with it in Table A1; the other 32 are listed apart from the archival extension's null and are reported here instead. Nothing about the windows themselves sets them apart: all 32 were searched with the full control ensemble, the largest statistic among them is 4.6, each carries its own phase centre and integration time from its measurement set, and the floor above admits every one of them – their lowest response is 0.23, and the window furthest from a pointing centre anywhere in the archival coverage is one of these, at $0.82\,\lambda/D$.

Nor does any block of this star carry an annual-parallax record. The displacement follows from its catalogued parallax, 310.6 mas, which is the largest in the sample: at the epoch of `A002_Xe539c7_X12189` it is 0.22″, or 0.20 of that block's synthesised beam, and 0.23″, or 0.21 of a beam, at the epoch of the block that sets the deepest limit on this star. That is smaller than the 0.25 of a beam at which the retained amplitude was measured below, so that measurement bounds the loss instead of being extrapolated to it, and the limit is optimistic by at most 9 per cent on this account. It is carried as a one-sided systematic in Table F1 and not folded in, because the retention every other window's correction uses is measured on that block's own baselines and these deliveries expose none. The flux-scale comparison of §4 is a separate continuum measurement on the same star, a block mean over the 4 compact-array executions that place it at the pointing centre, 0.22 h in Band 6, where no beam correction enters either.

The correction is applied exactly once. The offset is computed for every measurement set from its own `FIELD::PHASE_DIR` and the propagated stellar position, so no window carries an offset that is zero by assumption, and it enters in one place only: $S_{\min} = 5\sigma/A$ closes on all 1663 catalogued rows to 8.7×10^{-5} in relative terms. On 10 rows whose products are not retained A is back-solved from $5\sigma/S_{\min}$ and flagged as such, so there the closure is arithmetic; the test is the other 1653 rows, where A comes from the search's own stored geometry and closes to 8.7×10^{-5}. Nothing else needs correcting: the statistic of Eq. 1, the control ranks and the attribution are *ratios* formed inside one window, in which A divides out identically.

The star's apparent direction includes the annual parallax, and the amplitude a compact source retains when that is left out of the assumed phase centre is measured rather than modelled. A small fraction of a synthesised beam attenuates a compact source and does nothing else, while a displacement approaching a beam puts the phase wrong on the longest baselines, so the size of the displacement decides whether a scalar correction is legitimate at all. Re-extracting the 8 windows of 2 blocks furthest out at the full apparent position – proper motion, parallax and radial velocity where catalogued – leaves each window's noise unchanged to 0.20 per cent and alters no disposition. It cannot test the amplitude factor, which describes a source those windows do not hold: the statistic merely scatters from 0.78 to 1.15 about 0.98 instead of approaching the 0.49 a source would give. A carrier injected at the apparent position and recovered from the *same* visibilities at both does return the normalised dirty beam, on that block's own baselines and with no model in it. It measures 0.92 against a predicted 0.89 at 0.25 of a beam. At 0.67 of a beam it measures 0.58 against 0.49. It exceeds the model at the larger displacement because maximising over a plane of trajectories lets a partly decohered carrier still find a cell, so a limit corrected by the model is shallower than it need be.

A component of a bound pair does not travel in a straight line, and the propagation assumes it does. A Gaia DR3 five-parameter solution carries a star's position and its velocity at the catalogue epoch and nothing further, so a linear propagation omits the curvature of the

orbit: zero at that epoch, growing as the square of the interval, and reaching years across an archive spanning 2013–2025. Of the 91 searched positions – the 90 census stars and HD 139084 B, whose re-extracted windows the released per-window catalogue does not carry – 30 have a companion located inside 400 arcsec by Gaia DR3 or measured in the double-star catalogues (Mason et al. 2001; Matson et al. 2026), and 13 more have a disturbed astrometric solution with no companion found. Where the pair has a published visual orbit the displacement follows from it; where it has not, the projected separation is a lower bound on the true one and so puts a ceiling on the orbital acceleration and on the displacement, with the companion's mass taken from the main-sequence sequence of Pecaut & Mamajek (2013); and the Hipparcos–Gaia proper-motion anomaly, which is the departure from uniform motion measured directly over 24.75 years (Kervella et al. 2022), covers 23 of them. The largest displacement anywhere in the survey is 0.20 arcsec at UCAC2 20312880, 0.28 of that block's synthesised beam, against the 0.67 of a beam at which the response measured above falls to 0.49. The largest for which an orbit is available, and not merely a bound, is 0.10 arcsec at Sirius B, 0.11 of a beam. The deepest limit of any system moves by at most 16 per cent, those of LP 736-15 and UCAC2 20312880, and the median system limit does not move. A displacement can only attenuate: it can leave a limit optimistic and it can hide a carrier, but it cannot manufacture a crossing, and no crossing in the survey lies more than 0.007 of a beam from its predicted position. The displacement exceeds a tenth of the synthesised beam for 3 components in all – LP 736-15, UCAC2 20312880 and Sirius B – and for Barta 161 12, HD 155555C and LHS 1140 it cannot be computed at all, the solution being disturbed with no companion located and no Hipparcos epoch to measure one from; those 3 stars carry 24 of the survey's windows, and their limits are optimistic by an unmeasured amount.

The sets beside the primary census. The archival tail is 8 execution blocks and 32 windows, searched after the sweep that fixed the census was closed and toward stars the census already holds; its windows are not tabulated window by window and it enters no expectation. The archival extension is a larger harvest of the same kind, 115 blocks and 463 windows toward 16 of the same stars, run through the same pipeline and reported in §5.6.2. The hold-out and the control beyond 40 pc are reported there too, and the rank-calibration sample is drawn from blocks already counted (Appendix B). Rows 3.2 to 3.5 of Table A1 are the archival gap, over which no result is formed.

B. THE SPATIAL RANK, AND WHAT IT CAN SUPPORT

The spatial rank asks whether an excess sits at the star or anywhere else in the field. It is a diagnostic, no disposition rests on it, and this appendix is the measurement that settles why: The star-versus-control rank is an empirically useful way of deciding which crossing to examine first, but the stellar and control positions are not demonstrably exchangeable under the null. The rank therefore has no calibrated per-window false-alarm probability and is not used as evidence for detection.

The star lies at the phase centre, inside the control annulus, and the displacement is real: the star carries 1.05–1.15 times as many above-threshold cells as a control position at every level up to the trigger (§5.3). The chance expectation of §5.3.1 is not stepped by it, since the ratio at the trigger is formed from the cells under test; as a sensitivity test it gives 25.7 against the 24.6 adopted, at a probability of 0.18. The paragraphs below say what the displacement is *not*. *It is not a primary-beam effect.* Spectra are extracted from phase-rotated visibilities in apparent flux density and are never corrected for the primary-beam response (Appendix A), so the familiar rise of image noise with distance from the pointing centre cannot operate here; were it operating it would make the control probes noisier than the star and place the stellar rank *above* one half, the opposite of what is measured. The control ensemble does show a weak rise of its own level with radius, and the obvious repair is to remove it: each probe is standardised within its own radius bin, with no free parameter.

A calibration error that varied with frequency would explain the displacement, and it is the only mechanism on offer. An error identical in every channel is removed exactly by the per-integration median along frequency; a channel-dependent one is not, and two ordinary errors are channel-dependent: a residual in the bandpass, and a system temperature measured in the coarse correlator mode and interpolated onto the science channels, which near a telluric line leaves a narrow feature behind (Hunter et al. 2018). Both multiply the continuum, so both act at the star and not on empty sky, both stand still in topocentric frequency and both are coherent between integrations: they would stack at near-zero drift and put the stellar rank *below* one half, the sign observed.

Amplitude excludes them, measured where the search looks. Such an error multiplies the continuum at the extraction position, a dirty-map amplitude that includes any dust inside the synthesised beam; it must beat the noise of the de-drifted, block-combined spectrum, since a coherent residual stacks as a carrier does. A fractional error ϵ deposits $\epsilon S_{\rm c}$ at the star, so a crossing at the trigger needs $\epsilon = 5\sigma/S_{\rm c}$. That continuum is measured for each window in its own execution block, as the time average of the window's own retained spectra – the same estimator as the search, at the same position – over 1557 of the 1663 windows toward 86 stars, and over 214 blocks it agrees with independent continuum images of the same fields with no offset between the two (-0.001 mJy). No window in the survey requires as little as the 5–20 per cent a coarse system temperature is documented to leave at an ozone line. The smallest requirement anywhere is 21.4 per cent, near 103 GHz at AU Mic and in the one block (`X1240efd_X14140`) where a flare lifts that star's continuum by an order of magnitude; the tenth percentile is 747 per cent and the median 1067. The crossings' own windows ask more again: over the 51 of them with a measured continuum the least demanding needs 53 per cent, at BD05 1668, and the median 4268. AU Mic's 2 unattributed crossings, the brightest-continuum case of a star with any, need 887–3988 per cent.

The bandpass branch fails by amplitude everywhere, needing 21× the 1 per cent amplitude variation the observatory quotes for its worst bands (ALMA Science Portal 2026), from the bandpass stability campaign of Ka-

TABLE A1

The survey ledger, part one: execution blocks and windows. Each row carries its own number, and a group sums to the row whose number heads it. The primary census is the survey: unless a statement names one of the other sets, it is computed over that row; rows 3.2 to 3.5 are the archival gap, and no result is formed on them. Hours are window-hours: on-source time summed over spectral windows, so the classes partition the hours as they partition the windows. The telescope on-source time behind the primary census is 281 h, of which 153 h carries at least one Class A window and 277 h at least one Class B window – figures that overlap and must not be added. Confirmed planet counts here and in the target list come from the NASA Exoplanet Archive **pscomppars** table (Akeson et al. 2013), accessed 2026 October 4. Part two, Table A2, carries the events and the confirmation extents.

	Set of execution blocks	Blocks	Windows	Stars	Systems	Hours
1	Archival scope: blocks the query exposes	656	–	–	–	–
1.1	… processed with the same pipeline	479	–	–	–	–
1.2	… not processed at the freeze	177	–	–	–	–
2	Processed with the same pipeline[c]	484	–	–	–	–
2.1	**Primary census**	407	1663	90	83	1098
2.1.1	Class A, fine channels[a]	–	405	66	61	228
2.1.2	Class B, coarse channels[a]	–	1258	89	82	870
2.2	Reserved hold-out	77	315	36	–	214
3	Not processed at the freeze: what became of each	177	–	–	–	–
3.1	Calibrated and searched	144	–	–	–	–
3.1.1	**Archival extension**, reported	115	463	16	16	269
3.1.1.1	Class A, fine channels[a]	–	67	11	–	48
3.1.2	Searched after the extension was frozen	8	32	–	–	–
3.1.3	ϵ Eri, both pointings[d]	8	32	–	–	–
3.1.4	No window reached the export	13	0	–	–	–
3.2	No calibration in the ALMA delivery	18	–	–	–	–
3.3	Calibration failed	11	–	–	–	–
3.4	Not attempted	3	–	–	–	–
3.5	Working disk insufficient	1	–	–	–	–
4	Beyond 40 pc: out-of-sample control	47	198	22	–	88
5	Rank-calibration sample[b]	326	1322	56	–	–

[a] The two classes partition the census windows, not its blocks: 255 blocks carry a Class A window and 392 a Class B one, and most carry both, so the hours of the two rows partition the windows and not the observing time.
[b] Not an out-of-sample set and not an addition to this ledger: every one of its blocks is counted in a row above, in the primary census or in the hold-out, and it is listed apart only to keep it from being counted twice.
[c] Panels 1 and 2 overlap rather than add: 479 of these 484 blocks are row 1.1, and the other 5 carry no member observing unit in the archive metadata, so the scope query does not expose them; 4 of the 5 are in the primary census.
[d] Every one is an ϵ Eri Band 6 block. The archival pointings put the star either at the centre of the field, at a response of 1.00, or well off it, where it is 0.23–0.28 over the 16 of their 32 windows that lie off axis – inside the 0.22 floor of §3 in every case, which is where the documented Gaussian and the blocked-aperture form part company and is not a retention limit. All of them are searched, each from its own recorded phase centre, and their conventions are in Appendix A.
The same band's 12 windows that fall in the primary census are counted in row 2.1 and carry every census statistic; the on-axis pointings reach a response of 1.00.

meno (2014); and of either sample 1 crossing lies within 62.5 MHz – two of the 31.25 MHz channels on which the system temperature is measured – of a catalogued atmospheric transition (Appendix E), and it is counted as atmospheric. The shared noise scale raises one further question, and the same arithmetic settles it. Over the 1609 of the 1663 census windows that hold none of the adopted crossings, the stellar statistic divided by the median of the 512 control statistics is exactly the ratio of the two noises; its median is 1.001 (0.994–1.005), so the two scales agree.

B.1. *The reserved hold-out, and what it calibrates*

The statistic and the mask are set on the searched sample and cannot also calibrate themselves, so a further sample is separated by an identifier rule: a block is held out if and only if $\mathrm{sha256}(\mathrm{UID})_{[:8]} \bmod 5 = 0$, subject to two guards that use identifiers alone – a block on a named list stays in the survey, and no system loses every block – so that the assignment follows from the UIDs and that list alone. That this sample is independent of the choice of statistic and of mask is an assertion of ours and not something a reader can check. Of 484 blocks processed, 77 are reserved, 15.9 per cent of them. Those carry 315 windows toward 36 stars, with no star and no system lost from the headline sample. The rule hashes the execution block, and ALMA's identifier carries the observing session above it, so two executions of one session can fall on opposite sides of the split: 49 of the 77 reserved blocks share a session with a searched one, across 40 sessions, so 64 per cent of the reserved set shares calibrators, weather and correlator setup with the searched one. A session-level hash would have been the stronger choice. Every quantity below is therefore given twice, over the whole reserved set and over the 28 blocks that share no session with it. The rank displacement itself is quoted both ways in §4.2.

One further quantity is calibrated on those blocks and on nothing else, the radial behaviour of the control ensemble: the standardised level runs +0.12, +0.10, +0.12, +0.16 across the annulus, a mean per-window slope of $+0.067 \pm 0.033$, small and of the wrong sign to account for the displacement.

B.2. *The recurrence statistic under the null*

The exclusion in the crossing ledger is one formula. Writing S and σ_d for the discovery amplitude and its error, and $\bar{a}$ and $\bar{\sigma} = (\sum_i \sigma_i^{-2})^{-1/2}$ for the inverse-variance combination of the repeat readings at the predicted channel,

$$\sigma_\mathrm{excl} = \frac{S - \bar{a}}{\sqrt{\sigma_\mathrm{d}^2 + \bar{\sigma}^2}} = \frac{T_\mathrm{pers} - T_\mathrm{rep}}{\sqrt{1 + (T_\mathrm{pers}/T_\star)^2}}, \qquad \text{(B1)}$$

with $T_\mathrm{pers} = S/\bar{\sigma}$ and $T_\mathrm{rep} = \bar{a}/\bar{\sigma}$, so every exclusion follows from three columns of that table and nothing

TABLE A2
The survey ledger, part two: events, dispositions and confirmation. Every headline count in this paper is read from here, and **the itemisation accompanies this paper as data rather than as part of the article**, machine-readable and carrying all 21 rows, which is why the numbering here is not consecutive. Three counts are 52 over three different sets – the Class A crossings of the adopted ledger, those of the per-window extraction catalogue (§3), and the quality-passing crossings. The first two differ on 6 rows, where the statistic read over a narrower field falls on the other side of the trigger from the whole-primary-beam reading the ledger adopts, in both directions (Appendix D); the flag the accompanying table carries is the ledger's, 50 of them in the census. The last two hold the same rows, because the one block failing the quality criterion is also the survey's only block with a coarse-channel crossing. Row 3.1.1 is what this paper means by confirmation and the only such count it quotes: it applies the criterion of §4.2, a different observing session and more than a day, at a frequency the first block covered. Row 2.2.1 is decomposed system by system in Appendix D.

	Count	Value	What it counts
1	Threshold crossings at $T_\star \geq 5$	56	census and archival tail
1.1	Class A, fine channels	52	the carrier experiment
1.2	Class B, coarse channels	4	all in the one block failing §5.3.2
2	Quality-passing crossings	52	the same rows as 1.1
2.1	. . . attributed to a transition	18	of 21 coincidences inside the mask in the star's own frame
2.2	. . . unattributed	34	rows 2.2.1 to 2.2.3
2.2.1	. . . in the primary census	31	against 24.6 expected by chance at the 405 windows' own control positions – a count chance delivers with probability 0.13
2.2.2	. . . in archival-tail blocks	2	outside that comparison
2.2.3	. . . coincident with an atmospheric transition	1	within 62.5 MHz of one in the topocentric frame (Appendix E), and counted as atmospheric on both sides of that comparison
3	Class A systems	61	of 83 surveyed
3.1.1	Class A systems with a later block at a frequency the first covered, in another session	**40**	**the confirmable set, and the only one quoted**, for a carrier whose stellar-frame frequency at the two epochs agrees to within half a native channel; 13 of them over a year later. Over the census together with the repeat blocks the recurrence test reads
3.3	. . . with no separated second block at a covered frequency	21	confirmation impossible over the 166 blocks the recurrence test reads; 36 of all 83
4	Class A union bandwidth	47.7 GHz	26 disjoint islands
4.1	. . . removed by the line mask	6.4 %	3.05 GHz: the figure the text quotes, against 4.3 per cent of the 118.1 GHz archival union

TABLE A3
The 67 ALMA proposals the searched execution blocks belong to, in the standard ADS/JAO.ALMA#*code* form. Every block of the primary census belongs to one of these, and the blocks themselves are retrieved from the ALMA Science Archive by the execution-block identifiers the accompanying ledger file gives for every crossing.

2012.1.00142.S, 2012.1.00198.S, 2012.1.00268.S, 2012.1.00385.S, 2012.1.00437.S, 2012.1.00993.S, 2013.1.00359.S, 2013.1.00588.S, 2013.1.00645.S, 2013.1.01147.S, 2013.1.01214.S, 2015.1.00307.S, 2015.1.00633.S, 2015.1.00783.S, 2015.1.01015.S, 2015.1.01260.S, 2016.1.00104.S, 2016.1.00358.S, 2016.1.00515.S, 2016.1.00637.S, 2016.1.00803.S, 2016.1.00817.S, 2016.1.00878.S, 2016.1.00880.S, 2016.2.00015.S, 2016.2.00200.S, 2016.A.00013.S, 2017.1.00167.S, 2017.1.00200.S, 2017.1.00215.S, 2017.1.00561.S, 2017.1.00698.S, 2017.1.00704.S, 2017.1.00786.S, 2017.1.00986.S, 2017.1.01583.S, 2017.1.01644.S, 2018.1.00045.S, 2018.1.00072.S, 2018.1.00126.S, 2018.1.00461.S, 2018.1.00470.S, 2018.1.00947.S, 2018.1.01149.S, 2018.1.01545.S, 2018.1.01833.S, 2019.1.00189.S, 2019.1.00696.S, 2019.1.01443.T, 2019.1.01517.S, 2019.1.01603.S, 2019.1.01681.S, 2019.2.00208.S, 2021.1.00485.S, 2021.1.01209.S, 2021.2.00097.S, 2022.1.00134.S, 2022.1.00338.L, 2022.1.00793.S, 2022.1.01163.S, 2023.1.00390.S, 2023.1.00487.S, 2024.1.00165.S, 2024.1.00595.S, 2024.1.00722.S, 2024.1.01095.S, 2024.A.00040.S

else. $T_{\rm rep}$ is read at the carried drift and at one cell; the largest value over every drift trial at the same cell is tabulated beside it but is never compared with $T_{\rm pers}$, since a maximum is positive on average whatever is there. No exclusion can exceed the strength of the discovery it is built from, nor is it the strongest statement available: an amplitude selected as the largest of 10^2 to 10^6 cells at $T_\star \simeq 5$ is biased high, so a real carrier would be fainter than measured, and the limit $(T_{\rm rep}+5)/T_{\rm pers}$ survives the bias, the fraction of the discovery amplitude still ruled out at the trigger.

The same three numbers say where the test has power and where it has none. The criterion asks the repeat to reach the trigger, so a carrier radiating through both observations at the discovery amplitude satisfies it with probability $\Phi(T_{\rm pers} - 5)$, and at the trigger amplitude $5S/T_\star$ – the faintest carrier that could have produced the crossing at all – with $\Phi(5T_{\rm pers}/T_\star - 5)$. Where they are small the crossing is untested and not excluded, and is reported that way.

That probability is calculated, and on every crossing that carries an exclusion it is measured as well. A carrier at the trigger amplitude was injected at the predicted cell of every covering repeat block through the instrument's own channel response, at a sub-channel position drawn independently of the discovery's, and the readings combined as the exclusion combines them, each realisation taking its noise from a real reading of another channel of the same block. 2 are set out in full here, the rank-leading pair of 61 Vir and CP−72 2713. Over 1000 realisations the injection into 61 Vir's repeat blocks is recovered 1.00±0.01 of the time, against the 1.00 the expression gives. CP−72 2713's carries an error of 0.02. With the carrier removed the same readings recover it 0 times. Half a step of the window's own drift grid, carried into the reading, costs 0.4 per cent of the statistic inside one observation. The expression contains neither that error nor the sub-channel position of the predicted cell. Over all 33 crossings that carry an exclusion the same experiment returns a median 0.01 less than the expression gives, and two terms account for the difference. The first is where in a channel the carrier lands. The frame correction places it wherever it falls, and the channel

response there costs a median 4.8 per cent of the amplitude. Of the 191 block readings taken with a steady grid, 2.1 per cent lose more than 15.1 per cent, which is the most the shape of a correlator channel can cost on its own. The second is the drift grid itself, which is sized to the discovery block: carried into a repeat observation of a different length one of its steps walks a different distance, 4.1 channels in the one case where the repeat is much the longer (§5.6), and the reading is then taken off the predicted cell. The carriers are injected into the retained per-integration spectra of the discovery block and of every covering repeat block, 362 windows in all, before the baseline step, so the measurement is the recurrence reading. Those spectra are the survey's own extractions, all 482 block-roles the recurrence test reads, of 18 to 2940 integrations apiece; run with the carrier amplitude set to zero the same harness returns each block's published reading to the last digit, over 15 readings of 8 crossings. The step from calibrated visibilities to a spectrum – the phase shift to the stellar position, the visibility weighting and the primary-beam correction – is measured instead by the single-epoch injections of §5.1. Two harnesses built independently of each other agree to 0.03 on the crossings they share. The second returns 0.65 for CP−72 2713 where the two overlap, and the largest shortfall anywhere is 0.22 at AU Mic. The one case in which the repeat block is much the longer is η Crv: the injection is recovered 0.29 of the time (0.26–0.32) where the expression gives one, because that block is 8.3 times longer than the discovery and the rate fitted there lands on the predicted cell only 21 per cent of the time; no other repeat here is more than twice its own discovery block. Read instead at the largest of that repeat's own drift trials the injected carrier is recovered in every realisation, while the data themselves reach 1.50. Table B1 sets the measurement beside the expression for every one of them.

The reading at the predicted channel is centred on zero and scatters by one, measured and not merely designed. Over 511 single-cell readings of every repeat block in the census and the hold-out its mean is +0.014 and its scatter 1.046. At the control positions of the discovery windows it reads −0.036 and 1.034. Over the 2 crossings scanned that way, the largest value anywhere within $\pm 15\,\mathrm{km\,s^{-1}}$ of a predicted channel is 3.19 at the carried drift. That is a scan across frequency at one drift rather than a single cell; the 4.71 of Table 6 is its converse, one channel read at every trial of the grid, and that was recorded for each of the 444 repeat measurements at the time it was made. The search of the repeat epochs over whole windows at every drift is the survey's own search, one reading per window, which is where the 29 exceedances of §5.6 are counted. A maximum behaves as a maximum should. The largest of the 8 retained control probes of a repeat block averages 1.449 against 1.424 expected. The largest of the n_{rep} per-block readings averages +1.102 against +1.130. The combined T_{rep} of Eq. B1 averages −0.119. Widening the window costs trials and not calibration: the repeat-epoch exceedances match the same accounting, and the windows of 3 stars are narrower than the excursion an orbit at the grid ceiling could produce.

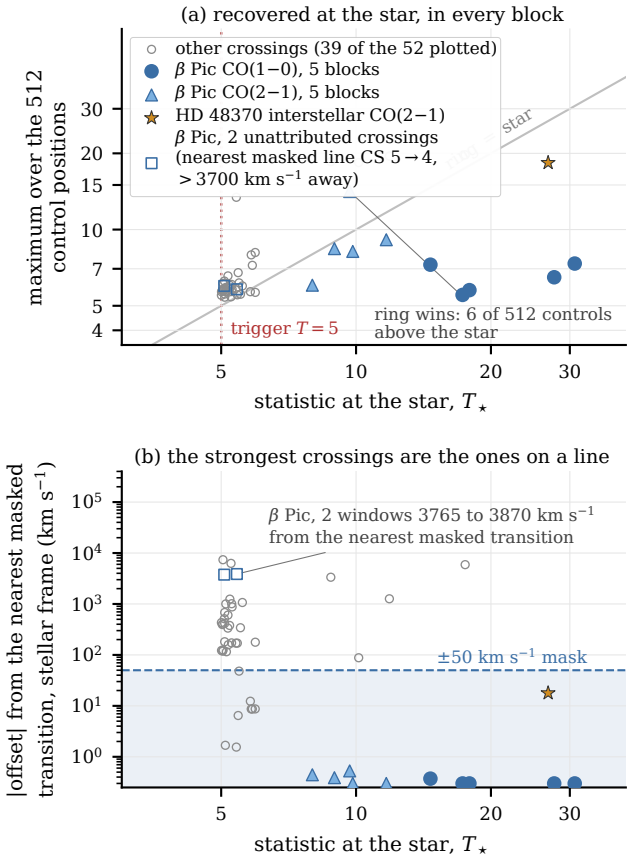


FIG. B1.— The β Pictoris positive control, in both directions. Both panels draw the same 52 of the 56 crossings, the remainder having no retained control ring. *(a)* The spatial screen: the stellar statistic against the maximum over that window's 512 control positions. Filled symbols are the blocks in which the search recovers circumstellar carbon monoxide at β Pictoris, 9 of them with the star above every control and one with the ring above it, block `X7116f1_X1785`, Band 6, where the ring reaches 14.13 against 9.68 at the star (§5.4). *(b)* The same crossings by velocity offset from the nearest masked transition *in their own star's rest frame*, with the adopted $\pm 50\,\mathrm{km\,s^{-1}}$ half-width marked: the 10 β Pictoris carbon-monoxide crossings land 0.1–0.5 $\mathrm{km\,s^{-1}}$ from CO, while its 2 unattributed crossings lie 3765–3870 $\mathrm{km\,s^{-1}}$ from the nearest transition the mask holds, CS(5→4). 39 further crossings are grey, HD 48370 marked.

C. THE LINE MASK

A crossing that falls on a known molecular transition has an astrophysical explanation to hand, and the mask exists to say which crossings those are. It reports the frequency coincidence and the judgement about the coinciding species separately, so the reproducible part can be recomputed and the arguable part argued with.

The list, and when it was made. The mask holds every transition of the 12 species these spectral setups cover – the three CO isotopologues, HCN, HCO^+, CS, CN, SiO, H_2CO, SO, [C I] and hydrogen recombination – that falls inside one of the 46 searched frequency intervals with an upper-state energy below 150 K. That rule is the whole of it, and it admits 111 transitions (Table C1). No strength cut is applied across species: any threshold admitting HCN's strong lines would discard the brightest lines in these fields. Within a species the main hyperfine components are kept and the satellites dropped. The list is a single Splatalogue query (Remijan et al. 2007) of CDMS and JPL (Pickett et al. 1998; Müller et al. 2005) made on

TABLE B1

What an injected carrier at the trigger amplitude is recovered at, for each of the 33 crossings that carry an exclusion, ordered by what is measured. Every realisation was deposited in the retained per-integration spectra of the real data − the discovery block and the $n_{\rm rep}$ covering repeat blocks, 362 windows in all − and none used synthetic noise. $\nu_\star$ is the crossing's frequency in the star's own rest frame, Φ the chance the expression above gives, and beside it the fraction of N realisations of the frozen reading that recovered the carrier, with its binomial interval. The leading 8 rows are the crossings reported untested and not excluded.

Star	$\nu_\star$ (GHz)	$n_{\rm rep}$	Φ	measured	N
η Crv	357.2475	1	1.00	0.29 (0.26–0.32)	200
β Pic	241.8594	1	0.39	0.35 (0.33–0.37)	700
ALMA J153702653−33192492	347.1890	1	0.50	0.49 (0.47–0.51)	700
β Pic	241.7739	1	0.60	0.55 (0.53–0.57)	700
CP−72 2713	344.2965	5	0.66	0.63 (0.61–0.65)	700
AU Mic	230.6705	2	0.86	0.64 (0.62–0.66)	700
CD−57 1054	346.5265	1	0.81	0.74 (0.73–0.76)	700
HD 110411	219.4357	1	0.89	0.84 (0.83–0.85)	700
HD 69830	322.4444	5	1.00	0.94 (0.93–0.96)	200
LHS 1140	230.6268	4	0.98	0.94 (0.93–0.96)	200
HD 207129	230.4070	4	1.00	0.98 (0.97–0.99)	200
AU Mic	230.8294	2	1.00	0.98 (0.97–0.99)	200
HD 23484	230.8006	4	1.00	0.98 (0.97–0.99)	200
HD 69830	321.2149	5	1.00	0.98 (0.97–0.99)	200
HD 14055	344.4515	30	1.00	0.98 (0.97–0.99)	200
61 Vir	346.3232	3	1.00	0.99 (0.98–0.99)	700
61 Vir	345.5678	3	1.00	0.99 (0.98–0.99)	200
ALMA J153702653−33192492	217.7269	7	1.00	0.99 (0.98–0.99)	200
HD 14055	345.0070	30	1.00	0.99 (0.98–0.99)	200
HD 31392	231.2231	8	1.00	0.99 (0.98–0.99)	200
ALMA J153702653−33192492	218.6688	7	1.00	0.99 (0.99–1.00)	200
ALMA J153702653−33192492	218.8923	7	1.00	0.99 (0.99–1.00)	200
HD 14055	345.5216	30	1.00	0.99 (0.99–1.00)	200
HD 161868	345.6491	13	1.00	0.99 (0.99–1.00)	200
HR 1010	230.8491	8	1.00	0.99 (0.99–1.00)	200
ALMA J153702653−33192492	216.2036	7	1.00	1.00 (1.00–1.00)	200
GJ 849	230.8581	5	1.00	1.00 (1.00–1.00)	200
HD 10647	331.6943	10	1.00	1.00 (1.00–1.00)	200
HD 14055	331.7035	30	1.00	1.00 (1.00–1.00)	200
HD 14055	344.6965	30	1.00	1.00 (1.00–1.00)	200
HD 14055	331.7664	30	1.00	1.00 (1.00–1.00)	200
HD 170773	331.0234	4	1.00	1.00 (1.00–1.00)	200
HD 14055	345.4303	30	1.00	1.00 (1.00–1.00)	200

2026 October 6, and the query is dated and repeatable, so the mask itself can be rebuilt. The attribution below rests on the species and the offsets reported beside each crossing.

The species are the ones the observers went to the telescope for, and that is a limitation. The spectral setups are tuned to them. 39 per cent of the 405 Class A windows are centred within $\pm50\,\mathrm{km\,s^{-1}}$ of a transition of the list and 88 per cent contain one somewhere. Displace the same windows bodily by 1 to 7 GHz and only 1–8 per cent remain centred on one. The list is therefore set by the science goals of the programmes that took the data and not by anything in this search, and *unattributed* means unattributed against it. A list chosen differently would say something different, and the measurement of how much is this: a query of CDMS and JPL (Pickett et al. 1998; Müller et al. 2005) on 2026-10-15 for every transition below 150 K of all 225 species with a published interstellar or circumstellar detection returns 24 976 transitions inside the searched intervals, and at the same $\pm50\,\mathrm{km\,s^{-1}}$ they cover 98 per cent of the Class A union against 6.4 per cent for the list adopted here. Such a list attributes every one of the 38 unattributed crossings and 100 per cent of the control positions' own exceedances, so it leaves the category empty for star and control alike: at $\pm1\,\mathrm{km\,s^{-1}}$, where it still covers only 15 per cent of the band, it would attribute 6 of the 38 and the unattributed count remains what the control positions predict (26 against 28.5 expected). The width of the mask buys its meaning, and the comparison of §5.3.1 survives either choice.

Attribution is decided in the star's own rest frame, and the sign is not the usual one. A crossing's frequency is carried from the topocentric value to the barycentre using its own block's pointing and mid-time, and then to the stellar rest frame using the star's systemic velocity. That frame, and no other, is where a coincidence is judged. The barycentric term alone spans $58\,\mathrm{km\,s^{-1}}$ across the survey, so a tolerance applied in the observed frame would spend most of its budget on the Earth's motion. The 3 HD 285968 epochs show the transform working in the only way that can be checked: $12\,\mathrm{km\,s^{-1}}$ apart as observed, they agree to $0.03\,\mathrm{km\,s^{-1}}$ in the star's frame. Throughout, $\Delta v_\star = c\,(\nu_\star - \nu_0)/\nu_0$, so a positive offset means the crossing lies *above* the transition in frequency, the opposite sign to the velocity convention of a spectral line.

Every crossing is read in the local standard of rest as well, where interstellar gas is at rest, and that second reading establishes which coincidences are interstellar. Of the 24 stars carrying crossings, the composite CO($1 \rightarrow 0$) survey of Dame & Thaddeus (2022), which contains Dame et al. (2001), covers 11 and holds emission toward 5 of them, between -18 and $+22\,\mathrm{km\,s^{-1}}$. Toward every other star, including the 13 outside the CfA footprint, the all-sky Planck CO map (Planck Collaboration et al. 2016) stays below $0.178\,\mathrm{K\,km\,s^{-1}}$. Toward ALMA J153702653−33192492, HD 139084 B, HD 285968 and HD 48370 the CO spectrum is non-zero at the crossing's own LSR velocity, within $3\,\mathrm{km\,s^{-1}}$ of its own peak and at 0.23 K or brighter against a 0.18 K chan-

TABLE C1
The line mask, by species: the number of transitions N it holds, the frequency range they span inside the searched intervals and their upper-state energies. Hydrogen recombination lines carry no tabulated E_u.

Species	N	ν_0 (GHz)	E_u (K)
H	5	92.03 – 231.90	–
SO	15	100.03 – 682.68	16–143
H_2CO	13	101.33 – 491.97	10–106
$C^{18}O$	3	109.78 – 329.33	5–32
CN	50	113.12 – 680.30	5–114
CO	3	115.27 – 345.80	6–33
CS	3	195.95 – 342.88	24–66
SiO	2	217.10 – 347.33	31–75
^{13}CO	8	220.40 – 330.59	16–32
HCN	7	354.50 – 354.51	43–43
HCO^+	1	356.73 – 356.73	43
[C I]	1	492.16 – 492.16	24

nel. So 6 of the 21 coincidences are interstellar gas and not circumstellar. Two are worth naming. ALMA J153702653−33192492 is a serendipitous field star toward the Lupus clouds, at 35.0 pc with the clouds far behind it. Its CO(2 → 1) crossing lies 0.5 km s^{-1} from a 6.6 K km s^{-1} cloud line at +4.9 km s^{-1}, so the field is a molecular sightline and not a disc. The velocity says so and the field does not: its brightest control position stands above the star in all six of its windows, but so it does in 393 of the 405 Class A windows of the survey, so that comparison distinguishes nothing. Its other 5 crossings lie at least 122 km s^{-1} from every transition of the list, and the mask disposes of none of them. On that sightline the wider chemistry a molecular cloud carries is a real limitation and is bounded here: of the species a Lupus line of sight would be expected to show, 1 reaches one of its crossings within the mask's own half-width, OCS at 15 km s^{-1} (E_u = 100 K). HD 48370 is the case where the two readings part: its coincidence lies −17.8 km s^{-1} from the star's own CO(2 → 1), while in the local standard of rest it sits at +23.9 km s^{-1}, on a catalogued cloud line peaking at +21.8. The offset follows the interstellar gas and not the star, so that window's extended emission is an interstellar cloud and not a resolved disc.

A coincidence is an attribution, and for most of them it is not the only evidence. 16 of the 18 attributions carry support beyond the frequency match: catalogued carbon monoxide on the sightline here, a second transition of the same species, or the crossing's own recurrence. The other two, toward HIP 10679 and HR 1010, rest on the coincidence alone – no catalogued gas on either sightline, no second transition and no repeat epoch that recovers them – so they are consistent with molecular emission and are not identified as it.

The coincidences, and the one species we do not read as emission. Of the 56 crossings, 21 fall within ±50 km s^{-1} of a transition of the list in their own star's frame; that count is the first layer and all the velocity criterion decides. Coincidences with sulphur monoxide account for 3 of them: CD−57 1054, CP−72 2713 and HD 161868, at −1.7, −12.3 and −48.0 km s^{-1} from transitions of upper-state energy 21–87 K, each inside a Band 7 window that also covers CO(3 → 2). Two things argue against reading those as emission. Sulphur monoxide has not been reported in a debris disc, where the second-generation gas is CO, C and O. And toward CP−72 2713 the CO(3 → 2) line falls in the same window and is absent to 10.5 mJy in a 0.42 km s^{-1} channel, while the crossing's fitted amplitude is 11.3 ± 2.0 mJy: an SO line there would have to be at least as bright as an undetected CO(3 → 2) line in the same data. That margin is only 0.43 of the fitted error, so it is the species and not the margin that carries the weight. Both arguments are about the species and not about any one crossing, so all 3 are treated alike: the ledger prints the offset beside each with a dagger and disposes of none of them, giving 18 attributed against 38 unattributed, of which two outrank every control position in their own windows. Reading them as attributions instead gives 21 and 35, and 1 such crossing; not one of the 3 is present again in any block covering it, and none is confirmed, so the choice moves counts and no disposition. The ledger forms the exclusion of Eq. B1 for each of them, at 4.1–5.5σ, but that significance is built on the discovery amplitude; read at the trigger amplitude instead, a crossing whose repeat leaves the criterion without power is reported in §5.6 as untested and not excluded, and that applies to these as to any other. The reserved hold-out is treated identically, leaving 1 attributed and 7 unattributed there (§5.6.2).

The half-width, and what it costs. A one-channel test is too tight for circumstellar gas, whose emission is shifted by many channels: the width follows from Keplerian motion where these molecules survive, $\lesssim$13 km s^{-1} beyond 10 au, and is ±50 km s^{-1}. At that width the mask removes 3.05 GHz of the 47.7 GHz Class A union, 6.4 per cent of it. Over both classes together it removes 5.04 GHz of 118.1 GHz, 4.3 per cent, so the cost falls on the primary experiment, because its windows are the line-tuned ones. Attributing on either frame instead would mask the union of two tubes separated by the star's LSR velocity, 3.72 GHz or 7.8 per cent of the Class A union – 22 per cent more bandwidth for no further attribution, which is the measurement behind deciding in one frame. Between ±20 and ±100 km s^{-1} the mask attributes 17 to 19 of the crossings. The 2 that outrank all 512 control positions of their own windows are the same two at every width.

The labelled crossings are tested for recurrence too. 15 of the 18 attributed crossings have a later block covering the predicted cell, and the 13 that are present again, toward β Pic and HD 285968, are the positive control of that stage. For the 2 not recovered a repeat window's own largest statistic bounds the predicted cell from above and the ledger prints the row as a bound; 3 have no covering block at all (HD 139084 B, HD 48370 and HIP 10679).

The 2 that are not recovered. HR 1010's crossing at 230.5287 GHz is −12.0 km s^{-1} from CO(2–1) and is bounded at $T_{rep} < 4.6$, against the $T_{pers} = 8.4$ that real carbon monoxide at the discovery flux density would have given – a disagreement of 2.1σ once the window's noise is measured over the 47 minutes it integrates, at 11.5 mJy, where its sibling blocks also lie. ALMA J153702653−33192492 is a serendipitous field star with no measured systemic velocity, so its frame is the barycentric one; its crossing at 230.5392 GHz is +1.6 km s^{-1} from the same transition and is bounded at $T_{rep} < 4.2$ against $T_{pers} = 6.2$, or 1.4σ. That attribution rests on a velocity rather than on the recurrence test: the sightline carries catalogued interstellar emission at +4.9 km s^{-1} in the local standard of rest, 0.5 km s^{-1} from the crossing's own velocity in that frame. Its re-

peat blocks show where that gas lies and not whether it can be recovered. All 6 share one 7 m configuration and beams of 4.8 to 5.3 arcsec, so an extended cloud is recovered alike in each, and in every one the ring reaches 10.9 to 13.8 against 4.2 to 4.8 at the phase centre, with between 193 and 496 of the 512 controls standing above the star: the gas is bright and it is not at the star. Toward HD 285968 it is, and over its 4 covering blocks the ring reaches only 6.4 to 8.1 where the star itself reaches 4.8 to 6.0.

D. THE FALSE-ALARM EXPECTATION AND THE CONTROL ENSEMBLE

This appendix gives the chance expectation for the unattributed crossings, what it is conditioned on, and what the control ensemble calibrating it can support.

D.1. *Conditions on the expectation*

One attribution set serves both sides of the comparison, the set of §5.3: the mask of Appendix C within $\pm 50\,\mathrm{km\,s^{-1}}$ in the star's own rest frame, sulphur monoxide excluded. So the 3 sulphur monoxide coincidences are unattributed crossings, and a control maximum on one of those transitions is an unattributed exceedance. A systemic velocity is measured for the stars of 277 of the census windows and the others are taken at rest, which moves the masked share by 3.0 points.

The expectation is built from the control positions, the only empirical null the data provide, and each Class A window contributes two quantities measured on itself. The fraction of its own 512 control positions that reaches the trigger is the probability that a position in that field does so – no cell count enters it and no form is assumed for the tail – and summed over the windows 36.2 are expected to trigger by chance. How many of those would be attributed is measured on the channels the control maxima land on, which the retained spectra record: 88 of 302 exceedances fall inside the mask, and the attribution therefore takes 29 per cent of the expectation. A maximum free to land anywhere on its own grid would take 20 per cent, the median window offering 6.0 per cent of its channels; the difference is that 15 narrow line-tuned windows lie wholly inside the mask and their fields, which hold real emission, are where a control position most often triggers, so a uniform landing would under-attribute them and leave the unattributed expectation too high. The atmospheric screen of Appendix E is then applied to these same positions, on the same radius and in the same topocentric frame as to the crossings, and takes 10 of the 214 exceedances the mask leaves, a further 5 per cent. That leaves 24.6 expected unattributed Class A crossings against 31 observed.

Both sides run over the 405 Class A windows of the primary census, so the expectation is a sum over rows and nothing is carried at an assumed rate. They are also read over the same field, and which field that is matters. The σ of Equation 1 is the median absolute deviation over the control positions, so $T_\star$ is a property of the annulus as much as of the cell: one product supplies both sides here, the statistic read over the whole primary beam with each block on its own antenna geometry, for the expectation and for the crossing count alike. A field narrower than the primary beam drops the integrations whose pointing falls outside it, and with them on-source time and real signal, so a statistic read over part of a beam is not the quantity §4.1 defines. The choice is not immaterial and is reported rather than assumed small: read over such a field on both sides, the same comparison gives 35 crossings against 24.1 expected at $P = 0.027$, where the whole beam gives 31 against 24.6 at $P = 0.13$; the bound above becomes 22 further crossings against 17 on the same resampling. The control side is a sum a reader can redo: over the 405 Class A rows of the accompanying table, the control exceedance counts against their own ensemble sizes give 36.2 windows reaching the trigger, which the mask shares and the atmospheric factors deposited beside them reduce to 24.6 – the two screens of this appendix and of Appendix E, row by row. The same comparison holds system by system, each system's expectation formed from its own control positions: 2 of the 61 systems fall below one chance in twenty, against 3 expected. The system holding the most crossings is HD 14055, toward which lie 8 of the 56 crossings and 8 of the unattributed ones. Over its 38 Class A windows it gives 6 against 3.5 expected. No system departs from its own controls by more than chance allows. A further 15 Class A windows arrived with the archival tail and lie outside it, together with the two unattributed crossings toward HD 14055 they hold, which are reported beside the comparison and enter neither side of it. No window holds more than one primary crossing, so both sides count windows; the weaker cells behind those maxima are counted at their own grain in §5.3.

The one assumption the searched data cannot test is that a star behaves like a control position in its own field, and two samples measure it. In the census the star carries 1.046 times as many above-threshold cells as a control position. The ratio is measured between $T = 3$ and $T = 4.5$, below the trigger, where no carrier claim is at stake, on 34 984 stellar cells over 399 windows, and resampling the execution blocks allows 1.033–1.059 at 95 per cent. Read level by level the ratio rises from 1.05 to 1.15 across that band. At the trigger itself it is 1.10, but there it is formed from the 31 above-threshold cells that are the crossings under test. No step is applied to the sum above: the expectation adopted is the control positions' own, the smaller of the two and the more demanding of the count under test. As a sensitivity test the sub-trigger ratio gives 25.7 against the 31 observed. The reserved blocks measure the same quantity on blocks the search did not reach. Their stars reach the trigger in 8 of 62 fine-channel windows, and the control positions of those same windows predict 7.5, a ratio of 1.07. Every one of those events lies in a reserved block that shares an observing session with a searched one, and the 28 reserved blocks that share none hold no trigger events between them, so they neither confirm the step nor contradict it (Appendix B.1). In the 253 coarse-channel reserved windows no star reaches the trigger, against 0.04 predicted. The same displacement is read in the body of the distribution, where the stellar rank sits below one half (Appendix B), and it runs the same way.

A count is compared with the distribution of counts chance delivers, and not with a range on its own mean. A rate is drawn from 20 000 resamplings – each window's rate redrawn from its own 512 probes, the attributed share from its own 302 exceedances, and the execution blocks resampled because windows share them and are

not independent trials – and the count scatters about that rate. A count of 31 or more then arises with probability 0.13 at the control positions alone, and that is the comparison adopted here; as a sensitivity test, applying the step gives 0.18. The same distribution bounds what the comparison could have missed: added to each resampled control-position rate, 17 further crossings still leave a count as low as 31 credible at 0.058, and one more does not, at 0.043 against the 5 per cent level. So 17 is the largest added population this comparison could not have rejected, the uncertainty in 24.6 carried inside it by the same resampling; it counts crossings and not transmitters, correcting for neither completeness nor duty cycle. Read at the grain of the individual cell rather than the window, the same accounting gives 28.1 expected from the 8 retained control probes of each of those 399 windows, 3 192 positions in all, 2.2 of them beside a window's largest. Against that it finds 34 and 3, with 26–36 covering 95 per cent of the same resampling.

D.2. *Execution-block quality*

The comparison with the control positions assumes each block's ensemble behaves like noise, and one block makes that false: every one of its 512 probes exceeds the trigger, the smallest at 6.6. A criterion for the condition is therefore applied to every block and not to the one that prompted it.

Two criteria are built from that statistic and they are not the same test. The statistic is the median of a window's own 512-position control ensemble, which separates a field with a source in it from a field that is no null at all. A *window* is flagged when that median exceeds twice the survey median of the same quantity, 5.5 against 2.74; the flag is a property of the window and cuts across attribution by design, firing on 5 of the 1663 windows in 2 blocks toward 2 stars (BD+05 1668 and HD 48370), every one of which carries a crossing. A *block* is judged on the same median divided by the median over the windows of its own class, since a maximum over a larger channel×drift grid sits higher; it fails where the median of that ratio over its windows exceeds twice the scale of its class. The factor is the same in both.

Neither criterion was fixed in advance. The condition was noticed in one block and the criterion written afterwards, so it is post hoc, and what makes it trustworthy is how little the position of the line matters. The failing block sits at 4.1 and the next most extreme at 1.12, a factor 3.7 apart. Over 41 thresholds spanning that interval the same single block is selected every time, in number and in identity, so the line may be drawn anywhere from 0.56 to 2.06 times where it is. Outside that interval the answer does change, which is how one can tell the rule discriminates: at 1.12 it selects 2 blocks, and at the median block, 1.00, 203 of the 407. Nor does the class treatment decide it. The same block is selected with each class normalised on its own windows and with the survey taken as one, and on this block's 4 coarse windows alone, which are all the windows it has; and no block in the census is failed on one class and passed on the other.

The median block of the primary census sits at 1.00, nothing else is close to the 1 the criterion fails, and the unscaled form of the rule fails the same 1 block. Its 4 windows are all coarse-channel and hold the survey's entire Class B crossing population, so it carries 4 of the 56 threshold crossings. 0 of those outrank all 512 control positions of their own window, and no Class A statistic here has ever contained it. Those 4 appear in the crossing ledger with their block named and enter no count elsewhere.

D.3. *The control ensemble: its calibration, and its resolution*

The ring geometry is set in §4.1; its calibration and its resolution are the subject here. Over the 1657 census windows that retain a control vector the stellar ranks are consistent with the one half exchangeability implies – median 0.49 with each star carrying equal weight, 0.47 to 0.51 at 95 per cent – in both array strata, with 1 053 of the 1 058 compact-array windows measured on each observation's own antenna table rather than on a 12 m beam; the 5 not so measured reach 3.8 at most and none of them holds a crossing. That is a weak statement and not a strong one, since a shift smaller than about 0.042 could not have been seen in a sample of this size. The reserved blocks lean further from one half when their windows are pooled, but those windows are concentrated on a few stars: 108 of 315 belong to five. The two samples agree once each star carries equal weight, 0.44 (0.33–0.52) against the 0.49 above. Compared star by star over the 36 stars both samples hold the difference is -0.115 with -0.17 to $+0.07$ allowed at the same coverage (Appendix B.1). That settles the gap between the two samples' pooled medians, which is an effect of how their windows are distributed over stars. It does not remove the displacement itself: that is small, it is measured on the census's own cells at every level up to the trigger, and the expectations above are not stepped by it – carried in as a sensitivity test it gives 25.7 against the 31 observed. The displacement is not flat in level, so it is also read where the comparison uses it. Between $T = 4$ and $T = 5$, over the 350 Class A windows whose star reaches the trigger nowhere and on the 1 612 stellar cells they hold, a star carries 1.094 times as many cells as a control position of the same field, 1.030 to 1.155 at 95 per cent with the execution blocks resampled, and level by level the ratio climbs from 1.03 to 1.13. It runs upward, so carried in as a sensitivity test it raises the expectation to 26.9 and the probability to 0.25 and lowers the bound to 15, the control positions' own rate being the smaller expectation, the larger bound, and the one adopted. A requirement takes the place of an assumption here, and it is stated: the 31 observed would stand above chance at the 5 per cent level only if stars reached the trigger at 0.90 times the rate of a control position or less, an expectation of 22.1, which is a deficit at the star of 10 per cent. Every ratio measured here is an excess instead. At the trigger itself the reserved blocks – only 28 of which share no observing session with a searched one – fix the same ratio to 0.48 to 1.79 on 8 events, and so neither establish that deficit nor exclude it; the band immediately below the trigger, where the cells are, excludes it.

The rank is not a significance test, and the reason is structural. Ranking the star against 512 control positions of the same field decides which windows deserve examination, but $1/(512 + 1) = 1.95 \times 10^{-3}$ is the *resolution* of the ensemble – the smallest rank it can express – and not the probability that noise produced the window. A survey of 1663 windows would need a rank 65 times finer,

and correlation makes it coarser still: controls inside the synthesised beam are not independent, so an annulus resolves no better than 0.023–0.032 however many probes are drawn into it, with the primary beam bounding how many a single pointing contains. That floor is the design's central limitation: no single window here can be significant on the rank alone, which is why confirmation requires independent recurrence.

D.4. *The trials budget*

The searched cells counted in §4.2 are not independent trials: channel smoothing and a drift grid stepped twice per channel correlate neighbouring ones, which alone would leave 0.19 of them independent. The control ensembles say otherwise, their maxima sitting at 0.995 of the prediction for the maximum of the full searched count treated as independent and Gaussian – a budget predicting 480 windows with a control above $T_{\rm trig}$ against the 459 observed – while fitting each window's effective count against its own ensemble returns 0.81 of its searched cells. Nothing turns on which is right, because the expectation above is summed from measured rates and not from a cell count: by the cell-count route the number reaching the trigger is 36.6 against 36.2, 3 per cent away.

The on-star crossings do exceed that budget, and the attribution accounts for it. A Gaussian tail over every searched cell predicts 37.9 stellar exceedances against the 56 found. But 18 are attributed by the set above, 4 come from the block of §D.2 and 2 fall outside the census windows, which leaves 31 against the 24.7 the same control positions predict over both classes together: the Class A census expectation above with every other block's coarse windows added, since every Class B crossing comes from that one block. No excess survives the attribution.

The remainder is bounded, and the bounds are null. The measured stellar ranks agree at $p = 0.38$ with the 220 672 ranks each control takes against the other probes of its own window; radial suppression is below 10 per cent in the mean; and scrambling a window so nothing coherent survives along a drift track carries 4 past the rank floor and changes no disposition.

D.5. *The registration of the predicted cell*

The recurrence test of §5.6 reads one channel at one drift rate, so what it can conclude depends on how well that cell is registered between two epochs. Every term of the budget is a frequency the reading knows to a fraction of a channel: half a channel for the predicted cell, half for the discovery channel it is referred to, and 0.25 for the drift, which the grid steps in half a channel across one observation. The systemic velocity is removed at one epoch and re-applied at the other, so it cancels to 0.000 of a channel, and the ephemeris term is 0.007. The total is 0.75 of a channel in every window, however wide that channel is, and $0.32\,\mathrm{km\,s^{-1}}$ in the median one.

That is a budget and not a measurement, and the measurement is larger where the channels are narrow: the stellar-frame centroid of a real recurring line sits a median $0.05\,\mathrm{km\,s^{-1}}$ from the predicted cell and 0.45 at worst (§5.6). Each exclusion was therefore re-read over every cell within $0.45\,\mathrm{km\,s^{-1}}$ of the predicted one – 2 channels at most, 5 cells – and formed from the worst of them, with the threshold raised to 5.30 so that the wider reading keeps the single cell's own false-alarm probability. Reading the same cell at every trial of the window's own drift grid instead, which bounds an error in the carried drift whatever its size, gives 0.72 and 7 ruling out nothing; over all 444 readings the largest value anywhere is 4.71. The wider reading costs depth: the amplitude ruled out falls from a median 0.43 to 0.53 of the discovery amplitude, and 2 of the 33 then rule out nothing – β Pic at 241.77 GHz, β Pic at 241.86 GHz, both of them already among the 8 crossings §5.6 reports as untested and not excluded. No disposition rests on the carried drift or on the width of the cell.

What the positive controls read, and how precisely. 2 of the 13 line-attributed recoveries fall below the trigger in every single block read alone, reaching 1.48 and 3.25 there, and all 13 reach it with their covering blocks combined, between 5.1 and 56.2; the 2 attributed crossings that are not recovered read 4.18 and −0.97. The frame correction is what makes that work: left out, only 5 of the 13 would have come back at all. The centroid distance above is measured over 61 block readings and is a velocity rather than a number of channels – a small fraction of a channel in the widest of these windows and 41 channels in the narrowest. These lines carry drifts of a median $48\,\mathrm{Hz\,s^{-1}}$ against 1682 in the crossings that carry an exclusion, so they test the registration and the reading; the largest carried drifts are covered by the injections.

The reading has also to be pinned somewhere along the track, because a drifting carrier occupies tens of channels and not one: each observation is de-drifted from its own first integration, so the predicted cell is the frequency a carrier would occupy at the start of the repeat block. Pinning it anywhere across the span the discovery track covers moves the cell by at most $16.9\,\mathrm{km\,s^{-1}}$, while the nearest above-trigger cell in any covering window lies $56\,\mathrm{km\,s^{-1}}$ away; over every anchor the strongest reading is +2.51 combined and +3.19 in a single block. The anchor does change the depth of an exclusion, and for 2 of them – CP−72 2713 and 61 Vir – that is measured: at CP−72 2713 the amplitude ruled out moves from 0.66 to 1.19, which rules out nothing. That crossing is one of the 7 whose exclusion rules out nothing once the carried drift is given up altogether, and every one of those is already reported as untested and not excluded, so no disposition turns on the anchor either.

Part of the power the single cell loses is the threshold and not the archive. A statistic of five was set for a window of 10^2 to 10^6 cells, and at one predicted cell it is a far stricter demand – a false-alarm probability of 9×10^{-7}, evaluated at the scatter measured for that reading in §B.2 rather than at unity. Holding the chance of a single false confirmation across all 444 readings at 1 per cent instead puts the bar at 4.3, where 27 of the crossings that carry an exclusion reach 90 per cent recovery at the trigger amplitude against 25 – and nothing would have been confirmed there either, the largest single-block reading at the carried drift being 2.48.

A carrier that keeps drifting across the gap is a different model and the numbers belong to it. At the rates fitted here such a carrier is a median 371 channels from the predicted cell by the nearest covering block, and 1.4×10^5 at AU Mic, whose nearest coverage lies 310 d away in 488 kHz channels.

D.6. *From the widened window's exceedances to the pairs*

Two counts of §5.6 are counts of different objects, and Table D1 takes one to the other. The 29 above-trigger readings are window readings, one for each pairing of an unattributed crossing with a window that covers its frequency in another block, so a pair whose two members are both unattributed crossings is read twice, once from each member: 11 of the readings are such a second reading, so they represent 18 distinct pairs. The covering window is also wider than the interval the pair test admits, which is the median such window's half-width, so 2 of those pairs lie outside it, one of them by 2105 km s^{-1}, and 16 remain. That is the set Table 5 is formed over.

The observatory's own line-of-sight acceleration is removed from both fitted drifts before they are compared, at most 40 Hz s^{-1} here. Of the 7 pairs whose rates the grid resolves, every one fails a drift changing at a constant rate, by 3.6 standard deviations in the mildest case; the offset $\frac{1}{2}(\dot{\nu}_1 + \dot{\nu}_2)\Delta t$ such a trajectory predicts has no freedom left once both rates and the interval are fixed. The pair the searched grid itself separates is HD 14055's, whose two crossings lie 717 km s^{-1} apart in blocks 2.2 hours apart: a single carrier would have had to drift at 25 times the grid's ceiling between them.

Nothing measured here bounds the curvature in between. The ceiling bounds the line-of-sight component of a carrier's acceleration, which is not the same as bounding how fast that component can turn over: an orbit whose periastron is deeper than the ceiling but directed across the plane of the sky keeps the component inside it throughout and still sweeps it 4.5 times faster than the circular orbit at the ceiling, and one nearer face-on faster still, without limit. The ceiling does bound the velocity excursion, to within 1.7 times the circular amplitude over every eccentricity and orientation, which is why the widened window of §5.6 can be set against that excursion and a reversal time cannot.

β Pictoris is the closest case. Its two crossings sit at 241.7739 and 241.8594 GHz, 106 km s^{-1} apart in the star's frame, in blocks 24.4 hours apart. Both fitted drifts are negative, −1611 and −679 Hz s^{-1}, so a single carrier would have had to rise in frequency between the epochs, and under constant acceleration it misses by 132 standard deviations. But the mean acceleration such a carrier needs, 1.21 m s^{-2}, is inside the 3.6 m s^{-2} ceiling. The pair fixes three numbers – the acceleration at each epoch and the frequency offset between them – and a circular orbit about a star of 1.75 $M_\odot$ (Crifo et al. 1997) has three free parameters, so the orbit is determined rather than merely allowed: one reproduces all three at once, at 1.54 d in period, 16.5° in inclination and 0.0314 au in radius, and inside the ceiling it is the only circular one that does. Eccentric orbits fit as well up to e of about 0.28, above which the periastron acceleration leaves the ceiling, so the pair is not separable.

The control row is a line that does not drift. The CO(2–1) line of HD 285968 is recovered at one stellar-frame frequency 5.0 days apart and 0.6 standard deviations from the prediction; it is 1 of the 23 pairs whose two members are both already attributed to a molecular line, and the only one of them for which a drift and an epoch were both recorded. Its fitted drifts are 0.40 of one 73 Hz s^{-1} grid step from zero, and the predicted offset, −12.6 MHz, is below the comparison's own resolution over the same interval, 22 MHz. Both scale with the interval, so the comparison resolves a mean drift to 103 Hz s^{-1} and no finer at any baseline. For the 12 pairs it cannot separate – 6 of them resolved and failed by the constant-acceleration comparison – 1.0 days to nearly two years apart, nothing measurable tells one carrier from two.

TABLE D1
The 29 above-trigger readings of the widened recurrence window, taken to the pairs the trajectory test is asked of. Each block of rows is the one above it with one effect removed; the counts in the lower block are those of Table 5.

Covering windows read, over the 37 unattributed crossings a later block covers	328
... holding a statistic above the trigger, in 17 windows of 15 blocks	29
Distinct crossing pairs those readings represent	18
... readings that are the second reading of such a pair	11
Pairs separated by more than ±833 km s^{-1} in the star's frame, to 2105 km s^{-1}	2
Pairs inside that interval	16
Pairs with at least one unattributed member	16
... with a fitted drift and a block epoch for both members	13
... separated by the bounded-trajectory criterion	1
... open	12

D.7. *Reading the crossing ledger*

The ledger carries one row for each of the 56 threshold crossings, of which Table D2 prints 9.

Its columns are the star, its execution block and the ALMA band; the crossing frequency as the correlator delivered it, or, where no peak frequency was measured, recovered from the line offset; the search statistic $T_\star$ and the largest statistic $T_{\rm ctrl}$ among the window's 512 control positions; the nearest masked transition and the stellar-frame offset $\Delta v_\star$ from it; and then the recurrence test – $n_{\rm rep}$ covering repeat blocks spanning Δt days, the statistic $T_{\rm pers}$ a carrier persisting at the discovery flux density would have produced at the predicted stellar-frame cell, the statistic $T_{\rm rep}$ measured there at the carried drift, and the exclusion of Eq. B1. Three of those columns reproduce that exclusion, and two of them also give the limit defined in Appendix B. The frequency in the star's own rest frame is not a column of either: it follows from the three columns beside it, as $\nu_\star = \nu_{\rm topo}(1 - \Delta v_\star/c)$ about the masked transition named in the same row, and it is given directly, to six decimal places, in the machine-readable ledger.

The last column is each crossing's disposition, coded: *A* attributed, *AR* attributed and recovered in a separated block, *AN* attributed and not recovered, *U* unattributed with no recurrence, *US* unattributed with no retained spectrum to test, and *AF* or *UF* where the window is data-quality flagged (§D.2). The first letter is the attribution, so the *A* and *U* rows are the two populations every count in this paper is formed over.

Two kinds of blank appear in that test, in the ledger's own columns. Where the emission is present again in a covering block it returns a recovery and not an exclusion, so $T_{\rm rep}$ and $\sigma_{\rm excl}$ are blank and the disposition says which it was. Rows whose window is data-quality flagged

TABLE D2
A sample of the crossing ledger, from which every disposition in this paper can be reproduced. 9 of the 56 threshold crossings, one to a window: the 2 unattributed crossings that lead their own control positions, the 3 with no covering repeat and the one with no retained spectrum, and enough of the rest to show every disposition. **The full ledger accompanies this paper as data rather than as part of the article**, machine-readable, with the execution block of each crossing, its rest-frame frequency and the recurrence readings this sample omits. The columns, the coded dispositions, the blanks and the dagger are defined in §D.7.

Star	B	$\nu_{\rm topo}$ (GHz)	$T_\star$	$T_{\rm ctrl}$	line	$\Delta v_\star$	disp.
61 Vir	7	346.3138	5.958	5.65	$SO(9_8–8_7)$	−177.6	U
CP−72 2713	7	344.2697	5.809	5.68	$SO(8_8–7_7)$	−12.3†	U
HD 48370	6	230.5117	26.834	18.36	CO(2–1)	−17.8	AF
HD 139084 B	7	345.8241	5.763	17.70	CO(3–2)	+24.4	A
HIP 10679	6	220.4185	5.456	6.76	^{13}CO(2–1)	+6.5	A
HD 14055	7	331.1559	5.082	6.26	^{13}CO(3–2)	+513.8	US
BD+05 1668	6	240.1016	73.398	74.69	CS(5–4)	−5911.9	UF
β Pic	3	115.2603	30.765	7.34	CO(1–0)	−0.2	AR
HR 1010	6	230.5133	5.410	5.65	CO(2–1)	−12.0	AN

(§D.2) are marked, and on those $T_{\rm pers}$ and the exclusion are withheld instead: both are built from the discovery amplitude, and in a window whose control ring stands as high as its star that amplitude cannot be assigned to a point source at the star.

Not every interval in the ledger's Δt column is measured from the blocks' own stamps: where the archive does not index a block under its own identifier it carries the earliest start time of its observing unit, so two blocks of one unit would return an interval of zero by construction. No such interval is given as a zero, and Table D3 marks a separation so measured with a double dagger. The blocks of each unit so marked fall inside one day of each other on their own barycentric terms, so an interval of tens of days is good to the day.

A crossing is attributed by the set defined above, and a dagger marks the 3 rows whose coinciding species that set excludes. The transition named is the nearest in the star's frame, so where it is far from every transition it is the magnitude of the offset and not the name that carries the information. The delivered and rest-frame frequencies agree to the digits printed in one row only, HD 139084 B, where the barycentric and systemic terms – 5.2 MHz each, in opposite senses – cancel to 39 kHz. That star's catalogued systemic velocity, $+4.55\,\mathrm{km\,s^{-1}}$ (Zúñiga-Fernández et al. 2021), and the barycentric correction at its epoch, $+4.5161\,\mathrm{km\,s^{-1}}$, agree to $0.0339\,\mathrm{km\,s^{-1}}$ – 3.4 times the $0.01\,\mathrm{km\,s^{-1}}$ uncertainty the catalogue gives that velocity, and the only such agreement in 1162 comparisons of every star's systemic velocity against the barycentric correction of every block it was observed in. Its offset from CO(3–2) is $+24.4\,\mathrm{km\,s^{-1}}$ either way, and $+23.7\,\mathrm{km\,s^{-1}}$ on the primary's radial velocity, $+3.90\,\mathrm{km\,s^{-1}}$, which SIMBAD takes from a different catalogue, so the attribution does not turn on the value. That star is also the fainter component of a 10.3 arcsec pair. The 4 readings taken at its position are second extractions of the same 4 spectral windows of one execution block and not windows of their own, so the per-window census counts them once, under HD 139084. Both components sit 5.21 and 5.09 arcsec inside a 18.1–18.8 arcsec beam, with a response no lower than 0.79.

D.8. *The drift grid's edges*

The drift trials of a window run from $-\dot\nu_{\rm max}$ in equal steps of half a native channel across the retained track, and the last trial is the first at or beyond $+\dot\nu_{\rm max}$; the interval searched is therefore asymmetric by up to one step. Each window's grid is recovered here from two quantities that enter no common calculation – the trial count recorded for it and the channel width and retained span its own crossing record carries – and the two agree on 56 of the 59 crossings that carry a fitted drift. On every one of them the fitted rate is a trial of that grid exactly, so no rate in this survey is interpolated between trials. 5 lie on an outermost trial (Table D4), toward BD+05 1668, HD 14055, HD 161868, TRAPPIST−1; for those the statistic was maximised at the boundary of the interval searched and the rate is reported as a lower bound. The remaining 2 are in archival-tail windows outside the census, for which no grid is recorded anywhere, so for them the question cannot be settled either way.

The ceilings themselves were computed window by window from Equation 2 and reproduce it to 1×10^{-4} over all 405 Class A windows. Because the last trial is the first at or beyond the positive ceiling, the upper edge of the interval searched reaches 12.68 nHz over the windows that hold a crossing, and in the coarsest windows, where one step spans the whole interval, no more than 40.0.

E. RADIO-FREQUENCY INTERFERENCE SCREENING

Interference is the first alternative explanation for an unattributed crossing, so the screening this survey has and the coverage it lacks are both set out here. Millimetre interferometry is an unusually poor environment for terrestrial interference, though not an empty one: ITU Radio Regulations Article 5 (International Telecommunication Union 2024) carries active allocations throughout 90–275 GHz, and WRC-19 identified parts of 275–450 GHz for land-mobile and fixed use. The specific hazard is the 94.0–94.1 GHz Earth exploration-satellite allocation used by spaceborne cloud radar (Stephens et al. 2002; Wehr et al. 2023), a documented contaminant of ALMA Band 3 (Emerson 2004). Of the 1663 census windows 1 overlaps it, carrying 0 threshold crossings and 0 rank-passing events, so no result here rests on it. Two things favour this band. Terrestrial transmitters are sparse compared with centimetre wavelengths and, decisively, an interferometer is not a total-power detector: a signal not in the far field at the phase centre acquires a rapidly varying geometric phase across the array, and is suppressed in the correlation rather than imaged.

The coverage of the screening is uneven. The allocation test above and the sky- and intermediate-frequency tests below are computed for every searched window of the census, as is the 512-position rank screen, which asks whether an excess is at the star or everywhere in the field. Recurrence reaches only the crossings with a covering repeat block, and the visibility fit is a consistency check rather than a filter. The correlator flagging is the observatory's own, replayed from each block's delivered calibration script, and was not retained window by window. *No per-channel interference flag of our own exists, and none can be recovered.* Five measurements follow,

TABLE D3

The recurrence stage's positive controls, and the baseline each was recovered over. The 13 line-attributed crossings the criterion of §4.2 recovers, toward 2 stars: for each, the block whose own crossing satisfies it at the same stellar-frame frequency, that block's epoch, its separation from the discovery block and the statistic it reaches there. Block identifiers omit the constant `A002_` prefix and are printed as session and block, so the first field is the observing session: every recovering block belongs to a different session from its discovery block and lies more than 1 day from it, as the criterion demands. T is the statistic of that block's own crossing at that frequency, which is a maximum over every channel and drift of its window; $T_{\rm rep}$ beside it is the single-cell reading at the cell the discovery predicts, taken in that same block through the frame correction, the start-of-block anchor and the carried drift, which is how the 33 ledger exclusions are formed; and $T_{\rm comb}$ is the same reading with all of that crossing's covering blocks combined, which is the quantity §5.6 quotes. 11 of the 13 reach the trigger in the single block named here and 13 with the blocks combined. Δc is the cell's distance in channels from the brightest channel of that block's own search, a median 0.06 of a channel over the readings that reach the trigger. On a resolved line that distance is not the line's position: the 10 rows whose line is spread over as many as 69 channels of their own window test the frame correction only to the width of that line – the largest entry is 0.82 km s^{-1} in channels of 15.3 kHz, inside a line 0.6–2.9 km s^{-1} wide – and the registration of a single cell is tested by the 3 rows of HD 285968 alone, whose line is confined to one channel and whose cells land within 0.06 of one. † marks a row involving the one window the archive delivers twice, once with the control annulus on this compact-array block's own antenna geometry and once on a nominal 12 m beam (§4.2). The reading on the block's own geometry is the one published, 5.9557; the 12 m annulus gives 6.1886 for the same cell, 3.9 per cent apart, 1.16 per cent of that in the window noise and the rest in the recovered amplitude. The mark appears where that window supplies the reading and where it supplies the discovery the cell is predicted from; in the two rows where it supplies the reading, that is why the single-cell reading there stands above the window maximum beside it, the two being measurements of different products of one window. ‡ marks a separation measured from an observing unit's stamp rather than from a block's own. Four of these crossings are present in blocks of their own observing unit as well, whose internal separations the archive does not record; the ledger file flags those rows in a column of its own.

Star	transition	$\nu_\star$ (GHz)	discovery block	recovering block	epoch	Δt (d)	T	$T_{\rm rep}$	$T_{\rm comb}$	Δc
β Pic	CO(1–0)	115.2711	`Xf5d76d_Xcc5`	`Xf4df6f_Xc7`	2022-01-09	53‡	17.91	17.91	56.16	−0.02
β Pic	CO(1–0)	115.2711	`Xf5d76d_Xe19`	`Xf4df6f_Xc7`	2022-01-09	53‡	17.91	17.91	52.44	−0.02
β Pic	CO(1–0)	115.2711	`Xf5d76d_Xeb8`	`Xf4df6f_Xc7`	2022-01-09	53‡	17.91	17.91	51.13	−0.02
β Pic	CO(1–0)	115.2711	`Xf4df6f_Xc7`	`Xf5d76d_Xeb8`	2022-03-02	53‡	30.76	30.76	56.11	+0.02
β Pic	CO(1–0)	115.2713	`Xf5d76d_X32f1`	`Xf4df6f_Xc7`	2022-01-09	53	17.91	14.69	51.38	+0.95
β Pic	CO(2–1)	230.5377	`Xa7a216_X23e4`	`Xd9668b_X3a90`	2019-03-12	1312	11.68	7.85	6.43	−7.93
β Pic	CO(2–1)	230.5377	`Xd9668b_Xa9df`	`X6f1341_X1484`	2013-10-06	1984	9.83	9.42	8.45	−0.88
β Pic	CO(2–1)	230.5378	`Xd9668b_X3a90`	`X6f1341_X1484`	2013-10-06	1983	9.83	9.42	8.19	−0.56
β Pic	CO(2–1)	230.5379	`X6f1341_X1484`	`Xd9668b_X3a90`	2019-03-12	1983	11.68	1.48	5.10	+8.91
β Pic	CO(2–1)	230.5384	`X7116f1_X1785`	`Xd9668b_X3a90`	2019-03-12	1958	11.68	3.25	5.49	+41.08
HD 285968	CO(2–1)	230.5447	`Xdb7ab7_X5b4e`	`Xd98580_X3bf6`	2019-03-15	44	5.96	6.19†	11.55	−0.06
HD 285968	CO(2–1)	230.5447	`Xdb4b9a_X108f`	`Xd98580_X3bf6`	2019-03-15	39	5.96	6.19†	11.61	−0.05
HD 285968	CO(2–1)	230.5447	`Xd98580_X3bf6`	`Xdb7ab7_X5b4e`	2019-04-28	44	5.87	5.87†	11.44	+0.06

TABLE D4

The crossings whose fitted drift lies on an outermost trial of their window's own grid, with the grid each is measured against. Rates are in the observatory frame, the frame the grid is stepped in; Table 5 gives them in the star's. A maximum at the boundary of the interval searched is a lower bound on the rate.

Star	block	ν (GHz)	$\dot\nu_{\rm obs}$ (Hz s^{-1})	trial	$\lvert a_{\rm los}\rvert$ (m s^{-2})
BD+05 1668	`X28a8`	242.195	+3452	2/2	$\geq$ 4.27
HD 14055	`X29ae`	331.156	−3976	0/135	$\geq$ 3.60
HD 14055	`X2c3bb`	345.526	+4184	141/141	$\geq$ 3.63
HD 161868	`X10ca8`	345.634	+4195	127/127	$\geq$ 3.64
TRAPPIST−1	`X259b`	346.245	−4623	0/137	$\geq$ 4.00

and each could have failed.

First, clustering in sky frequency. If interference dominated the crossing population, the crossings would pile up at particular sky frequencies. They do cluster, but on the survey's own molecular lines. The test needs a measured peak frequency in the frame an interfering carrier would stand still in, so it runs topocentrically over the 51 of the 56 crossings for which one was measured, against the transitions the search was tuned to. The other 5 are carbon monoxide toward β Pic, four in CO(1–0) and one in CO(2–1) of its ten such crossings; for those the catalogue carries only the velocity offset from the line, so the frequency the ledger prints is recovered from that offset and not measured, and a frequency reconstructed from a line offset cannot be used to ask whether frequencies crowd onto lines. Of the 51, 11 fall within ± 50 km s^{-1} of a searched transition. The expectation if each window's peak fell anywhere on that window's own channel grid is 4.8, $p = 0.0010$. The densest band, 230.422–230.723 GHz, holds 12 of them, 9 within that velocity window of CO(2→1). Whether the crossings crowd onto one another rather than onto the lines is a different question. The statistic for it is the number of pairs lying within a given separation in sky frequency, counted against the same peaks redrawn on their own channel grids, and it is read at 1 and 300 and the four decades between. The 13 crossings the mask attributes crowd together at every one of those separations, at $p < 10^{-4}$ where the test is weakest. The 34 unattributed crossings of the quality-passing data crowd at none of them, the smallest p of the six being 0.08. At the widest separation tried, 300 MHz, they make 32 pairs against 24.4 expected. Returning the 4 flagged crossings to the sample leaves that unchanged ($p \geq 0.10$). The two samples are divided by the same mask and measured by the same statistic, which is not itself a distance from a line, so the second could have crowded and did not. *The concentration is the attribution, not a residue left over after it.* The ledger counts the same band as 14 crossings with 11 attributed, over a different set – all 56 of them, each attributed in its own star's rest frame – and the two rows it adds are the ones this test cannot hold: one whose peak

frequency the release records only as an offset from the line and one whose window the release does not record as a crossing at all.

Second, the same test against the atmosphere. A residual of the atmospheric calibration stands still in topocentric frequency exactly as a transmitter does, and it is narrow where the atmosphere has lines. 570 transitions of ozone, water and oxygen lie within 89.6–873.1 GHz (Pickett et al. 1998), a median 0.9 GHz apart. An interpolated system-temperature residual occupies at least one of the 31.25 MHz channels on which that temperature is measured and reaches into the wings of the strong lines, so the crossings are screened at 2 such channels, 62.5 MHz, in the topocentric frame the atmosphere occupies, and not in the stellar frame the crossing frequencies are quoted in. 1 crossing of the 64 lies inside that radius: HD 69830 at 321.2005 GHz topocentric, 25.2 MHz from the H_2O line at 321.2257 GHz, which in the stellar frame it is quoted in appears 11 MHz away. It is treated as atmospheric and is not counted among the unattributed. The screen is not sharp where it fires, and that is the honest reading of it: this window is tuned on the line, so 95 per cent of its own channels lie within the same radius of the same transition, and the coincidence establishes not that this channel is atmospheric but that this window cannot separate the two. Removing the crossing is the conservative response. The same radius is taken off the control side for the same reason – in such a window every control channel is inside it too – although the two sides are screened by the same rule and not by the same estimator: each crossing is tested at its own frequency, while the control side enters the expectation of Appendix D as one surviving fraction formed over all windows and applied to every window's rate. That pooled fraction is the larger of the two, so the screen takes proportionally more from the expectation than from the observed count, and the residual asymmetry therefore runs against the null and not for it. The median separation is 0.81 GHz. 1 crossing lies inside the same $\pm 50\,\mathrm{km\,s^{-1}}$ window used above, against 2.1 expected from the windows' own channel grids. The one concentration a reader will notice is 15 crossings of the census and the hold-out between 344.2 and 346.4 GHz. It is not atmospheric either, because that interval holds 0 telluric transitions. It is instead the carbon monoxide tuning seen from the other side: 12 of the 43 unattributed crossings whose window the catalogue tabulates fall in the band, against 11.8 ± 0.9 expected from where the spectral windows were tuned.

Third, fixed intermediate frequency. An artefact of the signal chain would sit at a fixed position within the spectral window rather than at a fixed sky frequency. This test is run over every window for which the catalogue releases a peak frequency – 1401 of the 1663, of which only 51 carry a threshold crossing – because such an artefact would occupy the same intermediate frequency in all of them and not only where the trigger fires. It does not: the peaks' fractional positions within their own windows have median 0.50 and are uniform.

Fourth, a cross-target occupancy screen over every window the pipeline extracted, not only the crossings. That set is wider than the census: 3027 windows toward 133 stars, being the census together with the rows its audit removed, the reserved hold-out, the archival extension, the out-of-sample control, and 52 windows in 13 blocks that entered none of them. Interference belongs to the observatory and not to the star, so a terrestrial carrier should recur at the same sky frequency toward *unrelated* stars. It does not, and the sharpest place to look is the two unattributed crossings that outrank every control position in their own window, 61 Vir and CP−72 2713, both of which carry a measured frequency. Taking them in that order, 346.314 and 344.270 GHz are covered by 36 and 42 further windows, toward 24 and 8 other stars in 32 and 42 other blocks, and none of them carries a crossing in the same channel. With each window's peak randomised on its own channel grid, the number of cross-star coincidences within 5 MHz is 569 against 554 ± 13 expected ($p = 0.15$). Among the 109 windows whose peak exceeds the trigger it is 4 against 5.2 ± 1.9 – *below* the null. With the molecular-line mask applied as well it is 1 against 2.1 ± 1.3. There is no excess to explain. The mask has to be applied first, because 35 of those 109 windows sit on a masked transition (CO(2–1) ×17, CO(1–0) ×10, CO(3–2) ×4, ^{13}CO(2–1) ×3, SiO(5–4) ×1), and a carbon monoxide line falls at nearly the same sky frequency toward every nearby star, exactly the coincidence the screen is looking for.

Fifth, the image sideband. A sideband-separating receiver passes the image frequency $2\nu_{\mathrm{LO1}} - \nu$ at finite rejection, so a bright line in the other sideband can appear where no line is catalogued. The first local oscillator is not among the quantities the archive delivers with these products, but each block's own spectral windows bound it: every window must fall inside one sideband at an intermediate frequency the receiver passes, which pins ν_{LO1} to a median 0.40 GHz from a median of 4 windows, against 7.3 GHz from the crossing's own window alone. Reflecting each crossing about that interval places its image 8–24 GHz away, never inside the window the crossing was found in. Of the 36 unattributed crossings whose blocks the catalogue describes, 23 have no transition of the line catalogue anywhere in that interval, and 7 have one whose frequency the same block observed with nothing there bright enough to leak: ALMA specifies an image rejection of at least 10 dB, so a parent would have to be 10 times the crossing's own flux density. For the remaining 6 the candidate parent frequency was not observed, and the parent would be a hydrogen recombination or sulphur monoxide line of that strength, which no field in this survey shows. The screen rests on the receiver's quoted passbands: taken 1 GHz wider at each end it leaves 2 of the 36 excluded on frequency alone, so it is a bound and not a clearance. 0 unattributed crossings lie in Band 9 or Band 10, where the receivers are double-sideband and no such bound could be drawn. The resolved lines these fields do hold are many channels wide, and so would their images be, which the search retains; so the image position is measured instead of argued. Reflecting each Class A window about the interval its own block's windows leave for the first local oscillator, the image band of 145 of the 405 windows covers one of 14 carbon monoxide isotopologue transitions at the star's velocity or at rest in the local standard of rest. Those windows hold 19 crossings, of which 8 lie at the image position against 10.1 expected from the share of the window the image interval occupies – the oscillator is bounded and not known, and that share has a median

0.24. 2 of the 8 are unattributed: ALMA J153702653-33192492 at 218.8729 GHz, whose image the bounded oscillator localises only to within 2 190 MHz of CO(2–1), and HD 161868 at 345.6338 GHz, to within 424 MHz of ^{13}CO(3–2). Neither parent attributes its crossing, by the test above: each was observed at its own frequency in the same block, where no feature reaches the trigger – 4.84 at most – and the largest amplitude is 1.37 and 0.93 times the crossing's own, where a leak at the quoted rejection would need 10 times as much.

One execution block fails these tests, and it fails them as a block and not as a carrier. Its 4 crossings toward BD+05 1668 are the 4 highest statistics in the survey (§5.3.2, which identifies the failure). Whether interference accounts for them is what belongs here, and three readings say not. The visibility fit returns a large imaginary part, placing the emission away from the phase centre, and the block's 24 sibling executions of the same field, array and spectral setup produce no crossing at all. Each of its frequencies is covered by 127–137 further windows toward 17–22 other stars, of which 0 carries a crossing in the same channel.

None of this substitutes for the ON/OFF cadence a dedicated search uses. The claim is only that no identified interference mechanism accounts for the unattributed events: allocation tables cannot exclude harmonics, intermodulation products, unconventional emitters or near-field sources, and archival data without a simultaneous off-source strategy cannot do better. The first four tests establish that the crossing population carries none of the signatures interference leaves, and the fifth bounds the one mechanism that could put a crossing where no line is catalogued – statements about the population, not a clearance of any one crossing.

F. PER-SYSTEM SENSITIVITY

The limit Table F1 quotes for a system is that of its best window, so a system whose coverage is wide is not penalised for the shallowest window in it; the whole distribution, window by window, is the per-window result the sensitivity of §5.1 is drawn from.

The limits span more than two orders of magnitude, so no single survey depth describes the sample, and the frequency at which a system's best limit falls is set by what the archive happens to hold toward it and not by any property of the star. A limit also says nothing about whether a crossing there could have been followed up: for 21 of the 61 systems no sensitivity would have been enough, because no block of another observing session more than a day away covers a frequency the first one did.

The decorrelation column is measured where the delivery allows it and imputed where it does not. A system's best window lies in a block with no phase record of its own for 34 of the 61 systems, against 26 measured, and the deepest limits are no better served than the rest: of the 10 deepest, 4 carry a measured correction and 5 an imputed one, so the median quoted in §5.1 is a mixture of the two, 4 per cent over the measured and 3 over the imputed. The systems whose best limit carries more than a tenth are η Crv, GJ 14, HD 172555, TWA 3A, HIP 10679, HD 155555C, CD−57 1054, CP−72 2713, TWA 7, UCAC2 20312880, LP 353-51, 51 Eri and HD 377. The 128 blocks whose delivery exposes no record are not a random subset of the 407: of the 125 of them that carry a baseline, 54 per cent were taken with the 12 m array, against 26 per cent of the blocks that do expose a record, and it is that array's long baselines that lose coherence. Within each array the two sets are not distinguishable – 44 against 44 m in the median on the compact array and 456 against 448 m on the 12 m one – so the archive withholds a configuration rather than a baseline. Matching each of their windows to the measured blocks within a factor of two in baseline length puts the correction at 1.9 per cent in the median, rising to 21 per cent were their coherence as poor as the ninetieth percentile of the blocks matched to them, so those limits are conditional on a calibration performance the archive does not report.

The injected set is drawn by block, and that cost is measured. It spans both delivered response forms, 32 windows at the native channelisation and 17 with channels averaged online, and splitting it there moves the ninety per cent point by $0.20\,P_{\rm trig}$. The draw controlled the array, but the achieved mix still departs from the population by 22 per cent of the sample; re-weighting the pooled curve by the survey's own composition gives $2.98\,P_{\rm trig}$ against the adopted 3.03. Because a system's limit is its best window, a minimum over several estimates could favour windows whose own factor falls below the class average; drawing each window's factor from the 49 measured ones instead, over 4 000 trials, leaves the per-system median where it is – the shift is 0.0 per cent – and holds the count at 10^{15} W to 24–26 in nine trials in ten.

G. THE FIGURE OF MERIT, TERM BY TERM

The conventional measure for comparing programmes is the continuous-waveform transmitter figure of merit of Enriquez et al. (2017), CWTFM $= \zeta_{\rm AO}\,{\rm EIRP}_{\rm min}/(N_\star \nu_{\rm rel})$ with $\nu_{\rm rel} = \Delta\nu/\nu_{\rm mid}$, smaller being better. Those programmes quote ${\rm EIRP}_{\rm min}$ at their own detection thresholds, and the measure is linear in it. Put on the common footing of five (Table 1), 0.85 becomes 0.17 and 0.14 becomes 0.072. That runs against this survey, making the comparison programmes deeper than their own papers report – on one arithmetic convention and not at a matched false-alarm probability. Margot et al. (2023) publish no CWTFM; the value quoted here is the same expression on the three quantities they do give – 11 680 stars, 1.15–1.73 GHz and EIRP $> 1.35 \times 10^{13}$ W. This experiment takes no such factor.

The denominator used here modifies theirs: in place of a single survey-wide fractional bandwidth it is the star-bandwidth product $\sum_i \Delta\nu_i/\nu_i$, one term per system, in which $\Delta\nu_i$ is the union of that system's own searched windows and ν_i the midpoint of the span those windows cover. That is not the frequency of its best window, which is what Table F1 lists; formed with those the same sum is 0.70, and the smaller of the two is adopted. The two are one quantity for a survey in which every star is observed over the same band, which the programmes compared here are, while for an archive a single fractional bandwidth credits every system with frequencies searched toward another. Measured term by term it is

TABLE F1

Per-system sensitivity of the carrier search. For each of the 61 systems the fine-channel search covers, the $EIRP_{90}$ of its best window (Table 3), with that window's own primary-beam, annual-parallax and smearing retentions applied, beside that window's centre frequency ν and native channel width $\Delta\nu$ and the system's merged coverage $\Delta\nu_{cov}$. Rows are ordered by the limit, and a gravitationally bound pair is one system under its primary's name. **Every limit here is nominal in one respect: it excludes atmospheric decorrelation**, and *dec.* is the percentage by which that leaves it optimistic − measured from the phase calibrator of its own execution block, starred where the delivery exposes no phase record and the figure is the median of the blocks matched to it in baseline length, and dashed for ϵ Eri, the one system whose blocks offer neither (§5.1). It is reported and not folded in; ϵ Eri's window also omits the annual parallax and is optimistic by at most 9 per cent on that account (Appendix A.2). The last column records whether two execution blocks of different observing sessions cover a common frequency more than one day apart (d) or more than a year apart (yr), over the primary census together with the 166 repeat blocks the recurrence test of §5.6 reads, of which 42 lie outside the census (§5.6.1). Both are lower bounds, since two blocks can be shown to be separated but never shown to be simultaneous. **A dagger marks the 6 systems whose own best window was one of the 49 injected into, so that their completeness is measured**; the other 55 limits are transferred from the class average and carry instead the dispersion observed between injected windows, ×0.80 to ×1.58, wider than the calibration interval of Table 4. For the 4 systems whose best window lies in a band or channelisation that holds no injected window (§5.1) that dispersion is itself transferred, and is not a bound measured in their own configuration.

System	$EIRP_{90}$ (10^{14} W)	ν (GHz)	$\Delta\nu$ (kHz)	$\Delta\nu_{cov}$ (GHz)	dec. (%)	conf.	System	$EIRP_{90}$ (10^{14} W)	ν (GHz)	$\Delta\nu$ (kHz)	$\Delta\nu_{cov}$ (GHz)	dec. (%)	conf.
β Pic	0.71	230.52	15.3	4.4	2*	yr	HD 170773	17	230.13	488	7.4	2*	d
LP 876-10	0.87	345.83	488	1.7	4*	d	HD 61005	19	230.52	244	19	3*	d
AU Mic	1.2	230.56	488	3.5	3*	yr	HR 1010	19	230.53	977	2.8	1	d
τ Cet	1.2	230.55	488	1.8	0.4*	d	LHS 1140	19	230.00	488	1.8	0.4*	d
HD 69830	1.2	230.06	488	3.8	1	d	κ CrB	19†	230.57	977	1.7	2*	–
61 Vir	1.3	345.79	488	1.8	5*	yr	HD 155555C	21†	346.01	488	1.7	43	–
η Crv	1.3	345.80	61.0	7.2	28	yr	L 836-122	22	346.00	488	1.7	1	–
ϵ Eri	1.4	229.98	977	2.3	–	yr	CD−57 1054	23	345.99	488	1.7	20	d
TRAPPIST-1	1.6	346.58	488	1.7	5	d	CP−72 2713	24	345.12	488	5.8	64	d
AT Mic AB	1.7	346.04	488	1.7	3	–	HD 14055	24	230.11	488	7.2	3	yr
GJ 3379	2.2	229.96	488	1.9	0.4*	d	TWA 7	24	345.99	244	0.87	26*	d
GJ 14	3.1	227.22	488	3.5	19	d	GJ 2006A	24	346.02	488	1.7	1	–
HN Lib	3.9†	229.99	488	1.8	0.4*	d	Wolf 219	26	230.13	488	1.7	2*	d
HD 172555	4.0	230.53	15.3	5.3	40*	–	LP 736-15	28	345.82	977	1.7	4*	–
HD 10647	4.4	345.79	488	6.1	0.8	yr	TYC 2211-1309-1	28	346.00	488	1.7	5	–
HD 202628	5.4	230.51	488	1.8	4*	yr	UCAC3 176-23654	29	346.02	488	1.7	4	–
HD 53143	5.4	230.53	977	1.8	2*	yr	PM J03433+1958	31	230.13	488	1.7	2*	–
HD 207129	6.4	230.57	488	1.8	3*	yr	UCAC2 20312880	32	345.96	488	1.7	11	–
GJ 849	6.8	230.02	488	1.9	0.4*	d	LP 353-51	33	346.02	488	1.7	26	–
TWA 3A	7.6†	230.54	122	0.44	35*	d	Barta 161 12	34	346.02	488	1.7	4*	–
HD 139664	7.8	230.55	488	1.7	3*	–	51 Eri	37	230.53	1953	1.7	11*	–
HD 92945	8.3	355.11	488	3.7	6*	d	HD 23484	37	230.53	977	1.8	0.2	d
TX PsA	8.5	346.03	488	1.7	7	–	HD 139084	54	346.01	488	1.7	3	–
HD 285968	9.1	229.97	488	1.8	0.4*	d	WD 0407+179	84	230.13	488	1.7	2*	–
HIP 10679	9.4	230.55	15.3	0.11	31*	–	HD 48370	88†	230.73	122	3.3	0.4*	yr
HD 161868	9.9	227.22	488	7.4	5	d	HD 45184	93	230.53	977	1.7	2	d
HD 38858	13	230.50	488	1.8	3*	yr	ALMA J153702653-33192492	99	345.98	488	14	5*	d
HD 110411	13	219.58	244	4.7	2*	yr	HD 31392	130	230.52	977	1.7	0.9	d
HIP 490	13†	230.53	244	0.86	3*	–	HD 89452	260	230.53	977	1.7	0.7	d
HD 107146	14	355.14	488	3.7	6*	d	HD 377	370	230.54	977	1.7	68	d
LP 476-207	15	346.02	488	1.7	4	–							

0.64 over the 61 systems, against 5.9 on the uniform convention. That factor of 9.2 is there because the median system was covered over only 1.84 GHz of the 47.7 GHz union.

On that denominator the Class A experiment is 9.5×10^5 on the sample's largest threshold power and 3.7×10^4 on the median system. On the $EIRP_{90}$ adopted everywhere else here it is 2.9×10^6 and 1.1×10^5, a completeness rather than a threshold, so the comparison with the published values is made on the bare thresholds. The median here is a factor 4.4×10^4 above the published Enriquez value and 2.2×10^5 above it once that value is put on the common threshold of five. That gap is 2.76 decades of haystack and 2.58 decades of depth, so neither explains it alone: this is a far smaller haystack than a wideband centimetre survey of thousands of stars and a shallower one, and the measure compares surveys instead of estimating occurrence rates.

REFERENCES

Akeson R. L., et al., 2013, PASP, 125, 989
ALMA Science Portal, 2026, What is the tested capability of ALMA for bandpass calibration precision? ALMA Knowledgebase, General ALMA Queries, `https://help.almascience.org/kb/articles/what-is-the-tested-capability-of-alma-for-bandpass-calibration-precision`, accessed 2026 October 14
Anglada-Escudé G., et al., 2016, Nature, 536, 437
Astropy Collaboration, 2022, ApJ, 935, 167
Backus P. R., Project Phoenix Team, 2002, in Stanimirović S., Altschuler D., Goldsmith P., Salter C., eds, ASP Conf. Ser. Vol. 278, Single-Dish Radio Astronomy: Techniques and Applications. Astron. Soc. Pac., San Francisco, p. 525
Blomme R., et al., 2023, A&A, 674, A7
Bond H. E., et al., 2017, ApJ, 840, 70
Burton K., MacGregor M. A., Osten R. A., 2022, ApJ, 939, L6
CASA Team, 2022, PASP, 134, 114501
Cataldi G., et al., 2023, ApJ, 951, 111
Chen T. X., et al., 2022, ApJS, 260, 5
Choza C., et al., 2024, AJ, 167, 10
Cocconi G., Morrison P., 1959, Nature, 184, 844
Cortes P. C., et al., 2026, ALMA Cycle 13 Technical Handbook, ALMA Doc. 13.3, ver. 1.0, doi:10.5281/zenodo.18793803
Crifo F., Vidal-Madjar A., Lallement R., Ferlet R., Gerbaldi M., 1997, A&A, 320, L29
Czech D. J., et al., 2021, PASP, 133, 064502
Dame T. M., Hartmann D., Thaddeus P., 2001, ApJ, 547, 792
Dame T. M., Thaddeus P., 2022, ApJS, 262, 5
Dent W. R. F., et al., 2014, Science, 343, 1490
Drake F. D., 1961, Physics Today, 14, 40
Emerson D., 2004, The CloudSat radar and implications for ALMA, ALMA Memo 504
Enriquez J. E., et al., 2017, ApJ, 849, 104
Gaia Collaboration, Vallenari A., et al., 2023, A&A, 674, A1
Garrett M. A., 2026, MNRAS, 549, stag991
Ginsburg A., et al., 2019, AJ, 157, 98
Harris C. R., et al., 2020, Nature, 585, 357
Hunter T. R., Phillips N. M., Broguière D., González J., 2018, Improving the Calibration of Atmospheric Spectral Features in ALMA Data. NA ALMA 2016 Development Study Final Report, 2018 March 2
International Telecommunication Union, 2024, Radio Regulations, Edition of 2024. ITU, Geneva
Jacobson-Bell B., et al., 2025, AJ, 169, 206
Johnson O. A., et al., 2023, AJ, 166, 193
Kameno S., 2014, ALMA Cross Correlation Bandpass Stability. ALMA Technical Note 15 (draft), 2014 December 17
Kervella P., Thévenin F., Lovis C., 2017, A&A, 598, L7
Kervella P., Arenou F., Thévenin F., 2022, A&A, 657, A7
Li J.-K., Tao Z.-Z., Zhang T.-J., 2026, ApJ, 1006, 163
Ma P. X., et al., 2023, Nature Astronomy, 7, 492
MacGregor M. A., Osten R. A., Hughes A. M., 2020, ApJ, 891, 80
Manunza L., et al., 2025, Acta Astronautica, 233, 155
Margot J.-L., et al., 2023, AJ, 166, 206
Mason B. D., Wycoff G. L., Hartkopf W. I., Douglass G. G., Worley C. E., 2001, AJ, 122, 3466
Mason L. A., Garrett M. A., Wandia K., Siemion A. P. V., 2025, MNRAS, 536, 2127
Matrà L., et al., 2017, MNRAS, 464, 1415
Matson R. A., Williams S. J., Hartkopf W. I., Mason B. D., 2026, Sixth Catalog of Orbits of Visual Binary Stars, U.S. Naval Observatory, accessed 2026 October 14
Mauersberger R., Wilson T. L., Rood R. T., Bania T. M., Hein H., Linhart A., 1996, A&A, 306, 141
Morrison I. S., et al., 2023, PASA, 40, e019
Müller H. S. P., Schlöder F., Stutzki J., Winnewisser G., 2005, J. Mol. Struct., 742, 215
Pardo S., Poznanski D., Croft S., Siemion A. P. V., Lebofsky M., 2025, AJ, 170, 12
Pecaut M. J., Mamajek E. E., 2013, ApJS, 208, 9
Pickett H. M., Poynter R. L., Cohen E. A., Delitsky M. L., Pearson J. C., Müller H. S. P., 1998, J. Quant. Spectrosc. Radiat. Transfer, 60, 883
Planck Collaboration, Adam R., et al., 2016, A&A, 594, A10
Price D. C., et al., 2020, AJ, 159, 86
Privon G., et al., 2026, ALMA Cycle 13 Proposer's Guide, ALMA Doc. 13.2, ver. 1.0, doi:10.5281/zenodo.18793859
Remijan A. J., Markwick-Kemper A., ALMA Working Group on Spectral Line Frequencies, 2007, in American Astronomical Society Meeting Abstracts 211, 132.11
Richards A. M. S., Moravec E., Etoka S., Fomalont E. B., Pérez-Sánchez A. F., Toribio M. C., Laing R. A., 2022, Self-calibration and improving image fidelity for ALMA and other radio interferometers, ALMA Memo 620
Sheikh S. Z., 2020, International Journal of Astrobiology, 19, 237
Sheikh S. Z., Wright J. T., Siemion A. P. V., Enriquez J. E., 2019, ApJ, 884, 14
Sheikh S. Z., et al., 2021, Nature Astronomy, 5, 1153
Sheikh S. Z., Huston M. J., Fan P., Wright J. T., Beatty T., Martini C., Kopparapu R., Frank A., 2025, AJ, 169, 118
Steffes P. G., DeBoer D. R., 1994, Icarus, 107, 215
Stephens G. L., et al., 2002, Bull. Am. Meteorol. Soc., 83, 1771
Tarter J., 2001, ARA&A, 39, 511
Tremblay C. D., Tingay S. J., 2024, ApJ, 972, 76
Vidal C., et al., 2026, arXiv:2605.21093 (preprint)
Virtanen P., et al., 2020, Nature Methods, 17, 261
Wehr T., et al., 2023, Atmos. Meas. Tech., 16, 3581
Wenger M., et al., 2000, A&AS, 143, 9
Wright J. T., Kanodia S., Lubar E., 2018, AJ, 156, 260
Zhang Z.-S., et al., 2020, ApJ, 891, 174
Zúñiga-Fernández S., et al., 2021, A&A, 645, A30